



Affirmed Networks Lawful Intercept Module Specification and Provisioning Guide

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Preface

This document lists the specification for the Affirmed Networks Lawful Intercept (LI) Module. It defines the Affirmed Networks X1, X2, and X3 interfaces definition and describes how to provision and manage LI interfaces.

The Affirmed Networks LI implementation conforms to the standards developed by the European Telecommunications Standards Institute (ETSI) and the United States Communications Assistance for Law Enforcement Act (CALEA).

In this document, the term CALEA applies to both the LI and CALEA interface.

Audience

This guide is written for the following audiences:

- Developers of external devices supporting ADMF and MF/DF functionality that need to interoperate with the Affirmed Networks system.
- Operators configuring and managing CALEA interfaces.

As a reader of this guide, you should be familiar with Lawful Intercept and CALEA concepts.

How to Use This Guide

This guide contains the following chapters and appendices:

Chapter/Appendix	Describes
Chapter 1, Overview	Overview of the Affirmed Networks implementation of CALEA
Chapter 2, CALEA CLI Commands	Details of the CALEA CLI commands
Chapter 3, AN-FORMAT ASN.1 Definition	AN-FORMAT CALEA Module Definition
Chapter 4, AN-FORMAT IRI Event Details	AN-FORMAT Intercept Related Information (IRI) Events
Chapter 5, HI-FORMAT ASN.1 Definition	HI-FORMAT CALEA Module Definition
Chapter 6, HI-FORMAT IRI Event Details	HI-FORMAT Intercept Related Information (IRI) Events
Chapter 7, CALEA Alarms	CALEA alarms
Appendix A, CALEA Examples	Sample commands for configuring, monitoring, and deleting CALEA information

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management .
Variable parameters for which you supply values	<i><courier bold italic blue font></i>	configure_dr convert slave --masterhost <host IP>
User input	Courier bold blue font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.	 	Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<host name>
Asterisk after a CLI object indicates a required object	*	name*
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>Acuitas User's Guide</i> . <i>Workflow</i> is an application running on the network element.



CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

Note: Additional information that may apply to the subject text.

Tip: Helpful suggestions.

New Features and Enhancements

This release includes the following new features or enhancements.

Feature	Description
Added support to encrypt non-transient in-memory and inter-VM identities for Lawful Intercept (LI) targets	Added support to use AES-256 to encrypt non-transient in-memory and inter-VM identities for Lawful Intercept (LI) targets, including: ■ Identities: IMSI, IMEI, MSISDN ■ Lawful Intercept Identifier (LIID) ■ Correlation ID The LI Administrator (ADMF) configures the AES-256 encryption key using the Public Key Infrastructure (PKI) CLI: <pre>pki install type calea name target-key encrypt object <key-text></pre> <pre>pki remove type calea name target-key</pre> where <key-text> is always 32 bytes and can be specified in one of the following formats: ■ raw bytes – 32 characters ■ base64 – 44 characters ■ hex – 64 characters Issue the following command to verify that AES-256 encryption is enabled: <pre>show calea service-status</pre> The command displays the encryption statistics, including: <pre>target-encryption-type aes256</pre> which indicates that the encryption is enabled.

Feature	Description
Added support to use AES-256 to encrypt all Lawful Intercept (LI) information passed using PFCP over Sxa and Sxbc interfaces	Added support to use AES-256 to encrypt all LI information passed using PFCP over Sxa and Sxbc interfaces. These interfaces apply only to CUPS configurations. Previous Sxa and Sxbc interface encryption supported only MD5. The LI Administrator (ADMF) enables encryption: <pre>calea pfcplawful-intercept-ie admin-state enabled encryption [disabled md5 aes256]</pre> (Default: disabled) In addition: ■ To enhance security, configure a secret-key: <pre>calea pfcplawful-intercept-ie secret-key <key-text></pre> ■ If the secret-key <i>is not</i> configured, but an LI Target Encryption is configured, the specified AES-256 key is used as the secret-key: <pre>pki install type calea name target-key encrypt object <key-text></pre> ■ If neither is configured, a default hard-coded key is used.
Support for persistent and non-persistent multiple X2/X3 Delivery Destinations feature	Multiple X2/X3 Delivery Destinations feature is now supported with non-persistent and persistent targets using the following commands: ■ <code>calea createTarget</code> ■ <code>calea target-table</code>

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Overview

This section provides an overview of the Affirmed Networks system support for the *Communications Assistance for Law Enforcement Act* (CALEA).

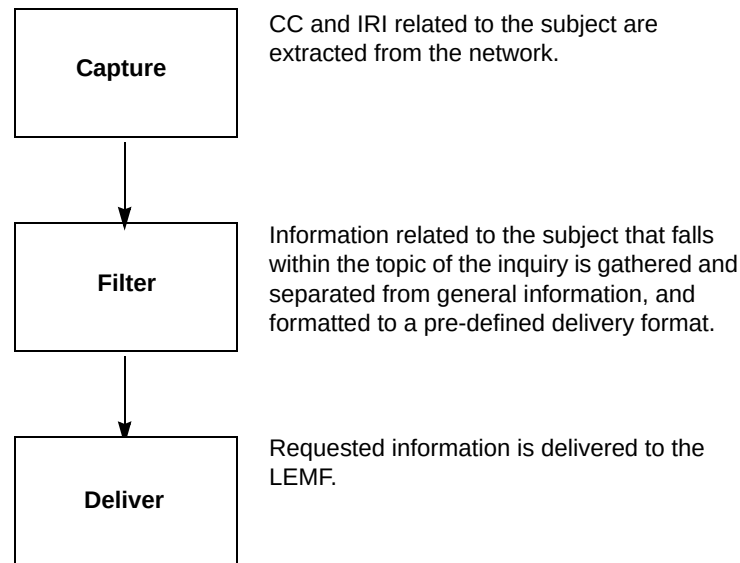
Introduction

The Affirmed Networks system supports the *Communications Assistance for Law Enforcement Act* (CALEA). On the system, the Lawful Intercept (LI) component of CALEA is a logical entity that spans application, infrastructure, and fastpath components and provides standards-based LI functionality. Lawful Intercept provides Operators with a mechanism to assist Law Enforcement Agencies (LEAs) in the monitoring of individuals (referred to as Targets). Lawful Intercept collects two types of data:

- *Intercept Related Information* (IRI) (referred to as Call Data or Call Identifying Information in the United States) consists of information about the Target communication itself such as signalling information, source and destination (telephone numbers, IP or MAC addresses, etc), frequency, duration, and time and date of communications.
- *Content of Communication* (CC) (sometimes referred to as Call Content) data may include voice, video, and text messages.

The act of LI, independent of the type of communication that is intercepted, involves a process with three distinct steps as shown in [Figure 1-1](#).

Figure 1-1 Lawful Intercept Process

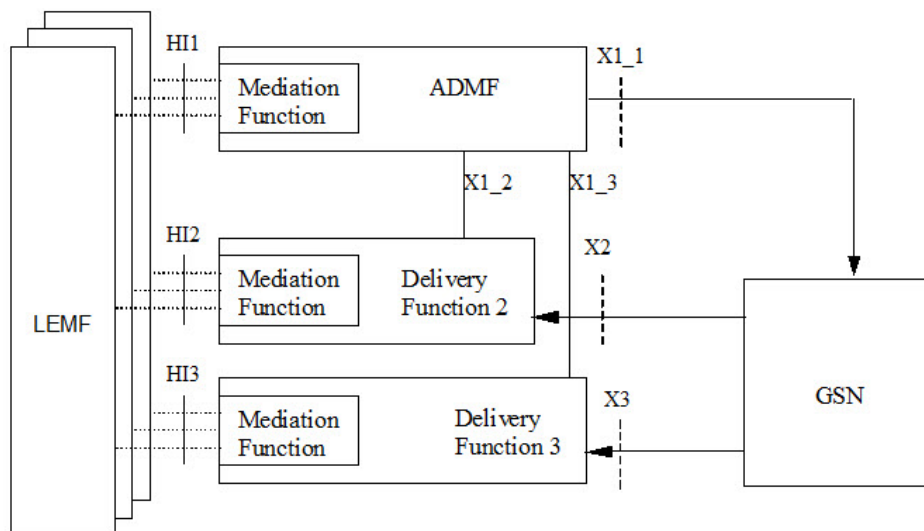


In this release, LI supports the following features and functions:

- Provides Target-based interception based on IMSI, IMEI, or MSISDN
- Supports all GW network elements: SGW, PGW, GGSN, WAG, and ePDG
- Supports IRI and IRI + CC
- Supports X1_1, X2, and X3 interfaces. Support for X2 and X3 interfaces includes AN Format and HI ASN.1 Format using BER and XER encoding.
- Supports IPsec on the X2 and X3 interfaces to provide encryption and security
- Supports TCP/IP transport with the system acting as the TCP/IP client
- Integrates with third-party ADMF, delivery, and mediation functions
- Supports logical separation of internal configuration and logging between LI and normal activity
- Supports 20,000 concurrent LI sessions
- Supports eight simultaneous LI sessions of the same user
- Provides full recovery of LI sessions under all redundancy scenarios

The following figure references 3GPP TS 33.107 (Lawful Interception Architecture and Functions) and shows the basic components in an LI environment:

Figure 1-2 Lawful Interception

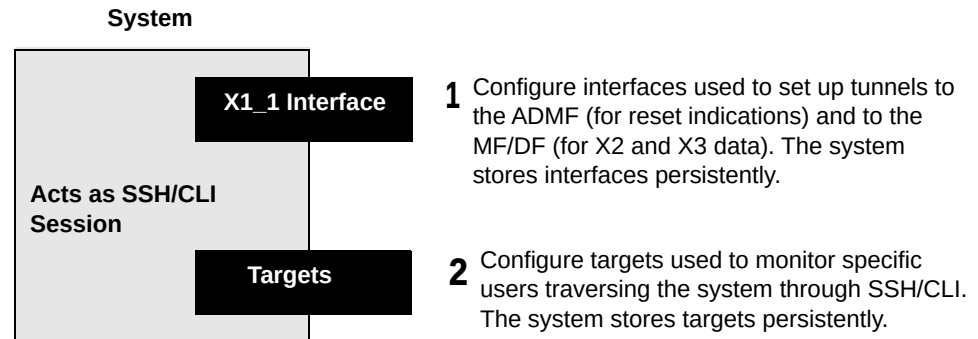


In [Figure 1-2](#), the GSN is the GW service that provides both control and bearer service to an ME. The X1_1 interface interacts with the administrative functions to provision, delete, and monitor LI sessions. The X2 interface is used to push IRI data to the MF/DF for further processing. The X3 interface is used to push CC data to the MF/DF for further processing. The MF/DF takes raw data generated by the GSN and converts the data to a standard format, dependent on laws and regulations standards of the regional authority issuing the LI order. Finally, the LEMF takes all the data from all of the ICEs (including the GW) and presents the data to the ultimate end user, the LEA.

In the Affirmed Networks implementation, the system supports the functions of the GSN, X1_1, X2, and X3 interfaces. The ADMF, MF, DF, and handover interfaces are part of an external package implemented by a third-party partner.

The system acts as the TCP/IP client and uses the X1_1 interface to detect reset events (when communication is lost with the ADMF) and provision, monitor, and remove interfaces and events. You must configure the interface prior to configuring the Target. [Figure 1-3](#) shows the configuration sequence. See [Chapter 2, CALEA CLI Commands](#) for details.

Note: The LI commands and activity are restricted to calea users. Users without CALEA access are not permitted to enter commands or view activity.

Figure 1-3 CALEA Persistent Configuration Sequence

Data and Transport

With the exception of the CLI, all data is defined in ASN.1 and is encoded in either BER or XER format. Data is encapsulated in a TPKT header (RFC 2126) for transport to the destination. In Release 9.4.0.0, only TCP/IP transport is supported, with the system acting as the TCP/IP client.

Security

IPsec is available to provide encryption and security for the X2 and X3 interfaces. The use of IPsec is optional and does not affect the CALEA functionality. However, if IPsec is configured, you must link the security policy rule with the CALEA interface.

For an example of this configuration, see [Example 2: Configuring IPsec to be used with a CALEA Interface](#).

Refer to the *Affirmed Networks CLI Reference Guide* for more information on configuring IPsec.

X1_1 Interface

The Affirmed Networks X1_1 interface contains the following components:

- An interface to detect possible system reset events that may require re-provisioning of target information.
- CLI commands to provision, monitor, and remove interface and target configuration. The reset indication is defined in ASN.1. It is expected that the ADMF use the reset indication to identify the need to re-provision the target. All status and provisioning commands are supported over the CLI interface.

Separate network-context, loopback-ip for the X1_1 Interface

In normal operation, the X1_1 interface uses the standard management-context, mgmt-ip address. In addition, CALEA supports use of a separate network-context, loopback-ip for the X1_1 interface.

This feature allows use of separate X1-1 Network Context, Loopback IP for:

- SSH CALEA CLI shell. It is the X1 CALEA provisioning interface used by the ADMF to provision CALEA on the MCC.
- X1_1 TCP connection used by the MCC to send reset indication messages to ADMF. This reset message notifies the ADMF when an event (such as redundancy switchover) occurs for which the ADMF should audit the configured Lawful Intercept targets.

The X1_1 TCP connection is established using that same Network Context, Loopback IP for the source side (MCC side) of the X1-1 TCP connection.

For details, see [Configuring a Separate network-context and loopback-ip for the X1_1 Interface](#).

Interfaces

The system requires interfaces to be provisioned prior to provisioning targets. The interfaces set up tunnels to the ADMF (for reset indications) and to the MF/DF (for X2 and X3 data). The system stores interfaces persistently.

In Release 9.4.0.0, the system supports up to 256 unique interfaces.

Only the CALEA Administrator can use the CLI to create/monitor/delete CALEA interfaces. There is no Acuitas (netconf) or other management interface support for CALEA configuration.

Interface Groups

X2/X3 Interface redundancy and load-balancing is provided by an LI interface group. An LI interface group is a group of LI interfaces, each of which must have identical interface type, encoding type, and output format type. The interface group sets up group of tunnels to the MF/DF (for X2 and X3 data). The system stores interface groups persistently.

Each interface group can be configured with the role of primary or secondary:

- **Primary:** The interface group also has IRI and CC buffers associated with it. The primary interface group configures the corresponding secondary interface group.
- **Secondary:** The interface group's interfaces must have the same values for interface type, encoding type, and output format type as those of the primary interface group.

An interface group is operationally up if *any* of its interfaces are operationally up. The group is operationally down if *all* its interfaces are operationally down.

When attempting to send data, the system tries the primary interface group first. Congestion may cause the send to fail even when the selected interface is operationally up. In this case, the system attempts to send using other interfaces in the group.

If all the primary interface group's connections are down or if the interfaces selected for sending the data are not available because the TCP window is closed, the primary interface group switches to using the secondary interface group (if available).

If the secondary interface group fails too, data is temporarily stored in interface group buffers. When the situation has been resolved, the data stored in the buffer is transmitted to the server and the system reverts back to using the primary interface group.

The system requires Interface Groups to be provisioned prior to provisioning Targets.

In Release 9.4.0.0, the system supports up to 256 unique Interface Groups.

Only the CALEA Administrator can use the CLI to create/monitor/delete CALEA Interface Groups. There is no Acuitas (netconf) or other management interface support for CALEA configuration.

Targets

The Target is uniquely identified by the Lawful Intercept ID (LIID). Each LIID resulting in an interception has a unique X2 and X3 data stream, even if multiple Targets map to the same subscriber.

The Targets monitor specific users traversing the system.

When the Target is configured as persistent (using persistent target commands), the target data is stored persistently in the local database on disk. When a Target is configured using non-persistent target commands, the targets are stored only in memory, and the ADMF must refresh the target if an event such as system restart occurs.

When a system reset occurs (such as an MCM switchover, chassis reboot, loss of IP connectivity, and so on), a reset indication is sent to the ADMF over the configured X1_1 interface, if any. The reset indication is used to indicate when the x1_1 connection transitions to UP. The ADMF queries and checks the target database. If the ADMF determines that inconsistencies are present, it can re-provision the target database. The ADMF may optionally send a reset response.

The system supports up to 32,768 unique Targets. A particular subscriber can be targeted up to 8 times (resulting in up to 8 unique X2 and X3 data streams).

Only the CALEA Administrator can use the CLI to create/monitor/delete CALEA interfaces. There is no Acuitas (netconf) or other management/policy (such as DIAMETER) interface support for CALEA configuration.

Target-based Interception Based on the MAC Address of a UE

Note: In release 9.0 and later, Lawful Intercept for ePDG and TWAG is no longer supported.

When TWAG is configured to either act as a RADIUS client (Session Learner interface **twag-radius-client**) or configured to provide in-band authentication (Session Learner interface **twag-inband-authentication**), CALEA supports target-based interception based on the MAC address of a UE.

To configure Lawful Intercept based on the MAC address of the UE:

- 1 Issue the following command to enable Lawful Intercept based on the MAC address on the Session Learner interface:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
gw-session-learner [twag-radius-client|twag-inband-authentication]
<name> derive-imei-from-mac true
```

- **true** – When the Session Learner interface receives a DHCP message, it creates a pseudo-IMEI from the MAC address received in the DHCP message and uses it as the IMEI in all communication messages going forward.
- **false** – The Session Learner interface does not create a pseudo-IMEI when it receives a DHCP message.

- 2 Issue the following command to configure the CALEA Target Table Lawful Intercept Identifier (LIID):

```
an3000-mcm-slot7cpu1(config)# calea target-table <liid> identityType
mac identityVal <mac address>
```

The WLAN Interworking section of *Handover Interface for Lawful Interception* (3GPP TS 33.108) does not specify IMEI or MAC address as a possible partyIdentity value in the IRI message. Thus, in the message sent by CALEA agent to the MF/DF for IRI, if no identity (IMSI/MSISDN/NAI) is filled for the partyIdentity value and UE MAC address is available, CALEA encodes the MAC address into the partyIdentity NAI field in string format (for example, **aa:bb:cc:dd:ee:ff**).

Target-based Interception Based on the IMEI of a Subscriber Device

IMEI (International Mobile Equipment Identity) is a unique number that identifies a device in the mobile network. GTP signaling may contain either the 15-digit IMEI or 16-digit IMEISV:

- The 15-digit IMEI includes a Type Allocation Code (TAC) of 8-decimal digits and a Serial Number (SNR) of 6-decimal digits plus a Spare decimal digit (which may be used to send a checksum value). When an LI target is provisioned for IMEI-based intercepts, the MCC gateway ignores the Spare Digit/Check Digit and performs Lawful Intercept based on a match of the 14 digits of TAC plus SNR.
- The 16-digit IMEISV includes the TAC and SNR plus a 2-decimal digit Software Version Number (SVN). When an LI target is provisioned for IMEI-based intercept, the MCC gateway ignores the Software Version Number field (SVN) and performs Lawful Intercept based on a match of the 14 digits of TAC plus SNR.

When provisioning a target for IMEI based Lawful Intercept, the target's identityVal field (identity-value) may be provisioned with any of the following values:

- The first 14 digits of IMEI on which to perform the intercept (TAC + SNR)
- The entire 15-digit IMEI (TAC + SNR + Spare Digit/Check Digit)
- The complete 16-digit IMEISV (TAC + SNR + Software Version)

In each of these cases, the MCC gateway performs Lawful Intercept on a match of the first 14 digits of the provisioned identity-value against the first 14 digits of the IMEI received in the GTP signaling. The MCC gateway ignores the 15th and 16th digits (if present in the provisioned identity-value and/or GTP signaling) when checking for a match of the target identity-value with the subscriber's IMEI.

In addition to this handling of *input* of IMEIs, the 14-digit-imei configuration setting controls how MCC CALEA handles the *output* of IMEIs. For details, see [14-digit-imei](#).

Example 1: Configuring Persistent Targets for Intercept by IMEI

The following three targets target the same subscriber (the lower 14 digits of the IMEI identityVal are identical). When provisioning the system, you only need to provision one of the three, not all three.

```
an3000-mcm-slot7cpu1# config
```

```
Entering configuration mode terminal
```

```

an3000-mcm-slot7cpu1(config)# calea target-table Mobster_1a
identityType imei identityVal 12345678901234 dataCollectionType
iri-plus-cc operator-id att gwType sae-gw x2InterfaceGroupName
x2-ber-hi-grp x3InterfaceGroupName x3-ber-hi-grp

an3000-mcm-slot7cpu1(config)# calea target-table Mobster_1b
identityType imei identityVal 123456789012345 dataCollectionType
iri-plus-cc operator-id att gwType sae-gw x2InterfaceGroupName
x2-ber-hi-grp x3InterfaceGroupName x3-ber-hi-grp

an3000-mcm-slot7cpu1(config)# calea target-table Mobster_1c
identityType imei identityVal 123456789012346 dataCollectionType
iri-plus-cc operator-id att gwType sae-gw x2InterfaceGroupName
x2-ber-hi-grp x3InterfaceGroupName x3-ber-hi-grp

an3000-mcm-slot7cpu1(config)# commit

Commit complete.

an3000-mcm-slot7cpu1(config)# exit

```

Example 2: Configuring Non-persistent Targets for Intercept by IMEI

The following three targets target the same subscriber (the lower 14 digits of the IMEI identityVal are identical). When provisioning the system, you only need to provision one of the three, not all three.

```

an3000-mcm-slot7cpu1# calea createTarget liid Mobster_1a identityType
imei identityVal 12345678901234 dataCollectionType iri-plus-cc gwType
sae-gw x2InterfaceName x2-ber-hi x3InterfaceName x3-ber-hi

result success

reason CALEA Director: target added

an3000-mcm-slot7cpu1# calea createTarget liid Mobster_1b identityType
imei identityVal 123456789012345 dataCollectionType iri-plus-cc gwType
sae-gw x2InterfaceName x2-ber-hi x3InterfaceName x3-ber-hi

result success

reason CALEA Director: target added

an3000-mcm-slot7cpu1# calea createTarget liid Mobster_1c identityType
imei identityVal 1234567890123456 dataCollectionType iri-plus-cc
gwType sae-gw x2InterfaceName x2-ber-hi x3InterfaceName x3-ber-hi

result success

reason CALEA Director: target added

```

Denying User Access to a Specific Target

The system supports banning users (denying user access) of a specific target. This feature is supported only for non-persistent targets.

When a session/bearer setup is rejected due to user ban enabled:

- MCC generates an IRI Bearer Activation Failure event with the cause code sent in the GTP Create Session Response
- MCC sends User authentication failed (GTP Cause Code 92)

When an established session/bearer is disconnected due to user ban enabled:

- MCC generates an IRI Bearer Deactivation event
- MCC clears the session (similar to an operator-initiated subscriber clear)

Geographic Redundancy Requirement

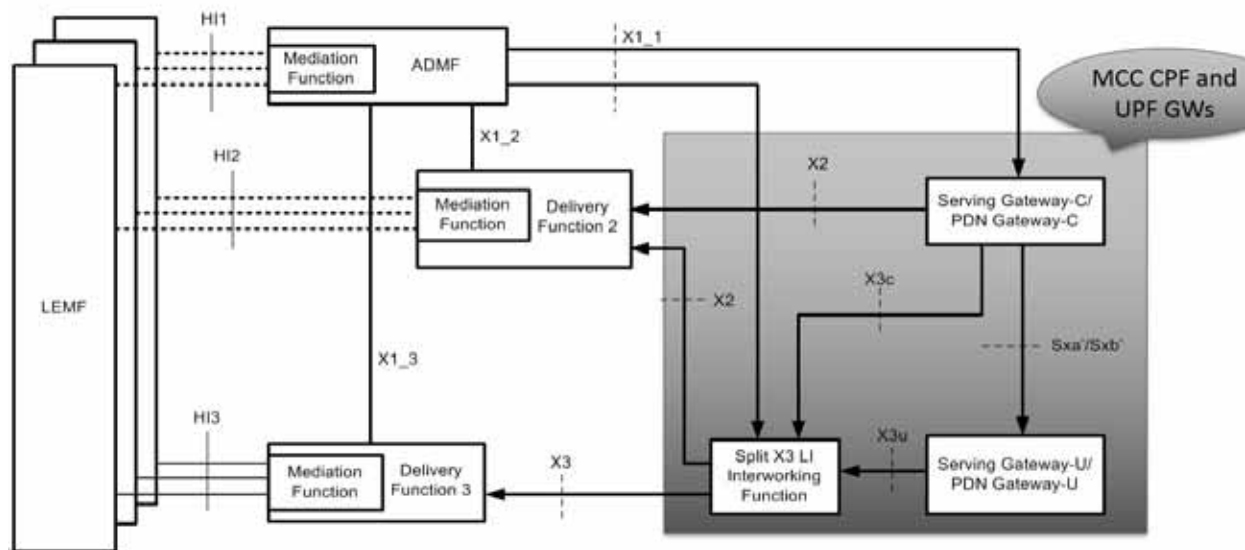
If Geographic Redundancy is configured, you must provision each node in the Geographic Redundancy Group with all the CALEA Targets expected for all the GW services that are protected.

Refer to the *Acuitas User's Guide* for more information on Geographic Redundancy.

Lawful Intercept for Control and User Plane Separation (CUPS)

The following figure references 3GPP TS 33.107 Section 12.9 Functional Requirements for LI in case of Control and User Plane Separation (Figure 12.1.4):

Figure 1-4 Lawful Intercept for Control and User Plane Separation



In Figure 1-4, the functionality in the gray box is provided by the MCC GW services. The Split X3 LI Interworking Function (SX3LIF) can be configured by either of the following options:

- **Call Content from CPF LI Targets:** admin user configures the Lawful Intercept Profile on both the Control Plane Function (CPF) and User Plane Function (UPF) MCC:
 - ADMF provisions LI targets only on the CPF
 - UPF delivers LI packets to the CPF via SX3LIF GTP-U tunnel
 - CPF delivers HI2 IRI and HI3 CC-data to DF/MF/LEMF
- **Call Content from UPF Static LI Targets:** calera user configures entities on CPF and UPF MCC:
 - ADMF provisions LI targets on both CPF and UPF
 - CPF delivers HI2 IRI to LEMF
 - UPF delivers HI3 CC-data directly to LEMF

■ **Call Content from UPF Dynamic LI Targets:**

- ADMF provisions LI targets only on the CPF
- CPF delivers HI2 IRI to the LEMF
- CPF signals target information to the UPF when the session is intercepted
- UPF creates UPF dynamic target based on target information sent from the CPF
- UPF delivers HI3 CC-data directly to the LEMF

To enable this configuration, the ADMF must configure the following objects on both CPF MCC and UPF MCC:

- pfcplawful-intercept-ie with secret-key
- interface-group-source-profiles

The provisioning requirements for these options are described in [Lawful Intercept Configuration for Control and User Plane Separation \(CUPS\)](#).

AES-256 Encryption of Non-transient In-memory and Inter-VM Identities for LI Targets

The system supports using AES-256 to encrypt non-transient in-memory and inter-VM identities for Lawful Intercept (LI) targets, including:

- Identities: IMSI, IMEI, MSISDN
- Lawful Intercept Identifier (LIID)
- Correlation ID

The LI Administrator (ADMF) configures the AES-256 encryption key using the Public Key Infrastructure (PKI) CLI:

```
pki install type calea name target-key encrypt object <key-text>  
pki remove type calea name target-key
```

where **<key-text>** is always 32 bytes and can be specified in one of the following formats:

- raw bytes – 32 characters
- base64 – 44 characters
- hex – 64 characters

Issue the following command to verify that AES-256 encryption is enabled:

```
show calea service-status
```

The command displays the encryption statistics, including:


```
target-encryption-type aes256
```

which indicates that the encryption is enabled.

AES-256 Encryption of all LI Information Using PFCP Over Sxa and Sxbc Interfaces

The system supports using AES-256 to encrypt all LI information passed using PFCP over Sxa and Sxbc interfaces. These interfaces apply only to CUPS configurations. Previous Sxa and Sxbc interface encryption supported only MD5.

The LI Administrator (ADMF) enables encryption:

```
calea pfcpl-lawful-intercept-ie admin-state enabled encryption
[disabled | md5 | aes256]
```

(Default: disabled)

In addition:

- To enhance security, configure a secret-key:

```
calea pfcpl-lawful-intercept-ie secret-key <key-text>
```
- If the secret-key is *not* configured, but an LI Target Encryption is configured, the specified AES-256 key is used as the secret-key:

```
pki install type calea name target-key encrypt object <key-text>
```
- If neither is configured, a default hard-coded key is used.

Behavior Change to Modify and Delete Persistent CALEA Targets

A change introduced in MCC release 9.3 affects modifying and deleting persistent CALEA targets. As part of the upgrade to the ConfD database (MCC release 9.3) specifying the cleartext name (LIID) to modify or delete a Persistent calea target is no longer supported.

The name (LIID) is stored in encrypted form in the ConfD database:

- When creating the persistent target using the following config command, you must specify the name (LIID) in cleartext:

```
calea target-table <cleartext-name> ...
```
- After the target has been created, to modify or delete the target using the config command, you must specify the encrypted form of the name by using the `$7$<encrypted-name>` format of the name (LIID).

However, to avoid having to know the encrypted name of the target, it is highly recommended to instead use the action commands below to modify or delete the target:

- Recommended procedure is to delete the target using its cleartext name via the following action command:

calea deleteTarget liid <cleartext-name>

- To modify a target, first delete the target using the action command:

calea deleteTarget liid <cleartext-name>

Then, re-create the target using the new configuration.

Notes:

- In MCC releases 9.2 and earlier, the action command **calea deleteTarget liid <name>** applied only to non-persistent targets.
- In MCC release 9.3 and later the action command **calea deleteTarget liid <name>** now applies to both non-persistent and persistent targets.

Standards

Affirmed Networks uses the following standards in this feature implementation:

- *3GPP TS 33.106 V15.1.0: 3G security – Lawful Interception Requirements (Release 15).*
- *3GPP TS 33.107 V15.4.0: 3G security – Lawful Interception Architecture and Functions (Release 15). The MCC provides support for Control and User Plane Separation (CUPS) as described in TS 33.107 Release 15.*
- *3GPP TS 33.108 V12.7.0: 3G security – Handover Interface for Lawful Interception (LI) (Release 12). The MCC X2 and X3 HI format interfaces are compliant with the TS 33.108 Release 12 standards.*
- *RFC 2126 – ISO Transport Service on Top of TCP (ITOT).*
- *ETSI TS 101 671 V3.14.1 – Lawful Interception (LI); Handover Interface for Lawful Interception of Telecommunications Traffic: V3.5.1.*

CALEA CLI Commands

For details on how to use the CLI, see the *Affirmed Networks CLI Reference Guide*.

The CLI is text over an established SSH session to the system.

CALEA CLI User Account

To execute any CALEA CLI commands, you must be logged in as the CALEA user:

- Username: calea
- Default Password: calea

Changing Password for User Accounts

The installation and initialization script prompts you to enable password aging for all user accounts. When you enable password aging, you are prompted to change the password on the first login and after 45 days.

Execute the script (`an3000-init`) through the console port on an MCM.

It is *strongly encouraged* to enable password aging by answering Y to the following prompt:

Enable password aging? <Y/n>

You make password changes on the active MCM. Password changes are synchronized from the active MCM to all other cards.

When modifying passwords, you must enter a password that differs from the last three password entries. The password must be at least eight characters and must include upper and lower case letters and at least one special character.

System-level Configuration Options

This section describes the following system-level configuration options:

- 14-digit-imei — Populates X2/HI2 IMEI field with 14-digit IMEI followed by Spare Digit 0 and end marker F
- packet-data-header-summary — Configure Packet Data Summary Reports
- pfcplawful-intercept-ie — PFCP Lawful Intercept IE from CUPS CPF to UPF
- user-ban — CALEA User Ban functionality

14-digit-imei

IMEI can be received in GTP signaling in the following formats:

- 15-digit IMEI, where the first 14 digits are the 8-digit TAC plus the 6-digit SNR (as defined in *3GPP TS 23.003*) followed by a 1-digit Check Digit/Spare Digit field
- 16-digit IMEISV, where the first 14 digits are the 8 digit TAC plus the 6 digit SNR (as defined in *3GPP TS 23.003*) followed by a 2-digit Software Version field

The calea global configuration setting calea 14-digit-imei specifies how MCC CALEA reports the IMEI and IMEISV to the LEMF.

Command: calea 14-digit-imei [enabled | disabled]

- **disabled**: Send to LEMF exact digits received in GTP signaling, including SV digits, if present.

- **enabled**:

When sending IMEI to LEMF:

- Convert 16-digit IMEISV to 15-digit IMEI representation with Check Digit of 0 and terminating digit of all ones (0xF).
- Send 15-digit IMEI using 15-digit IMEI representation with Check Digit of 0 and terminating digit of all ones (0xF).

Affirmed recommends setting 14-digit-imei to **enabled**. The setting defaults to **disabled** for backward compatibility with older MCC releases.

packet-data-header-summary

Packet Data Summary Reports provide periodic summary information for the number of packets and bytes transmitted or received by the target for each unique packet flow within a PDP context.

On CUPS systems, the Packet Data Summary Reports (and per-packet PDHIR reports) are delivered by the UPF MCC to the LEMF. Hence the PDHIR features can be used when CUPS systems are configured for *Call Content from UPF Dynamic LI Targets* or *Call Content from UPF Static LI Targets*.

Command: `calea packet-data-header-summary admin-state [enabled | disabled]report-interval <interval> byte-count-threshold <threshold>`

- **admin-state** must be enabled to generate Packet Data Summary Reports on an intercepted session.
- **report-interval** specifies how often (in seconds) to generate Packet Data Summary Reports on an intercepted session.
- **byte-count-threshold** specifies how many kilo-bytes are transmitted or received by the intercepted session before generating a Packet Data Summary Report.

Both **report-interval** and **byte-count-threshold** are applied simultaneously on the intercepted session. Whichever condition is met first determines when to generate the Packet Data Summary Report. Once a report is generated, both option values are reset to begin counting for the next report generation interval. If no packets are received/transmitted during a given interval, no report is generated.

pfcp-lawful-intercept-ie

The PFCP Lawful Intercept IE is a vendor-specific PFCP Information Element that the MCC uses in CUPS deployments to send Lawful Intercept target information from the CPF MCC to the UPF MCC. It is recommended to configure the PFCP Lawful Intercept IE to use an encryption algorithm along with a secret-key. Configure the same algorithm and secret-key on both the CPF MCC and UPF MCC.

For details, see [calea pfcp-lawful-intercept-ie](#).

user-ban

CALEA User Ban functionality allows you to ban (deny user access) or cancel the ban for users of a specific LI target. To enable/disable CALEA User Ban functionality, you must be in **Config** mode.

Command: `calea user-ban admin-state [enabled | disabled]`

For details, see [calea banTarget](#).

Non-Persistent Target Commands

Non-persistent targets are targets that *are not* stored on disk, but reside only in MCC memory. As a result, if a disruptive event occurs (such as node reboot or switchover) the ADMF must re-program the targets in MCC memory. The Affirmed Networks proprietary X1_1 reset indication message is provided to aid in this situation. The X1_1 reset indication message is sent to the ADMF whenever such a disruptive event occurs.

Note: Unless you have an explicit requirement to avoid storing the target data on disk, it is recommended that you skip this section and use persistent targets. See [Persistent Target Commands](#).

Non-Persistent targets are created and deleted using the following commands:

- `calea createTarget`
- `calea deleteAllTargets`

To delete a single target, use [calea deleteTarget](#).

No modify command exists for non-persistent targets. To modify an existing target, first delete the target and then re-add it using the `calea createTarget` command.

Additional commands supported only for non-persistent targets are:

- `calea banTarget`
- `calea user-ban admin-state [enabled|disabled]`

The following table lists the non-persistent target commands:

Table 2-1 Non-Persistent Target Commands

Syntax	Description
<code>calea createTarget</code>	Configures a target
<code>calea deleteAllTargets</code>	Deletes all targets
<code>calea banTarget</code>	Bans (denies user access) or cancels the ban for users of a specific LI target
<code>calea user-ban admin-state [enabled disabled]</code>	Enables/disables CALEA User Ban functionality

calea createTarget

To configure a target you must be in Exec mode.

The X3 interface is conditional:

- If the data collection type is iri-only, the X3 interface is not required (it is ignored if specified).
- If the data collection type is iri-plus-cc, the X3 interface is required.

Command: calea createTarget

Parameter	Description	Value or Range	Type
dataCollectionType	Type of data collected	iri-only iri-plus-cc	enumeration
gwType	Gateway Type	sgw-only pgw-only sgw-plus-pgw ggsn-only pgw-plus-ggsn all-gw sae-gw saegw-plus-ggsn	enumeration
identityType	Type of user identifier	imsi imei msisdn	enumeration
identityVal	Target identifier	-	numeric
liid	Lawful Intercept Identifier	-	string
x2InterfaceName	Name of X2 interface	-	-
x2-primary-ip-address	Specifies the IP address of the Primary LEMF for the dynamic X2 calea interface	IPv4 or IPv6 address	union
x2-primary-port	Specifies the port of the Primary LEMF for the dynamic X2 calea interface		numeric
x2-secondary-ip-address	Specifies the IP address of the Secondary LEMF for the dynamic X2 calea interface	IPv4 or IPv6 address	union
x2-secondary-port	Specifies the port of the Secondary LEMF for the dynamic X2 calea interface		numeric
x3-primary-ip-address	Specifies the IP address of the Primary LEMF for the dynamic X3 calea interface	IPv4 or IPv6 address	union
x3-primary-port	Specifies the port of the Primary LEMF for the dynamic X3 calea interface		numeric
x3-secondary-ip-address	Specifies the IP address of the Secondary LEMF for the dynamic X3 calea interface	IPv4 or IPv6 address	union
x3-secondary-port	Specifies the port of the Secondary LEMF for the dynamic X3 calea interface		numeric

x3InterfaceName	Name of X3 interface	-	-
-----------------	----------------------	---	---

calea deleteAllTargets

To delete all targets you must be in Exec mode.

To delete a single target, use [calea deleteTarget](#).

Command: `calea deleteAllTargets`

calea banTarget

To ban (deny user access) or cancel the ban for users of a specific LI target, you must be in **Exec** mode.

The command requires the user-ban feature to be enabled (see [calea user-ban](#)).

The command is supported only for non-persistent targets.

When a session/bearer setup is rejected due to user ban enabled:

- MCC generates an IRI Bearer Activation Failure event with the cause code sent in the GTP Create Session Response
- MCC sends User authentication failed (GTP Cause Code 92)

When an established session/bearer is disconnected due to user ban enabled:

- MCC generates an IRI Bearer Deactivation event
- MCC clears the session (similar to an operator-initiated subscriber clear)

Command: `calea banTarget liid <liid> userBan [enabled | disabled]`

Parameter	Description	Value or Range	Type
liid*	Lawful Intercept Identifier	-	string
userBan*	Whether or not user ban (deny access) is enabled/disabled	disabled enabled	enumeration

calea user-ban

To enable/disable CALEA User Ban functionality, you must be in Config mode.

Command: calea user-ban admin-state [enabled | disabled]

Persistent Target Commands

The following table list commands applicable only to Persistent Targets.

Table 2-2 Persistent Target Commands

Syntax	Description
calea interface-group <name>	Configure a CALEA interface group
calea target-table <liid>	Create (or modify) persistent target with specified LIID

calea interface-group <name>

To configure the CALEA Interface Group, you must be in Config mode.

Command: `calea interface-group <name>`

Parameter	Description	Value or Range	Type
buffer-utilization	Percentage buffer utilization before raising alarm	1 .. 100(70)	numeric
cc-buffer-size	Size of CC buffer in KB	0 .. 10485760(102400)	numeric
failover-buffer-threshold	Percentage buffer utilization before switching over to secondary interface group	1 .. 100(50)	numeric
interface	Name of interface(s) contained in the interface group <i>Note: One or more interfaces should be configured in an interface group. When multiple interfaces are present in the interface group, the traffic to be sent for the targets associated with the interface group are load-balanced among the interfaces listed in the interface group.</i>	-	string
interface-type*	Type of interface: enumerated list	x2 x3 x2-and-x3	enumeration
interfaceFormat-type*	Type of Format expected on this interface group	AN_format HI_for mat	enumeration
iri-buffer-size	Size of IRI buffer in KB	0 .. 1048576(102400)	numeric
name*	CALEA Interface Group Name	-	string
output-type*	Type of output encoding: enumerated list	xer ber	enumeration
role	CALEA interface group role: primary/secondary	primary secondary	enumeration

secondary-interface-group	Name of secondary interface group for this interface group	-	-
	<i>Note: See Interface Groups (page 1-5) for a description of when data will be sent via interfaces in the secondary-interface-group.</i>		

calea target-table <liid>

To configure the CALEA Target Table, you must be in Config mode.

Command: `calea target-table <liid>`

Parameter	Description	Value or Range	Type
cc-data-time-stamp-type	Specifies the CC Data Time Stamp Type of the target user.	utc-time local-time	enumeration
dataCollectionType*	Type of data collected	iri-only iri-plus-cc	enumeration
gwType	3GPP Gateway Type	sgw-only pgw-only sgw-plus-pgw ggsn-only pgw-plus-ggsn all-gw sae-gw saegw-plus-ggsn none For descriptions of these values, see Description of gwType .	enumeration
identityType*	Type of user identifier	imsi imei msisdn mac	enumeration
identityVal*	Target identifier	-	string
iri-event-time-stamp-type	Specifies the IRI Events Time Stamp Type of the target user.	utc-time local-time	enumeration
liid*	Lawful Intercept Identifier	-	string
location-info	Specifies the Location Info of the target user.	false true	enumeration
operator-id	Specifies the Operator Id of the target user.	-	string
packet-data-header-info-report	Specifies the Packet Data Header Info Report of the target user. When a target is configured with a setting other than PDHIR_DISABLE, the appropriate X2 PDHIR events are delivered by the UPF MCC to the LEMF.	PDHIR_DISABLE PDHIR_PER_PACKET_ MAPPED_HEADER PDHIR_PER_PACKET_ RAW_HEADER PDHIR_SUMMARY	enumeration
wlanGwType	WLAN Access Gateway Type Note: In release 9.0 and later, Lawful Intercept for ePDG and TWAG is no longer supported.	wag-only epdg-only wag-p lus-epdg	enumeration

x2-interface-group-source-profile	Specifies the Interface Group Source Profile to use when establishing dynamic calea X2 interfaces	-	string
x2-primary-ip-address	Specifies the IP address of the Primary LEMF for the dynamic X2 calea interface	IPv4 or IPv6 address	union
x2-primary-port	Specifies the port of the Primary LEMF for the dynamic X2 calea interface		numeric
x2-secondary-ip-address	Specifies the IP address of the Secondary LEMF for the dynamic X2 calea interface	IPv4 or IPv6 address	union
x2-secondary-port	Specifies the port of the Secondary LEMF for the dynamic X2 calea interface		numeric
x3-interface-group-source-profile	Specifies the Interface Group Source Profile to use when establishing dynamic calea X3 interfaces	-	string
x3-primary-ip-address	Specifies the IP address of the Primary LEMF for the dynamic X3 calea interface	IPv4 or IPv6 address	union
x3-primary-port	Specifies the port of the Primary LEMF for the dynamic X3 calea interface		numeric
x3-secondary-ip-address	Specifies the IP address of the Secondary LEMF for the dynamic X3 calea interface	IPv4 or IPv6 address	union
x3-secondary-port	Specifies the port of the Secondary LEMF for the dynamic X3 calea interface		numeric
x2InterfaceGroupName	Name of X2 interface group	-	-
x3InterfaceGroupName	Name of X3 interface group	-	-

Description of gwType

Gateway type specifies the type of gateway(s) for which X2 and X3 events are reported to LEMF. The supported values are described below:

Value	Description
all-gw	Applies to 2G, 3G, and 4G EPC sessions. When MCC functions as SAEGW for the session, both SGW and PGW records are reported to LEMF. When MCC functions as standalone SGW for the session, SGW records are reported to LEMF. When MCC is functions as standalone PGW for the session, PGW records are reported to LEMF. When MCC functions as GGSN for the session, GGSN records are reported to LEMF.
ggsn-only	Applies only to 2G and 3G sessions. When MCC functions as GGSN for the session, the GGSN records are reported to LEMF.
none	Disables the target - no X2 or X3 data are reported to LEMF.
pgw-only	Applies only to 4G EPC sessions. When MCC functions as SAEGW or standalone PGW for the session, the PGW records are reported to LEMF.
pgw-plus-ggsn	Applies to 2G, 3G, and 4G EPC sessions. When MCC functions as SAEGW or standalone PGW for the session, the PGW records are reported to LEMF. When MCC functions as GGSN for the session, the GGSN records are reported to LEMF.
sae-gw	Applies only to 4G EPC sessions. When MCC functions as SAEGW for the session, only one set of X2 and X3 records are reported to LEMF. When MCC functions as standalone SGW for the session, SGW records are reported to LEMF. When MCC functions as standalone PGW for the session, PGW records are reported to LEMF.
saegw-plus-ggsn	Applies to 2G, 3G, and 4G EPC sessions. When MCC functions as SAEGW for the session, only one set of X2 and X3 records are reported to LEMF. When MCC functions as standalone SGW for the session, SGW records are reported to LEMF. When MCC functions as standalone PGW for the session, PGW records are reported to LEMF. When MCC functions as GGSN for the session, GGSN records are reported to LEMF.

sgw-only	Applies only to 4G EPC sessions. When MCC functions as SAEGW or standalone SGW for the session, the SGW records are reported to LEMF.
sgw-plus-pgw	Applies only to 4G EPC sessions. When MCC functions as SAEGW for the session, both SGW and PGW records are reported to LEMF When MCC functions as standalone SGW for the session, SGW records are reported to LEMF. When MCC functions as standalone PGW for the session, PGW records are reported to LEMF.

Use the **no** form of this command to remove the entire CALEA Target Table.

Command: no calea target-table <name>

For information on a change introduced in MCC release 9.3 that affects modifying and deleting individual persistent CALEA targets, see [Behavior Change to Modify and Delete Persistent CALEA Targets](#).

Common Commands

The following table lists calea commands applicable to both Non-Persistent Targets and Persistent Targets.

Table 2-3 Common Commands

Syntax	Description
calea deleteTarget	Deletes a target
calea interface-group <name> interface <name>	Add interfaces to a CALEA interface group
calea interface-group-source-profile	Configures the CALEA Interface Group Source Profile
calea interface-table	Configures the CALEA Interface Table
calea packet-data-header-summary report-interval <interval> byte-count-threshold <threshold>	Configures Packet Data Summary Reports
calea pfcplawful-intercept-ie	Configures a vendor-specific PFCP Information Element
calea tacacs-plus state [enabled disabled]	Enables/disables TACACS AAA for a CALEA group
infrastructure logging event-log policy liaudit admin-state [enabled disabled]	Enables/disables CALEA calea user command logging
show calea instance dispatcher interface-status <CALEA dispatcher service name>	Displays the CALEA instance dispatcher Interface Table status
show calea instance dispatcher service-status <CALEA dispatcher service name>	Displays the CALEA instance dispatcher service status
show calea instance encoder interface-group-status <CALEA encoder service name>	Displays the CALEA instance encoder Interface Group status
show calea instance encoder interface-status <CALEA encoder service name>	Displays the CALEA instance encoder Interface Table status
show calea instance encoder service-status <CALEA encoder service name>	Displays the CALEA instance encoder service status
show calea instance encoder target-status <CALEA encoder service name>	Displays the CALEA instance encoder target status
show calea instance event-agent interface-group-status <CALEA event agent service name>	Displays the CALEA instance event agent Interface Group status
show calea instance event-agent interface-status <CALEA event agent service name>	Displays the CALEA instance event agent Interface Table status

Table 2-3 Common Commands

Syntax	Description
show calea instance event-agent service-status <CALEA event agent service name>	Displays the CALEA instance event agent service status
show calea instance event-agent target-status <CALEA event agent service name>	Displays the CALEA instance event agent target status
show calea interface-group-status <name>	Displays the CALEA Interface Group status
show calea interface-status	Displays the CALEA Interface Table status
show calea service-status	Displays the CALEA service status
show calea target-status	Displays the CALEA Interface target status
show image-management status	Displays the Image Management current status running on the MCC
show system uptime	Displays the system uptime in seconds

In the following tables, a parameter listed with * indicates a required parameter.

calea deleteTarget

To delete a target you must be in Exec mode.

Command: calea deleteTarget

Parameter	Description	Value or Range	Type
liid	Lawful Intercept Identifier	-	string

calea interface-group-source-profile

Configure the calea interface-group-source-profile to be used by the MCC when establishing a dynamic calea interface (and interface-group) for use with a Dynamic LI Target on the MCC. The profile specifies the source IP address and source port for the MCC to use when connecting to the LEMF.

Command: `calea interface-group-source-profile <name>`

Parameter	Description	Value or Range	Type
buffer-utilization	Percentage buffer utilization before raising alarm	1 .. 100(70)	numeric
cc-buffer-size	Configured size of CC buffer (bytes)	-	numeric
dscp-value	Differentiated Services Code Point (DSCP) value with which to mark all IP packets on this TCP connection	dscp_0_be dscp_01 through dscp_09 dscp_10_af11 dscp_11 dscp_12_af12 dscp_13 dscp_14_af13 dscp_15 through dscp_17 dscp_18_af21 dscp_19 dscp_20_af22 dscp_21 dscp_22_af23 dscp_23 through dscp_25 dscp_26_af31 dscp_27 dscp_28_af32 dscp_29 dscp_30_af33 dscp_31 through dscp_33 dscp_34_af41 dscp_35 dscp_36_af42 dscp_37 dscp_38_af43 dscp_39 through dscp_43 dscp_44_va dscp_45 dscp_46_ef dscp_47 through dscp_63	enumeration

echo-attempt-count	Number of echos sent before declaring link down (if no response received)	2 .. 10(3)	numeric
echo-interval	Number of seconds between echo attempts	1 .. 360(5)	numeric
echo-state	Whether or not echo support is enabled/disabled	disabled enabled	enumeration
failover-buffer-threshold	Percentage buffer utilization before switching over to secondary interface group	1 .. 100(50)	numeric
interfaceFormat-type	Type of Interface Format <i>Note: Default interfaceFormat-type is AN_format. Be sure to set to HI_format if your system is deploying HI_format interface-groups.</i>	AN_format HI_format	enumeration
interface-type*	Type of interface	x2 x3 x2-and-x3	enumeration
iri-buffer-size	Configured size of IRI buffer (bytes)	-	numeric
lemf ip-address port	Explicit list of individual LEMF IP address and port pairs to which this profile should be applied. IP address and port in the list must match IP address and port of LEMF configured in calea interface on the CPF. <i>Note: When the list is not specified, the profile is the default profile for dynamically created interfaces (and groups) of the specified interface-type. See below for discussion of Default versus Explicit profiles</i>		
name*	Profile name	-	string
network-context*	Name of a network-context provisioned on the MCC.	-	-
number-of-source-ports	Number of TCP source ports in the range from which source port can be selected. <i>Note: Default value of number-of-source-ports is 1, for consistency with calea interface default value. It is recommended to set this to a larger value (such as 4 or 8) to enable rapid reconnect of TCP connections to the LEMF when the connection is lost for some reason</i>	1..16(1)	numeric
output-type	Type of output encoding	xer ber	enumeration
source-ip-address-name*	Loopback IP Address Name used as source IP. Can be a IPv4 or IPv6 address.	-	-

source-port*	Base TCP source port of the range from which source port can be selected	-	numeric
tcp-connection-retry-interval	Number of seconds between TCP connection retry attempts	1..30(10)	numeric
tcp-keep-alive	Whether or not TCP keep alive support is enabled/disabled.	disabled enabled	enumeration
tcp-keep-alive-probes	The number of unacknowledged probes to send before considering the connection dead and notifying the application layer.	1 .. 127(6)	numeric
tcp-keep-alive-time	Time interval in seconds between subsequent keepalive probes, regardless of what the connection has exchanged in the meantime.	1 .. 32767(60)	numeric
tcp-send-sock-buffer-size	TCP send socket buffer size, in kilobytes. This setting governs how much data is buffered at send time. For high volume connections, increase the value to ensure that the data is transferred at the desired rate. Tuning this parameter is useful in scenarios where a mismatch exists in the rate at which the sender can send data and the receiver can handle received data. The configured kilobyte value is converted internally into bytes before setting the actual size of the buffer on the socket.	128 .. 10240(10240)	numeric
tcp-user-timeout	The maximum amount of time in seconds that transmitted data may remain unacknowledged before TCP forcefully closes the corresponding connection and returns ETIMEDOUT to the application. Increasing this value allows a TCP connection to survive extended periods without end-to-end connectivity. Decreasing this value allows applications to <i>fail fast</i> , if so desired. This option can be made during any state of a TCP connection, but is only effective during the synchronized states of a connection (ESTABLISHED, FIN-WAIT-1, FIN-WAIT-2, CLOSE-WAIT, CLOSING, or LAST-ACK). When tcp-keep-alive is set to enabled, this setting takes precedence to specify when to close a connection due to keep alive failure.	30 .. 900(60)	numeric
tpkt-keep-alive	Whether or not tpkt keep alive support is enabled.	disabled enabled	enumeration

tpkt-keep-alive-time-interval	Number of minutes between tpkt keep alive attempts	1 .. 5(5)	
transport-type	Type of transport used by this interface	tcp	enumeration

Use the **no** form of this command to remove the CALEA Interface Group Source Profile.

**Command: no calea interface-group-source-profile
<explicit-profile-name>**

The options applied to the dynamically created interface-group and interface can be configured in the interface-group-source-profile as described in the following examples.

Default X3 interface-group-source-profile

The ADMF should configure a default X3 interface-group-source-profile for use by the UPF MCC when establishing X3 connections to the LEMF.

The default X3 interface-group-source-profile can be configured as follows:

```
calea interface-group-source-profile <x3-source-profile-name>
network-context          <network-context-name>
source-ip-address-name   <source-ip-address-name>
  source-port            <source-port>
number-of-source-ports   8
interface-type            x3
interfaceFormat-type     HI_format
transport-type           tcp
output-type              ber
buffer-utilization        70
failover-buffer-threshold 50
iri-buffer-size          102400
cc-buffer-size           102400
tcp-keep-alive           disabled
tcp-connection-retry-interval 10
tcp-user-timeout         60
tcp-send-sock-buffer-size 1024
tpkt-keep-alive          disabled
dscp-value              dscp_0_be
```

Explicit X3 interface-group-source-profile for LEMFs Requiring Special Configuration

Create an explicit X3 interface-group-source-profile for any LEMF for which you want configuration that differs from the default X3 interface-group-source-profile.

For example, if you want to use a different source IP address and source port for X3 connections to a specific LEMF, create an explicit X3 interface-group-source-profile. An explicit X3 interface-group-source-profile is defined by adding a list of individual LEMF IP address and port pairs to the calea interface-group-source-profile as follows:

```
calea interface-group-source-profile <x3-source-profile-name>
network-context          <network-context-name>
source-ip-address-name   <source-ip-address-name>
    source-port           <source-port>
number-of-source-ports   8
interface-type            x3
interfaceFormat-type     HI_format
transport-type            tcp
output-type              ber
buffer-utilization        70
failover-buffer-threshold 50
iri-buffer-size           102400
cc-buffer-size            102400
echo-state                disabled
tcp-keep-alive            disabled
tcp-connection-retry-interval 10
tcp-user-timeout          60
tcp-send-sock-buffer-size 1024
dscp-value                dscp_0_be
lemf ip-address 10.32.5.11 port 8888
lemf ip-address 10.33.39.49 port 7463
!
!
```

Default X2 interface-group-source-profile

The ADMF should configure a default X2 interface-group-source-profile for use by the UPF MCC only if the UPF MCC needs an X2 connection to the LEMF. This requirement applies if the calea targets are configured for Packet Data Header Information Reports or Packet Data Summary Reports. If the PDHIR/PDSR features are not used, no X2 interface-group-source-profile is required on the UPF MCC.

Configure the default X2 interface-group-source-profiles in the same manner as default X3 interface-group-source-profiles (see [Default X3 interface-group-source-profile](#)), but specify the interface-type as x2.

Explicit X2 interface-group-source-profiles for LEMFs Requiring Special Configuration

If the PDHIR/PDSR features are used by your system, the ADMF should configure explicit X2 interface-group-source-profiles for any LEMF for which you want configuration that differs from the default X2 interface-group-source-profile.

Configure the explicit X2 interface-group-source-profiles in the same manner as explicit X3 interface-group-source-profiles (see [Explicit X3 interface-group-source-profile for LEMFs Requiring Special Configuration](#)), but specify the interface-type as x2.

calea interface-table

To configure the CALEA Interface Table, you must be in Config mode.

Command: `calea interface-table <name>`

Parameter	Description	Value or Range	Type
admin-state	Controls whether CALEA interface establishes a TCP connection to the specified LEMF IP address and port.	enabled disabled	enumeration
dscp-value	Differentiated Services Code Point (DSCP) value with which to mark all IP packets on this TCP connection	dscp_0_be dscp_01 through dscp_09 dscp_10_af11 dscp_11 dscp_12_af12 dscp_13 dscp_14_af13 dscp_15 through dscp_17 dscp_18_af21 dscp_19 dscp_20_af22 dscp_21 dscp_22_af23 dscp_23 through dscp_25 dscp_26_af31 dscp_27 dscp_28_af32 dscp_29 dscp_30_af33 dscp_31 through dscp_33 dscp_34_af41 dscp_35 dscp_36_af42 dscp_37 dscp_38_af43 dscp_39 through dscp_43 dscp_44_va dscp_45 dscp_46_ef dscp_47 through dscp_63	enumeration
echo-attempt-count	Number of echos sent before declaring link down (if no response received)	2 .. 10(3)	numeric
echo-interval	Number of seconds between echo attempts	1 .. 360(5)	numeric

echo-state	Whether or not echo support is enabled/disabled	disabled enabled	enumeration
interface-type*	Type of interface	x1-1 x2 x3 x2-and-x3	enumeration
interfaceFormat-type	Type of Format expected on this interface	AN_format HI_format	enumeration
ip-address*	Can be a IPv4 or IPv6 address	-	union
name*	Interface Name	-	string
network-context	- <i>Note: Required parameter for all interfaces except x1-1.</i>	-	-
number-of-source-ports	Number of TCP source ports in the range from which source port can be selected. <i>Note: Default value of number-of-source-ports is 1. It is recommended to set this to a larger value (such as 4 or 8) to enable rapid reconnect of TCP connections to the LEMF when the connection is lost for some reason</i>	1..16(1)	numeric
output-type	Type of output expected on this interface	xer ber	enumeration
port*	-	-	numeric
source-ip-address-name	Loopback IP Address Name used as source IP. Can be a IPv4 or IPv6 address. <i>Note: Required parameter for all interfaces except x1-1.</i>	-	-
source-port	- <i>Note: Required parameter for all interfaces except x1-1.</i>	-	numeric
tcp-connection-retry-interval	Number of seconds between TCP connection retry attempts.	1..30(10)	numeric
tcp-keep-alive	Whether or not TCP keep alive support is enabled/disabled.	disabled enabled	enumeration
tcp-keep-alive-interval	The interval in seconds between subsequent TCP keep alive probes, regardless of what the connection has exchanged in the meantime.	1 .. 32767(10)	numeric
tcp-keep-alive-probes	The number of unacknowledged probes to send before considering the connection dead and notifying the application layer.	1 .. 127(6)	numeric

tcp-keep-alive-time	The interval in seconds between the last data packet sent (simple ACKs are not considered data) and the first keep alive probe; after the connection is marked to need keep alive, this counter is not used any further.	1 .. 32767(60)	numeric
tcp-send-sock-buffer-size	TCP send socket buffer size, in kilobytes. This setting governs how much data is buffered at send time. For high volume connections, increase the value to ensure that the data is transferred at the desired rate. Tuning this parameter is useful in scenarios where a mismatch exists in the rate at which the sender can send data and the receiver can handle received data. The configured kilobyte value is converted internally into bytes before setting the actual size of the buffer on the socket.	128 .. 10240(10240)	numeric
tcp-user-timeout	The maximum amount of time in seconds that transmitted data may remain unacknowledged before TCP forcefully closes the corresponding connection and returns ETIMEDOUT to the application. Increasing this value allows a TCP connection to survive extended periods without end-to-end connectivity. Decreasing this value allows applications to <i>fail fast</i> , if so desired. This option can be made during any state of a TCP connection, but is only effective during the synchronized states of a connection (ESTABLISHED, FIN-WAIT-1, FIN-WAIT-2, CLOSE-WAIT, CLOSING, or LAST-ACK). When tcp-keep-alive is set to enabled, this setting takes precedence to specify when to close a connection due to keep alive failure.	30 .. 900(60)	numeric
tcpk-keep-alive	Whether or not TCPK keep alive support is enabled/disabled. Note: Applies to interfaceFormat-type HI_format only.	disabled enabled	enumeration
tpkt-keep-alive-time-interval	Number of minutes between TCPK keep alive attempts. Note: Applies to interfaceFormat-type HI_format only.	1 .. 5(5)	numeric
transport-type	Type of transport to use for interface	tcp	enumeration

Use the **no** form of this command to remove the CALEA Interface Table. All targets must be removed prior to removing the CALEA Interface Table.

Command: no calea interface-table <name>

calea interface-group <name> interface <name>

To add interfaces to a CALEA Interface Group, you must be in Config mode.

Command: calea interface-group <name> interface <name>

Parameter	Description	Value or Range	Type
name*	-	-	-

Use the **no** form of this command to remove interfaces from the CALEA Interface Group.

Command: no calea interface-group <name> interface <name>

calea pfcplawful-intercept-ie

The PFCP Lawful Intercept IE is a vendor-specific PFCP Information Element that the MCC uses in CUPS deployments to send Lawful Intercept target information from the CPF MCC to the UPF MCC. It is recommended to configure the PFCP Lawful Intercept IE to use an encryption algorithm along with a secret-key. Configure the same algorithm and secret-key on both the CPF MCC and UPF MCC.

When configured for encryption type aes256, the pfcplawful-intercept-ie can be used in conjunction with PKI installed AES-256 keys as described in [AES-256 Encryption of all LI Information Using PFCP Over Sxa and Sxbc Interfaces](#) (page 1-13).

Command: calea pfcplawful-intercept-ie

Parameter	Description	Value or Range	Type
admin-state*	Controls whether PFCP Lawful Intercept IE is sent by CPF	enabled disabled	enumeration
dummy-ie-frequency	Probability of sending dummy Lawful Intercept IEs on sessions that are not intercepted (number of IEs per 1000 PFCP messages)	0 .. 1000	numeric

encryption	Specifies whether PFCP Lawful Intercept IE is sent in encrypted form	disabled md5 aes256	enumeration
secret-key	String used to encrypt/decrypt PFCP Lawful Intercept IE Is configured in cleartext Is stored and displayed in encrypted form	-	string

Use the **no** form of this command to remove the PFCP Lawful Intercept IE.

Command: no pfcplawful-intercept-ie

calea tacacs-plus

To enable/disable TACACS AAA for a CALEA group, you must be in Config mode.

Command: calea tacacs-plus state [enabled | disabled]

When the **tacacs-plus** feature is enabled (default), when a user logs in, Lawful Intercept sends an authentication request to the TACACS+ server. Upon successful authentication, the server responds with the user's group assignment.

When the **tacacs-plus** feature is disabled for a CALEA group, the user *is not* authenticated by the TACACS+ server. User login and all commands entered are not validated.

infrastructure logging event-log policy liaudit

To enable/disable CALEA calea user command logging, you must be in Config mode.

Command: infrastructure logging event-log policy liaudit admin-state [enabled | disabled]

When enabled, logs CALEA calea user commands related to configure/modify/display of LI targets to a CALEA audit file. The logs can be output directly, such as using syslog, snmp, or sftp to a secure logging server. Logging is supported for both persistent and non-persistent targets.

When the **liaudit** feature is enabled, MCC configures the following default logging policy:

```
sinks liaudit
  type      file
  file-name  liaudit
  file-format ascii
```

```
display-field displaystring
```

The calea user cannot create or delete these logging filters, but can modify `file-management profile` parameters to set different file transfer policies and set username/password. For example:

```
infrastructure file-management profile eventlog liaudit
  purge-thr          100
  purge-low-thr      0
  file-rotate-size-bytes 10485760
  file-transfer-enabled disabled
  upload-frequency   1hour
```

And, the calea account can issue the `trigger-file-rotate` and `trigger-file-transfer` commands.

The audit log uses the following file format:

```
2017-10-09T03:00:06.789917543-0400 <userid> <LIID> <operation> <result>
[<attributename> <attributeValue>]
```

For example:

```
2017-11-24T05:23:59.282318654-0500 Sev=4 [CALEA] [CALEATarget] [ID=00]
[00012c0700011fe4] [28407 28724] [CaleaDirector_CaleaDirector_0]
"[CaleaUserName:calea Liid:mobster OperationType:createTarget
Operationresult:success]: dataCollectionType: iri-plus-cc gwType:
all-gw wlangwtype: none x2InterfaceName: X2-AN-V4 x3InterfaceName:
X3-AN-V4"
2017-11-24T05:24:21.029159989-0500 Sev=4 [CALEA] [CALEATarget] [ID=00]
[00012c0700011fe4] [28407 28724] [CaleaDirector_CaleaDirector_0]
"[CaleaUserName:calea Liid:NA OperationType:target-status
Operationresult:success]: ."
```

show calea instance dispatcher interface-status <CALEA dispatcher service name>

To display the CALEA instance dispatcher Interface Table status, you must be in Exec mode.

Command: show calea instance dispatcher interface-status <CALEA dispatcher service name>

Parameter	Description	Value or Range	Type
interface-format-type	Type of interface Format	AN_format HI_for mat	enumeration
interface-type	Type of interface	x2 x3 x2-and-x3	enumeration

number-of-tpkt-keep-alive-ber-encode-fail-count	Number of TPKT Keep Alive BER Encode Fail Count	-	numeric
number-of-tpkt-keep-alive-messages-sent	Number of TPKT Keep Alive Messages Sent	-	numeric
number-of-tpkt-keep-alive-xer-encode-fail-count	Number of TPKT Keep Alive XER Encode Fail Count	-	numeric
number-of-transmit-fail-bad-interface	Number of transmit failures bad interface	-	numeric
number-of-transmit-fail-interface-down	Number of transmit failures interface down	-	numeric
number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric
operational-status	Operational Status	up down partially-up maint dryup	enumeration
transport-type	Type of transport used by this interface	tcp	enumeration
iri-buffer-current-size	Total number of bytes currently in the IRI buffer	-	numeric
iri-buffer-high-water-mark	Maximum value reached for the total number of bytes present in the IRI buffer	-	numeric
iri-buffer-num-buffer-size-exceeded	Number of times IRI buffer size was exceeded requiring packets to be dropped	-	numeric
iri-buffer-num-dropped-packets	Number of packets dropped as a result of IRI buffer size being exceeded	-	numeric
iri-buffer-num-packets	Total number of packets currently in the IRI buffer	-	numeric
iri-buffer-size	Configured size of IRI buffer (bytes)	-	numeric
cc-buffer-current-size	Total number of bytes currently in the CC buffer	-	numeric
cc-buffer-high-water-mark	Maximum value reached for the total number of bytes present in the CC buffer	-	numeric
cc-buffer-num-buffer-size-exceeded	Number of times CC buffer size was exceeded requiring packets to be dropped	-	numeric
cc-buffer-num-dropped-packets	Number of packets dropped as a result of CC buffer size being exceeded	-	numeric
cc-buffer-num-packets	Total number of packets currently in the CC buffer	-	numeric
cc-buffer-size	Configured size of CC buffer (bytes)	-	numeric

show calea instance dispatcher service-status <CALEA dispatcher service name>

To display the CALEA instance dispatcher service status, you must be in Exec mode.

**Command: show calea instance dispatcher service-status
<CALEA dispatcher service name>**

Parameter	Description	Value or Range	Type
number-of-echo-requests-received	Number of ECHO Requests Received	-	numeric
number-of-echo-requests-sent	Number of ECHO Requests Sent	-	numeric
number-of-echo-responses-received	Number of ECHO Responses Received	-	numeric
number-of-echo-responses-sent	Number of ECHO Responses Sent	-	numeric
number-of-echo-timeouts	Number of ECHO Timeouts	-	numeric
number-of-transmit-fail-bad-interface	Number of transmit failures bad interface	-	numeric
number-of-transmit-fail-interface-down	Number of transmit failures interface down	-	numeric
number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric
number-of-x1-1-interfaces	Number of configured X1_1 interfaces	-	numeric
number-of-x2-interfaces	Number of configured X2 interfaces	-	numeric
number-of-x3-interfaces	Number of configured X3 interfaces	-	numeric
operational-status	Operational Status	up down partially-up maintenance dryup	enumeration

show calea instance encoder interface-group-status <CALEA encoder service name>

To display the CALEA instance encoder Interface Group status, you must be in Exec mode.

Command: show calea instance encoder interface-group-status <CALEA encoder service name>

Parameter	Description	Value or Range	Type
currently-active-li-calls	Currently Active LI Calls using this interface group	-	numeric
interface-group-type	Type of interface group	x2 x3 x2-and-x3	enumeration
interfaceFormat-type	Type of Interface Format	AN_format HI_for mat	enumeration
iri-buffer-current-size	Total number of bytes currently in the IRI buffer	-	numeric
iri-buffer-high-water-mark	Maximum value reached for the total number of bytes present in the IRI buffer	-	numeric
iri-buffer-num-buffer-size-exceeded	Number of times IRI buffer size was exceeded requiring packets to be dropped	-	numeric
iri-buffer-num-dropped-packets	Number of packets dropped as a result of IRI buffer size being exceeded	-	numeric
iri-buffer-num-packets	Total number of packets currently in the IRI buffer	-	numeric
iri-buffer-size	Configured size of IRI buffer (bytes)	-	numeric
number-of-targets	Number of configured monitored targets using this interface group	-	numeric
number-of-transmit-fail-bad-interface-group	Number of transmit failures bad interface group	-	numeric
number-of-transmit-fail-interface-group-down	Number of transmit failures interface group down	-	numeric
number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric
number-of-x2-ber-encode-fail	Number of X2 BER Encode Fail	-	numeric
number-of-x2-encode-fail-no-buffer	Number of X2 Encode Fail no buffer	-	numeric
number-of-x2-events-for-current-active-calls	Number of X2 Events For current Active Calls using this interface group	-	numeric

number-of-x2-events-offered	Number of X2 Events Offered for transmission	-	numeric
number-of-x2-xer-encode-fail	Number of X2 XER Encode Fail	-	numeric
operational-status	Operational Status	up down partially-up maint dryup	enumeration
output-type	Type of output encoding	xer ber	enumeration
role	Calea interface group role	primary secondary	enumeration
total-cumulative-LI-Sessions	Total Cumulative LI Sessions using this interface group	-	numeric
total-number-of-all-terminated-LI-sessions	Total cumulative Number of All Terminated LI Sessions	-	numeric
total-number-of-intercepted-packets-for-current-active-calls	Total Number of intercepted Packets for current Active Calls using this interface group	-	numeric
total-number-of-terminated-LI-sessions-due-to-call-disconnect	Total cumulative Number of Terminated LI Sessions due to Call Disconnect	-	numeric
total-number-of-terminated-LI-sessions-due-to-deprovision	Total cumulative Number of Terminated LI Sessions due to Deprovisioning of Target.	-	numeric

show calea instance encoder interface-status <CALEA encoder service name>

To display the CALEA instance encoder Interface Table status, you must be in Exec mode.

Command: show calea instance encoder interface-status <CALEA encoder service name>

Parameter	Description	Value or Range	Type
interface-format-type	Type of interface Format	AN_format HI_for mat	enumeration
interface-type	Type of interface	x2 x3 x2-and-x3	enumeration
number-of-transmit-fail-bad-interface	Number of transmit failures bad interface	-	numeric
number-of-transmit-fail-interface-down	Number of transmit failures interface down	-	numeric

number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric
operational-status	Operational Status	up down partially-up maint dryup	enumeration
transport-type	Type of transport used by this interface	tcp	enumeration

show calea instance encoder service-status <CALEA encoder service name>

To display the CALEA instance encoder service status, you must be in Exec mode.

Command: show calea instance encoder service-status <CALEA encoder service name>

Parameter	Description	Value or Range	Type
currently-active-li-calls	Currently Active LI Calls using this interface group	-	numeric
currently-camp-on-triggers	Currently Camp on LI Triggers: (Total Configured LIIDs - Current Active Calls)	-	numeric
number-of-bad-events-received	Number of Bad Events Received	-	numeric
number-of-rx-events-dropped-no-target	Number of Received Events Dropped No Target	-	numeric
number-of-targets	Number of configured monitored targets	-	numeric
number-of-transmit-fail-bad-interface	Number of transmit failures bad interface	-	numeric
number-of-transmit-fail-interface-down	Number of transmit failures interface down	-	numeric
number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric
number-of-x1-1-interfaces	Number of configured X1_1 interfaces	-	numeric
number-of-x2-ber-encode-fail	Number of X2 BER Encode Failures	-	numeric

number-of-x2-encode-fail-no-buffer	Number of X2 Encode Fail No Buffer	-	numeric
number-of-x2-events-dropped-no-interface	Number of X2 Events Dropped No Interface	-	numeric
number-of-x2-events-for-current-active-calls	Number of X2 Events For current Active Calls	-	numeric
number-of-x2-events-received	Number of X2 Events Received	-	numeric
number-of-x2-interfaces	Number of configured X2 interfaces	-	numeric
number-of-x2-output-events	Number of X2 Encode Output Events	-	numeric
number-of-x2-output-events-failed	Number of X2 Encode Output Events Failed	-	numeric
number-of-x2-xer-encode-fail	Number of X2 XER Encode Failures	-	numeric
number-of-x3-ber-encode-fail	Number of X3 BER Encode Failures	-	numeric
number-of-x3-event-encode-failures-no-buffer	Number of X3 Event Encode Failures due to No Buffer	-	numeric
number-of-x3-events-dropped-no-interface	Number of X3 Events Dropped No Interface	-	numeric
number-of-x3-events-for-current-active-calls	Number of X3 Events For current Active Calls	-	numeric
number-of-x3-events-received	Number of X3 Events Received	-	numeric
number-of-x3-interfaces	Number of configured X3 interfaces	-	numeric
number-of-x3-output-events	Number of X3 Encode Output Events	-	numeric
number-of-x3-output-events-failed	Number of X3 Encode Output Events Failed	-	numeric
number-of-x3-xer-encode-fail	Number of X3 XER Encode Failures	-	numeric
operational-status	Operational Status	up down partially-up maintenance dryup	enumeration
total-cumulative-LI-Sessions	Total Cumulative LI Sessions	-	numeric
total-number-of-all-terminated-LI-sessions	Total cumulative Number of All Terminated LI Sessions	-	numeric
total-number-of-intercepted-packets-for-current-active-calls	Total Number of intercepted Packets for current Active Calls	-	numeric
total-number-of-terminated-LI-sessions-due-to-call-disconnect	Total cumulative Number of Terminated LI Sessions due to Call Disconnect	-	numeric

total-number-of-terminated-LI-sessions-due-to-deletion-of-context	Total cumulative Number of Terminated LI Sessions due to Deletion of Interface Context	-	numeric
total-number-of-terminated-LI-sessions-due-to-deprovision	Total cumulative Number of Terminated LI Sessions due to Deprovisioning of Target	-	numeric

show calea instance encoder target-status <CALEA encoder service name>

To display the CALEA instance encoder target status, you must be in Exec mode.

Command: show calea instance encoder target-status <CALEA encoder service name>

Parameter	Description	Value or Range	Type
data-collection-type	Type of data collected	iri-only iri-plus-cc	enumeration
gateway-type-3gpp	Gateway Type	sgw-only pgw-only sgw-plus-pgw ggsn-only pgw-plus-ggsn all-gw sae-gw saegw-plus-ggsn none	enumeration
gateway-type-wlan	WLAN Access Gateway Type	none wag-only	enumeration
identity-type	Type of user identifier	imsi imei msisdn	enumeration
identity-value	Target identifier	-	numeric
number-of-x2-ber-encode-fail	Number of X2 BER Encode Fail	-	numeric
number-of-x2-encode-fail-no-buffer	Number of X2 Encode Fail no buffer	-	numeric
number-of-x2-events-dropped-no-interface	Number of X2EventsDroppedNoInterface	-	numeric
number-of-x2-events-for-current-active-calls	Number of X2 Events For current Active Calls	-	numeric
number-of-x2-events-sent	Number of X2 Events Sent	-	numeric
number-of-x2-pdhir-per-packet-raw-events-sent	Number of X2 Packet Data Header Information Reporting packet raw events sent This count is incremented on the UPF MCC, because the UPF MCC delivers the X2 PDHIR events to the LEMF.	-	numeric

number-of-x2-pdhir-per-packet-mapped-events-sent	Number of X2 Packet Data Header Information Reporting packet mapped events sent This count is incremented on the UPF MCC, because the UPF MCC delivers the X2 PDHIR events to the LEMF.	-	numeric
number-of-x2-packet-data-summary-report-events-sent	Number of X2 Packet Data Summary Report events sent This count is incremented on the UPF MCC, because the UPF MCC delivers the X2 PDHIR events to the LEMF.	-	numeric
number-of-x2-xer-encode-fail	Number of X2 XER Encode Fail	-	numeric
number-of-x3-ber-encode-fail	Number of X3 BER Encode Fail	-	numeric
number-of-x3-event-ber-encode-failures	Number of X3 Event BER Encode Failures	-	numeric
number-of-x3-event-encode-failures-no-buffer	Number of X3 Event Encode Failures due to No Buffer	-	numeric
number-of-x3-event-xer-encode-failures	Number of X3 Event XER Encode Failures	-	numeric
number-of-x3-events-dropped-no-interface	Number of X3 Events Dropped - No Interface	-	numeric
number-of-x3-events-for-current-active-calls	Number of X3 Events For current Active Calls	-	numeric
number-of-x3-events-sent	Number of X3 Events Sent	-	numeric
number-of-x3-xer-encode-fail	Number of X3 XER Encode Fail	-	numeric
storage-type	Storage Type: persistent non-persistent	non-persistent persistent	enumeration
total-cumulative-LI-Sessions	Total Cumulative LI Sessions	-	numeric
total-number-of-all-terminated-LI-sessions	Total cumulative Number of All Terminated LI Sessions	-	numeric
total-number-of-intercepted-packets-for-current-active-calls	Total Number of intercepted Packets for current Active Calls	-	numeric
total-number-of-terminated-LI-sessions-due-to-call-disconnect	Total cumulative Number of Terminated LI Sessions due to Call Disconnect	-	numeric
total-number-of-terminated-LI-sessions-due-to-deletion-of-context	Total cumulative Number of Terminated LI Sessions due to Deletion of Interface Context	-	numeric
x2-interface-group-name	Name of X2 interface group	-	string

x3-interface-group-name	Name of X3 interface group	-	string
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show calea instance event-agent interface-group-status <CALEA event agent service name>

To display the CALEA instance event agent Interface Group status, you must be in Exec mode.

Command: show calea instance event-agent interface-group-status <CALEA event agent service name>

Parameter	Description	Value or Range	Type
currently-active-li-calls	Currently Active LI Calls using this interface group	-	numeric
interface-group-type	Type of interface group	x2 x3 x2-and-x3	enumeration
interfaceFormat-type	Type of Interface Format	AN_format HI_for mat	enumeration
iri-buffer-current-size	Total number of bytes currently in the IRI buffer	-	numeric
iri-buffer-high-water-mark	Maximum value reached for the total number of bytes present in the IRI buffer	-	numeric
iri-buffer-num-buffer-size-exceeded	Number of times IRI buffer size was exceeded requiring packets to be dropped	-	numeric
iri-buffer-num-dropped-packets	Number of packets dropped as a result of IRI buffer size being exceeded	-	numeric
iri-buffer-num-packets	Total number of packets currently in the IRI buffer	-	numeric
iri-buffer-size	Configured size of IRI buffer (bytes)	-	numeric
number-of-targets	Number of configured monitored targets using this interface group	-	numeric
number-of-transmit-fail-bad-interface-group	Number of transmit failures bad interface group	-	numeric
number-of-transmit-fail-interface-group-down	Number of transmit failures interface group down	-	numeric
number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric

number-of-x2-ber-encode-fail	Number of X2 BER Encode Fail	-	numeric
number-of-x2-encode-fail-no-buffer	Number of X2 Encode Fail no buffer	-	numeric
number-of-x2-events-for-current-active-calls	Number of X2 Events For current Active Calls using this interface group	-	numeric
number-of-x2-events-offered	Number of X2 Events Offered for transmission	-	numeric
number-of-x2-xer-encode-fail	Number of X2 XER Encode Fail	-	numeric
operational-status	Operational Status	up down partially-up maint dryup	enumeration
output-type	Type of output encoding	xer ber	enumeration
role	Calea interface group role	primary secondary	enumeration
total-cumulative-LI-Sessions	Total Cumulative LI Sessions using this interface group	-	numeric
total-number-of-all-terminated-LI-sessions	Total cumulative Number of All Terminated LI Sessions	-	numeric
total-number-of-intercepted-packets-for-current-active-calls	Total Number of intercepted Packets for current Active Calls using this interface group	-	numeric
total-number-of-terminated-LI-sessions-due-to-call-disconnect	Total cumulative Number of Terminated LI Sessions due to Call Disconnect	-	numeric
total-number-of-terminated-LI-sessions-due-to-deprovision	Total cumulative Number of Terminated LI Sessions due to Deprovisioning of Target.	-	numeric

show calea instance event-agent interface-status <CALEA event agent service name>

To display the CALEA instance event agent Interface Table status, you must be in Exec mode.

Command: show calea instance event-agent interface-status <CALEA event agent service name>

Parameter	Description	Value or Range	Type
interface-format-type	Type of interface Format	AN_format HI_for mat	enumeration
interface-type	Type of interface	x2 x3 x2-and-x3	enumeration
number-of-transmit-fail-bad-interface	Number of transmit failures bad interface	-	numeric
number-of-transmit-fail-interface-down	Number of transmit failures interface down	-	numeric
number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric
operational-status	Operational Status	up down partially-up maint dryup	enumeration
transport-type	Type of transport used by this interface	tcp	enumeration

show calea instance event-agent service-status <CALEA event agent service name>

To display the CALEA instance event agent service status, you must be in Exec mode.

Command: show calea instance event-agent service-status <CALEA event agent service name>

Parameter	Description	Value or Range	Type
currently-active-li-calls	Currently Active LI Calls using this interface group	-	numeric
currently-camp-on-triggers	Currently Camp on LI Triggers: (Total Configured LIIDs - Current Active Calls)	-	numeric
number-of-bad-events-received	Number of Bad Events Received	-	numeric
number-of-rx-events-dropped-no-target	Number of Received Events Dropped No Target	-	numeric
number-of-targets	Number of configured monitored targets	-	numeric
number-of-transmit-fail-bad-interface	Number of transmit failures bad interface	-	numeric
number-of-transmit-fail-interface-down	Number of transmit failures interface down	-	numeric
number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric
number-of-x1-1-interfaces	Number of configured X1_1 interfaces	-	numeric
number-of-x2-ber-encode-fail	Number of X2 BER Encode Failures	-	numeric
number-of-x2-encode-fail-no-buffer	Number of X2 Encode Fail No Buffer	-	numeric
number-of-x2-events-dropped-no-interface	Number of X2 Events Dropped No Interface	-	numeric
number-of-x2-events-for-current-active-calls	Number of X2 Events For current Active Calls	-	numeric
number-of-x2-events-received	Number of X2 Events Received	-	numeric
number-of-x2-interfaces	Number of configured X2 interfaces	-	numeric
number-of-x2-output-events	Number of X2 Encode Output Events	-	numeric
number-of-x2-output-events-failed	Number of X2 Encode Output Events Failed	-	numeric

number-of-x2-xer-encode-fail	Number of X2 XER Encode Failures	-	numeric
number-of-x3-ber-encode-fail	Number of X3 BER Encode Failures	-	numeric
number-of-x3-event-encode-failures-no-buffer	Number of X3 Event Encode Failures due to No Buffer	-	numeric
number-of-x3-events-dropped-no-interface	Number of X3 Events Dropped No Interface	-	numeric
number-of-x3-events-for-current-active-calls	Number of X3 Events For current Active Calls	-	numeric
number-of-x3-events-received	Number of X3 Events Received	-	numeric
number-of-x3-interfaces	Number of configured X3 interfaces	-	numeric
number-of-x3-output-events	Number of X3 Encode Output Events	-	numeric
number-of-x3-output-events-failed	Number of X3 Encode Output Events Failed	-	numeric
number-of-x3-xer-encode-fail	Number of X3 XER Encode Failures	-	numeric
operational-status	Operational Status	up down partially-up maint dryup	enumeration
total-cumulative-LI-Sessions	Total Cumulative LI Sessions	-	numeric
total-number-of-all-terminated-LI-sessions	Total cumulative Number of All Terminated LI Sessions	-	numeric
total-number-of-intercepted-packets-for-current-active-calls	Total Number of intercepted Packets for current Active Calls	-	numeric
total-number-of-terminated-LI-sessions-due-to-call-disconnect	Total cumulative Number of Terminated LI Sessions due to Call Disconnect	-	numeric
total-number-of-terminated-LI-sessions-due-to-deletion-of-context	Total cumulative Number of Terminated LI Sessions due to Deletion of Interface Context	-	numeric
total-number-of-terminated-LI-sessions-due-to-deprovision	Total cumulative Number of Terminated LI Sessions due to Deprovisioning of Target	-	numeric

show calea instance event-agent target-status <CALEA event agent service name>

To display the CALEA instance event agent target status, you must be in Exec mode.

Command: show calea instance event-agent target-status <CALEA event agent service name>

Parameter	Description	Value or Range	Type
data-collection-type	Type of data collected	iri-only iri-plus-cc	enumeration
gateway-type-3gpp	Gateway Type	sgw-only pgw-only sgw-plus-pgw ggsn-only pgw-plus-ggsn all-gw sae-gw saegw-plus-ggsn none	enumeration
gateway-type-wlan	WLAN Access Gateway Type	none wag-only	enumeration
identity-type	Type of user identifier	imsi imei msisdn	enumeration
identity-value	Target identifier	-	numeric
number-of-x2-ber-encode-fail	Number of X2 BER Encode Fail	-	numeric
number-of-x2-encode-fail-no-buffer	Number of X2 Encode Fail no buffer	-	numeric
number-of-x2-events-dropped-no-interface	Number of X2EventsDroppedNoInterface	-	numeric
number-of-x2-events-for-current-active-calls	Number of X2 Events For current Active Calls	-	numeric
number-of-x2-events-sent	Number of X2 Events Sent	-	numeric
number-of-x2-xer-encode-fail	Number of X2 XER Encode Fail	-	numeric
number-of-x3-ber-encode-fail	Number of X3 BER Encode Fail	-	numeric
number-of-x3-event-ber-encode-failures	Number of X3 Event BER Encode Failures	-	numeric
number-of-x3-event-encode-failures-no-buffer	Number of X3 Event Encode Failures due to No Buffer	-	numeric
number-of-x3-event-xer-encode-failures	Number of X3 Event XER Encode Failures	-	numeric
number-of-x3-events-dropped-no-interface	Number of X3 Events Dropped - No Interface	-	numeric
number-of-x3-events-for-current-active-calls	Number of X3 Events For current Active Calls	-	numeric

number-of-x3-events-sent	Number of X3 Events Sent	-	numeric
number-of-x3-xer-encode-fail	Number of X3 XER Encode Fail	-	numeric
storage-type	Storage Type: persistent non-persistent	non-persistent persistent	enumeration
total-cumulative-LI-Sessions	Total Cumulative LI Sessions	-	numeric
total-number-of-all-terminated-LI-sessions	Total cumulative Number of All Terminated LI Sessions	-	numeric
total-number-of-intercepted-packets-for-current-active-calls	Total Number of intercepted Packets for current Active Calls	-	numeric
total-number-of-terminated-LI-sessions-due-to-call-disconnect	Total cumulative Number of Terminated LI Sessions due to Call Disconnect	-	numeric
total-number-of-terminated-LI-sessions-due-to-deletion-of-context	Total cumulative Number of Terminated LI Sessions due to Deletion of Interface Context	-	numeric
x2-interface-group-name	Name of X2 interface group	-	string
x3-interface-group-name	Name of X3 interface group	-	string

show calea interface-group-status <name>

To display the CALEA Interface Group status, you must be in Exec mode. The interface-group-status statistics that display differ depending on whether the interface-group-type is X2 or X3 and whether the interface role is primary or secondary.

Command: `show calea interface-group-status <name>`

Parameter	Description	Value or Range	Type
cc-buffer-current-size	Total number of bytes currently in the CC buffer	-	numeric
cc-buffer-high-water-mark	Maximum value reached for the total number of bytes present in the CC buffer	-	numeric
cc-buffer-num-buffer-size-exceeded	Number of times CC buffer size was exceeded requiring packets to be dropped	-	numeric
cc-buffer-num-dropped-packets	Number of packets dropped as a result of CC buffer size being exceeded	-	numeric
cc-buffer-num-packets	Total number of packets currently in the CC buffer	-	numeric
cc-buffer-size	Configured size of CC buffer (bytes)	-	numeric
currently-active-li-calls	Currently Active LI Calls using this interface group	-	numeric
interface-group-type	Type of interface group	x2 x3 x2-and-x3	enumeration
interfaceFormat-type	Type of Interface Format	AN_format HI_format	enumeration
iri-buffer-current-size	Total number of bytes currently in the IRI buffer	-	numeric
iri-buffer-high-water-mark	Maximum value reached for the total number of bytes present in the IRI buffer	-	numeric
iri-buffer-num-buffer-size-exceeded	Number of times IRI buffer size was exceeded requiring packets to be dropped	-	numeric
iri-buffer-num-dropped-packets	Number of packets dropped as a result of IRI buffer size being exceeded	-	numeric
iri-buffer-num-packets	Total number of packets currently in the IRI buffer	-	numeric
iri-buffer-size	Configured size of IRI buffer (bytes)	-	numeric
name*	Interface Group Name	-	-
number-of-agent-instances	Number of CALEA Agent instances supporting this interface group	-	numeric
number-of-targets	Number of configured monitored targets using this interface group	-	numeric

number-of-transmit-fail-bad-interface-group	Number of transmit failures bad interface group	-	numeric
number-of-transmit-fail-interface-group-down	Number of transmit failures interface group down	-	numeric
number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric
number-of-x2-ber-encode-fail	Number of X2 BER Encode Fail	-	numeric
number-of-x2-encode-fail-no-buffer	Number of X2 Encode Fail no buffer	-	numeric
number-of-x2-events-for-current-active-calls	Number of X2 Events For current Active Calls using this interface group	-	numeric
number-of-x2-events-offered	Number of X2 Events Offered for transmission	-	numeric
number-of-x2-xer-encode-fail	Number of X2 XER Encode Fail	-	numeric
number-of-x3-ber-encode-fail	Number of X3 BER Encode Fail	-	numeric
number-of-x3-encode-fail-no-buffer	Number of X3 Encode Fail no buffer	-	numeric
number-of-x3-event-ber-encode-failures	Number of X3 Event BER Encode Failures	-	numeric
number-of-x3-event-encode-failures-no-buffer	Number of X3 Event Encode Failures due to No Buffer	-	numeric
number-of-x3-event-xer-encode-failures	Number of X3 Event XER Encode Failures	-	numeric
number-of-x3-events-for-current-active-calls	Number of X3 Events For current Active Calls using this interface group	-	numeric
number-of-x3-events-offered	Number of X3 Events Offered for transmission	-	numeric
number-of-x3-xer-encode-fail	Number of X3 XER Encode Fail	-	numeric
operational-status	Operational Status	up down partially-up maint dryup	enumeration
output-type	Type of output encoding	xer ber	enumeration
role	Calea interface group role	primary secondary	enumeration
total-cumulative-LI-Sessions	Total Cumulative LI Sessions using this interface group	-	numeric
total-number-of-all-terminated-LI-sessions	Total cumulative Number of All Terminated LI Sessions	-	numeric

total-number-of-intercepted-packets-for-current-active-calls	Total Number of intercepted Packets for current Active Calls using this interface group	-	numeric
total-number-of-terminated-LI-sessions-due-to-call-disconnect	Total cumulative Number of Terminated LI Sessions due to Call Disconnect	-	numeric
total-number-of-terminated-LI-sessions-due-to-deprovision	Total cumulative Number of Terminated LI Sessions due to Deprovisioning of Target.	-	numeric

show calea interface-status

To display the CALEA Interface Table status, you must be in Exec mode.

Command: `show calea interface-status <name>`

Parameter	Description	Value or Range	Type
currently-active-li-calls	Currently Active LI Calls using this interface	-	numeric
interface-format-type	Type of interface Format	AN_format HI_for mat	enumeration
interface-type	Type of interface	x1-1 x2 x3 x2-and-x 3	enumeration
name*	Interface Name	-	-
number-of-agent-instances	Number of CALEA Agent instances supporting this interface	-	numeric
number-of-connection-attempts	Number of times socket connect() has been called	-	numeric
number-of-connection-attempt-fail	Number of times socket connect() failed to connect to LI server	-	numeric
number-of-connection-congested-high	Number of times TCP connection entered congestion state HIGH	-	numeric
number-of-connection-congested-low	Number of times TCP connection entered congestion state LOW	-	numeric
number-of-connection-congested-medium	Number of times TCP connection entered congestion state MEDIUM	-	numeric
number-of-connection-disconnect	Number of times a successfully established connection was subsequently disconnected	-	numeric

number-of-connection-success	Number of times socket connect() successfully established connection to LI server	-	numeric
number-of-echo-requests-received	Number of ECHO Requests Received	-	numeric
number-of-echo-requests-sent	Number of ECHO Requests Sent	-	numeric
number-of-echo-responses-received	Number of ECHO Responses Received	-	numeric
number-of-echo-responses-sent	Number of ECHO Responses Sent	-	numeric
number-of-echo-timeouts	Number of ECHO Timeouts	-	numeric
number-of-reset-indication-send-failures	Number of Reset Indication send failures	-	numeric
number-of-reset-indications-received	Number of Reset Indications received	-	numeric
number-of-reset-indications-sent	Number of Reset Indications sent	-	numeric
number-of-reset-response-send-failures	Number of Reset Response send failures	-	numeric
number-of-reset-responses-received	Number of Reset Responses received	-	numeric
number-of-reset-responses-sent	Number of Reset Responses sent	-	numeric
number-of-targets	Number of configured monitored targets using this interface	-	numeric
number-of-tpkt-keep-alive-ber-encode-fail-count	Number of TPKT Keep Alive BER Encode Fail Count	-	numeric
number-of-tpkt-keep-alive-messages-sent	Number of TPKT Keep Alive Messages Sent	-	numeric
number-of-tpkt-keep-alive-xer-encode-fail-count	Number of TPKT Keep Alive XER Encode Fail Count	-	numeric
number-of-transmit-fail-bad-interface	Number of transmit failures bad interface	-	numeric
number-of-transmit-fail-interface-down	Number of transmit failures interface down	-	numeric
number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric

number-of-x2-ber-encode-fail	Number of X2 BER Encode Fail	-	numeric
number-of-x2-encode-fail-no-buffer	Number of X2 Encode Fail no buffer	-	numeric
number-of-x2-events-for-current-active-calls	Number of X2 Events For current Active Calls using this interface	-	numeric
number-of-x2-events-offered	Number of X2 Events Offered for transmission	-	numeric
number-of-x2-xer-encode-fail	Number of X2 XER Encode Fail	-	numeric
number-of-x3-ber-encode-fail	Number of X3 BER Encode Fail	-	numeric
number-of-x3-encode-fail-no-buffer	Number of X3 Encode Fail no buffer	-	numeric
number-of-x3-event-ber-encode-failures	Number of X3 Event BER Encode Failures	-	numeric
number-of-x3-event-encode-failures-no-buffer	Number of X3 Event Encode Failures due to No Buffer	-	numeric
number-of-x3-event-xer-encode-failures	Number of X3 Event XER Encode Failures	-	numeric
number-of-x3-events-for-current-active-calls	Number of X3 Events For current Active Calls using this interface	-	numeric
number-of-x3-events-offered	Number of X3 Events Offered for transmission	-	numeric
number-of-x3-xer-encode-fail	Number of X3 XER Encode Fail	-	numeric
operational-status	Operational Status	up down partially-up maint dryup	enumeration
total-cumulative-LI-Sessions	Total Cumulative LI Sessions using this interface	-	numeric
total-number-of-all-terminated-LI-sessions	Total cumulative Number of All Terminated LI Sessions	-	numeric
total-number-of-intercepted-packets-for-current-active-calls	Total Number of intercepted Packets for current Active Calls using this interface	-	numeric
total-number-of-terminated-LI-sessions-due-to-call-disconnect	Total cumulative Number of Terminated LI Sessions due to Call Disconnect	-	numeric
total-number-of-terminated-LI-sessions-due-to-deprovision	Total cumulative Number of Terminated LI Sessions due to Deprovisioning of Target.	-	numeric
transport-type	Type of transport used by this interface	tcp	enumeration

show calea service-status

To display the CALEA service status, you must be in Exec mode.

Command: `show calea service-status <name>`

Parameter	Description	Value or Range	Type
actual-max-calea-traces-per-subscriber-water-mark	Largest number of CALEA traces per subscriber observed on the system. This statistic tracks the actual maximum observed during system operation. It represents the maximum value high water mark.	-	numeric
currently-active-li-calls	Currently Active LI Calls	-	numeric
currently-camp-on-triggers	Currently Camp on LI Triggers: (Total Configured LIIDs - Current Active Calls)	-	numeric
decrypt-block-bytes	Number of bytes decrypted as blocks	-	numeric
decrypt-block-operations	Number of block decryptions	-	numeric
decrypt-stream-bytes	Number of bytes decrypted as streams	-	numeric
decrypt-stream-fail	Number of stream decryption failures	-	numeric
decrypt-stream-operations	Number of stream decryptions	-	numeric
encrypt-block-bytes	Number of bytes encrypted as blocks	-	numeric
encrypt-block-operations	Number of block encryption	-	numeric
encryption-instances-activated	Number of encryption instances activated	-	numeric
encryption-key-disabled	Number of times encryption key was disabled	-	numeric
encryption-key-enabled	Number of times encryption key was enabled	-	numeric
encrypt-stream-bytes	Number of bytes encrypted as streams	-	numeric
encrypt-stream-operations	Number of stream encryptions	-	numeric
max-calea-traces-per-subscriber-overflow	Total number of targets that exceed the recommended maximum of CALEA traces per subscriber. This counter is incremented every time a subscriber, which has already reached the maximum CALEA traces (8), is targeted for interception.	-	numeric
name*	-	all	enumeration
number-of-agent-instances	Number of CALEA Agent instances running	-	numeric

number-of-bad-events-received	Number of Bad Events Received	-	numeric
number-of-echo-requests-received	Number of ECHO Requests Received	-	numeric
number-of-echo-requests-sent	Number of ECHO Requests Sent	-	numeric
number-of-echo-responses-received	Number of ECHO Responses Received	-	numeric
number-of-echo-responses-sent	Number of ECHO Responses Sent	-	numeric
number-of-echo-timeouts	Number of ECHO Timeouts	-	numeric
number-of-rx-events-dropped-no-target	Number of Received Events Dropped No Target	-	numeric
number-of-targets	Number of configured monitored targets	-	numeric
number-of-transmit-fail-bad-interface	Number of transmit failures bad interface	-	numeric
number-of-transmit-fail-interface-down	Number of transmit failures interface down	-	numeric
number-of-transmit-fail-no-space	Number of transmit failures no space	-	numeric
number-of-transmit-ok	Number of transmits succeeded	-	numeric
number-of-user-ban-enabled	Total number of calea targets on which userBan feature is enabled	-	numeric
number-of-user-ban-session-disconnect	Total number of established sessions/bearers which were disconnected due to userBan enabled	-	numeric
number-of-user-ban-session-reject	Total number of session/bearer setup requests which were rejected due to userBan enabled	-	numeric
number-of-x1-1-interfaces	Number of configured X1_1 interfaces	-	numeric
number-of-x2-ber-encode-fail	Number of X2 BER Encode Failures	-	numeric
number-of-x2-encode-fail-no-buffer	Number of X2 Encode Fail No Buffer	-	numeric
number-of-x2-events-dropped-no-interface	Number of X2 Events Dropped No Interface	-	numeric
number-of-x2-events-for-current-active-calls	Number of X2 Events For current Active Calls	-	numeric
number-of-x2-events-received	Number of X2 Events Received	-	numeric
number-of-x2-interfaces	Number of configured X2 interfaces	-	numeric

number-of-x2-output-events	Number of X2 Encode Output Events	-	numeric
number-of-x2-output-events-failed	Number of X2 Encode Output Events Failed	-	numeric
number-of-x2-xer-encode-fail	Number of X2 XER Encode Failures	-	numeric
number-of-x3-ber-encode-fail	Number of X3 BER Encode Failures	-	numeric
number-of-x3-event-encode-failures-no-buffer	Number of X3 Event Encode Failures due to No Buffer	-	numeric
number-of-x3-events-dropped-no-interface	Number of X3 Events Dropped No Interface	-	numeric
number-of-x3-events-for-current-active-calls	Number of X3 Events For current Active Calls	-	numeric
number-of-x3-events-received	Number of X3 Events Received	-	numeric
number-of-x3-interfaces	Number of configured X3 interfaces	-	numeric
number-of-x3-output-events	Number of X3 Encode Output Events	-	numeric
number-of-x3-output-events-failed	Number of X3 Encode Output Events Failed	-	numeric
number-of-x3-xer-encode-fail	Number of X3 XER Encode Failures	-	numeric
operational-status	Operational Status	up down partially-up maint dryup	enumeration
target-encryption-type	Target identity encryption type in use	-	numeric
total-cumulative-LI-Sessions	Total Cumulative LI Sessions	-	numeric
total-number-of-all-terminated-LI-sessions	Total cumulative Number of All Terminated LI Sessions	-	numeric
total-number-of-intercepted-packets-for-current-active-calls	Total Number of intercepted Packets for current Active Calls	-	numeric
total-number-of-terminated-LI-sessions-due-to-call-disconnect	Total cumulative Number of Terminated LI Sessions due to Call Disconnect	-	numeric
total-number-of-terminated-LI-sessions-due-to-deletion-of-context	Total cumulative Number of Terminated LI Sessions due to Deletion of Interface Context	-	numeric
total-number-of-terminated-LI-sessions-due-to-deprovision	Total cumulative Number of Terminated LI Sessions due to Deprovisioning of Target	-	numeric

show calea target-status

To display the CALEA Interface target status, you must be in Exec mode.

Command: `show calea target-status <liid>`

Parameter	Description	Value or Range	Type
data-collection-type	Type of data collected	iri-only iri-plus-cc cc-only	enumeration
gateway-type-wlan	WLAN Access Gateway Type	none wag-only	enumeration
gateway-type	3GPP Gateway Type	sgw-only pgw-only sgw-plus-pgw ggsn-only pgw-plus-ggsn all-gw sae-gw saegw-plus-ggsn none	enumeration
identity-type	Type of user identifier	imsi imei msisdn	enumeration
identity-value	Target identifier. IMEI values include leading zeros.	-	numeric
identity-value-mac	Deprecated field that is always empty. Provided for backward compatibility in the show calea target-status display.	-	-
liid*	Lawful Intercept Identifier	-	string
number-of-agent-instances	Number of CALEA Agent instances supporting this interface	-	numeric
number-of-user-ban-session-disconnect	Number of established sessions/bearers for this target which were disconnected due to userBan enabled	-	numeric
number-of-user-ban-session-reject	Number of session/bearer setup requests for this target which were rejected due to userBan enabled	-	numeric
number-of-x2-ber-encode-fail	Number of X2 BER Encode Fail	-	numeric
number-of-x2-encode-fail-no-buffer	Number of X2 Encode Fail no buffer	-	numeric
number-of-x2-events-dropped-no-interface	Number of X2EventsDroppedNoInterface	-	numeric
number-of-x2-events-for-current-active-calls	Number of X2 Events For current Active Calls	-	numeric
number-of-x2-events-sent	Number of X2 Events Sent	-	numeric
number-of-x2-xer-encode-fail	Number of X2 XER Encode Fail	-	numeric
number-of-x3-ber-encode-fail	Number of X3 BER Encode Fail	-	numeric

number-of-x3-event-ber-encode-failures	Number of X3 Event BER Encode Failures	-	numeric
number-of-x3-event-encode-failures-no-buffer	Number of X3 Event Encode Failures due to No Buffer	-	numeric
number-of-x3-event-xer-encode-failures	Number of X3 Event XER Encode Failures	-	numeric
number-of-x3-events-dropped-no-interface	Number of X3 Events Dropped - No Interface	-	numeric
number-of-x3-events-for-current-active-calls	Number of X3 Events For current Active Calls	-	numeric
number-of-x3-events-sent	Number of X3 Events Sent	-	numeric
number-of-x3-xer-encode-fail	Number of X3 XER Encode Fail	-	numeric
storage-type	Storage Type	non-persistent persistent dynamic	enumeration
total-cumulative-LI-Sessions	Total Cumulative LI Sessions	-	numeric
total-number-of-all-terminated-LI-sessions	Total cumulative Number of All Terminated LI Sessions	-	numeric
total-number-of-intercepted-packets-for-current-active-calls	Total Number of intercepted Packets for current Active Calls	-	numeric
total-number-of-terminated-LI-sessions-due-to-call-disconnect	Total cumulative Number of Terminated LI Sessions due to Call Disconnect	-	numeric
total-number-of-terminated-LI-sessions-due-to-deletion-of-context	Total cumulative Number of Terminated LI Sessions due to Deletion of Interface Context	-	numeric
user-ban-enabled	True/false indicating whether userBan feature is enabled for this calea target	false true	enumeration
x2-interface-group-name	Name of X2 interface group	-	string
x2-interface-name	Name of X2 interface	-	string
x3-interface-group-name	Name of X3 interface group	-	string
x3-interface-name	Name of X3 interface	-	string

show image-management status

To display the Image Management current status running on the MCC.

Command: `show image-management status <state> <id> <slot> <cpu>`

Parameter	Description	Value or Range	Type
activation-status	Activation status	-	string
admin-version	Current upgrade admin version	-	string
cpu*	-	-	numeric
id*	-	-	numeric
operational-version	Version currently running	-	string
prepared-version	Version extracted and ready for upgrade	-	string
rollback-version	Version available for rollback	-	string
slot*	-	-	numeric
state*	-	active standby n/a	enumeration

show system uptime

To display the system uptime in seconds.

Command: `show system uptime`

Parameter	Description	Value or Range	Type
uptime	-	-	numeric

user-access Commands

MCC supports the following calea user-access command to manage user accounts:

```
an3000-mcm-slot7cpu1# user-access
```

Possible completions:

<code>add-user</code>	Add a new management user
<code>change-password</code>	Change password
<code>change-sftp-dir</code>	Change the directory sftp/scp starts in for this user
<code>delete-user</code>	Delete a management user
<code>reset-password</code>	Reset password
<code>reset-sftp-dir</code>	Reset sftp dir
<code>ssh-password-authentication</code>	Enable/Disable the use of passwords for ssh authentication

Configure the parameters as follows:

Table 2-4 user-access Parameters

Object	Description
add-user	Adds a new CALEA user and password. Specify the cli-calea group for the new user.
change-password	Changes a CALEA user password.
change-sftp-dir	Changes the directory in which sftp/scp starts for this CALEA user.
delete-user	Deletes a CALEA user.
reset-password	Resets the password for a CALEA user.
reset-sftp-dir	Resets the default SFTP directory of a CALEA user.
ssh-password-authentication	Enables/disables the use of passwords for ssh on a per-user basis.

Use the show user-access users command to list calea user details:

```
an3000-mcm-slot7cpu1# show user-access users
```

Group	Username	Type	HomeDir	Session Limits	Pki	Last Passwd Chg	Last Login	Active/Standby
cli-calea	CaleaUser1	local	/home/calea/CaleaUser1	100:100	cli	100 ems 100 scp	**Never changed**	**Never
cli-calea	calea	default	/home/calea/calea	100: 100	cli	100 ems 100 scp	**Never changed**	Wed Apr 11 16:50:26 -0400 2018 / none

Security Recommendation

Only the calea user can manage user accounts for calea users, the admin user cannot. However, the root account can manage user accounts for calea users.

It is recommended for security purposes that you disable the root account, so that root cannot manage user accounts for calea users.

To disable the root account:

- 1 Login as **root**.
- 2 Issue the following command:

```
usermod --lock root
```

WARNING: This command permanently disables the **root** password login for this release and any release to which you upgrade. There is no recovery from this command; it requires that you roll back to a previous release in which the **root** account was not disabled or re-commission the node.

ADMIN CLI User Account

To execute any ADMIN CLI commands, you must be logged into the CLI shell as user **admin**.

Note: The majority of CALEA configuration commands are executed from the CALEA user account. This section documents the few CALEA related commands that are executed from the ADMIN user account.

Configuring a Separate network-context and loopback-ip for the X1_1 Interface

In normal operation, the X1_1 interface uses the standard management-context, mgmt-ip address. In addition, CALEA supports use of a separate network-context, loopback-ip for the X1_1 interface.

This feature allows use of separate X1-1 Network Context, Loopback IP for:

- SSH CALEA CLI shell. It is the X1 CALEA provisioning interface used by the ADMF to provision CALEA on the MCC.
- X1_1 TCP connection used by the MCC to send reset indication messages to ADMF. This reset message notifies the ADMF when an event (such as redundancy switchover) occurs for which the ADMF should audit the configured Lawful Intercept targets.

The X1_1 TCP connection is established using that same Network Context, Loopback IP for the source side (MCC side) of the X1-1 TCP connection.

To configure the CALEA Management IP, you must be in Config mode.

Command: `infrastructure calea mgmt-ip network-context <network context name> loopback-ip <loopback ip name>`

Parameter	Description	Value or Range	Type
network-context*	Name of a network-context provisioned on the MCC.	-	-
loopback-ip*	Name of a loopback-ip address in the network-context		

This command enables the CALEA SSH CLI shell access using the specified network-context, loopback-ip.

Security Recommendation

Once the shell is enabled, it is recommended for security purposes that you disable CALEA ssh access using the standard management-context, mgmt-ip address from the following command issued from the **maint** user:

To disable CALEA ssh access:

- 1 On the master MCM console, login as user **maint**.
- 2 Issue the command **sshdisable calea**.

It is the recommended mode of operation when CALEA SSH CLI access is enabled for a separate network-context, loopback-ip.

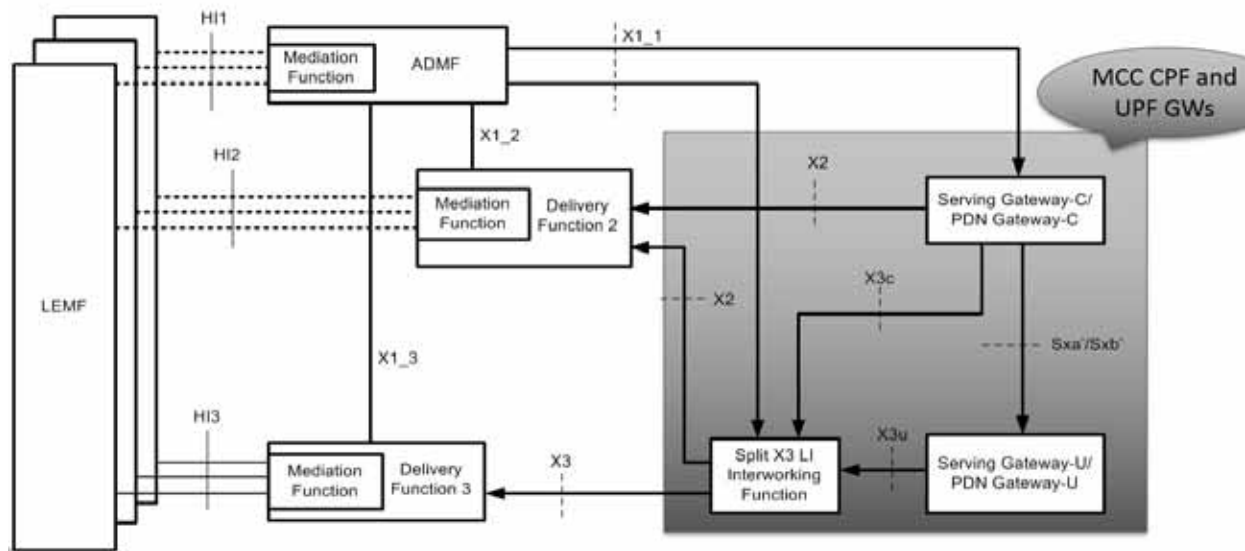
To re-enable CALEA SSH CLI access, login to the master MCM console as user **maint** and issue the command **sshenable calea**.

When this feature is enabled as described above, the X1_1 TCP interface is established with its source address being the same network-context, loopback-ip as specified in the **infrastructure CALEA mgmt-ip** command for the CALEA SSH CLI shell.

Lawful Intercept Configuration for Control and User Plane Separation (CUPS)

The following figure references 3GPP TS 33.107 Section 12.9 Functional Requirements for LI in case of Control and User Plane Separation (Figure 12.1.4):

Figure 2-1 Lawful Interception for Control and User Plane Separation



In Figure 2-1, the functionality in the gray box is provided by the MCC gateway services.

X2 IRI events are always delivered by the CPF. For X3 Call Content, configuration of the Split X3 LI Interworking Function (SX3LIF) determines whether X3 Call Content is delivered by the CPF or UPF. Choose one of these configurations for your deployment. Configure it on the CPF and UPF MCC gateways:

- **Call Content from CPF LI Targets:**

- ADMF provisions LI targets only on the CPF
- UPF delivers LI packets to the CPF via SX3LIF GTP-U tunnel
- CPF delivers HI2 IRI and HI3 CC-data to the DF/MF/LEMF

- **Call Content from UPF Static LI Targets:**

- ADMF provisions LI targets on both the CPF and UPF
- CPF delivers HI2 IRI to the LEMF
- UPF delivers HI3 CC-data directly to the LEMF

■ **Call Content from UPF Dynamic LI Targets:**

- ADMF provisions LI targets only on the CPF
- CPF delivers HI2 IRI to the LEMF
- CPF signals target information to the UPF when the session is intercepted
- UPF creates UPF dynamic target based on target information sent from the CPF
- UPF delivers HI3 CC-data directly to the LEMF

To enable this configuration, the ADMF must configure the following objects on both CPF MCC and UPF MCC:

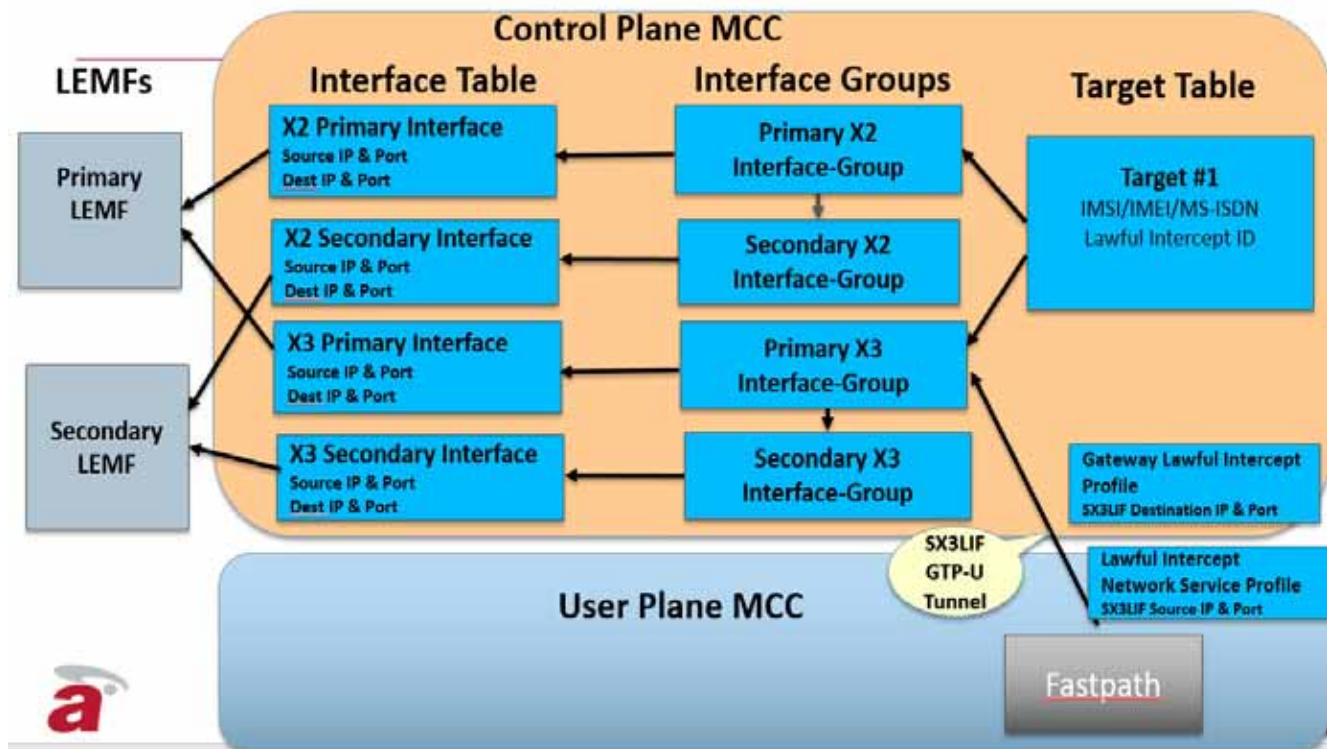
- pfcplawful-intercept-ie with secret-key
- interface-group-source-profiles

The provisioning requirements for these options are described in the following sections.

Call Content from CPF LI Targets

Configure the CPF and UPF MCC gateways as described in this section if you want the CPF to deliver Call Content (X3/HI3) to the LEMF.

Figure 2-2 Call Content from CPF LI Targets



The administrator account (admin) should perform the following configuration:

To provision SX3LIF:

- 1 Create the gateway lawful-intercept-profile on the CPF MCC:

```
admin% zone <name> gateway profile lawful-intercept-profile  
<LI-profile-name> network-context <name> sx3lif-ip-address-name  
<loopback-ip-name> sx3lif-port <udp-port-number>
```

where:

network-context is the Network Context in which the SX3LIF IP address resides.

sx3lif-ip-address-name and **sx3lif-port** are the IP address and port at which the CPF MCC receives duplicated packets from UPF MCC. The **sx3lif-port** is optional. If you omit it, the CPF uses the default GTP-U port number.

For example:

```
admin% # zone DEFAULT gateway profile lawful-intercept-profile  
GW_LI_PROFILE network-context GN sx3lif-ip-address-name CALEALoop  
sx3lif-port 7777
```

- 2 Associate the lawful-intercept-profile with the desired Gateway(s) using the following CLI command:

```
admin% zone <name> gateway pgw <name> lawful-intercept-profile  
<LI-profile-name>
```

For example, to associate the lawful-intercept-profile GW_LI_PROFILE with both the PGW and SGW:

```
admin% # zone default gateway pgw MYPGW lawful-intercept-profile  
GW_LI_PROFILE;top
```

```
admin% # zone default gateway sgw MYSGW lawful-intercept-profile  
GW_LI_PROFILE;top;commit
```

- 3 On the UPF MCC, create a network-profile of type lawful-intercept:

```
admin% service-construct network-profile <name> interface-type  
lawful-intercept network-context <context-name> li-source-ip-address  
<loopback-ip-address-name> li-source-port <udp-port-number>
```

where:

interface-type is **lawful-interface**.

network-context is the Network Context that the UPF should use to reach the SX3LIF IP address on the CPF.

li-source-ip-address is the source IP address that the UPF fastpath should populate in GTP-U packets sent to SX3LIF IP address.

li-source-port is the source UDP port number that the UPF fastpath should populate in GTP-U packets sent to SX3LIF IP address. The **li-source-port** is optional. If you omit it, the UPF uses the default GTP-U port number.

For example:

```
admin% # service-construct network-profile UPS1-LI-PROF interface-type
lawful-intercept network-context SXAB li-source-ip-address li-sx3-IP
li-source-port 9019
```

- 4 Associate the network-profile with the desired user-plane-service using the following CLI command:

```
admin% zone default gateway user-plane-service <name> network-profile
<profile-name>
```

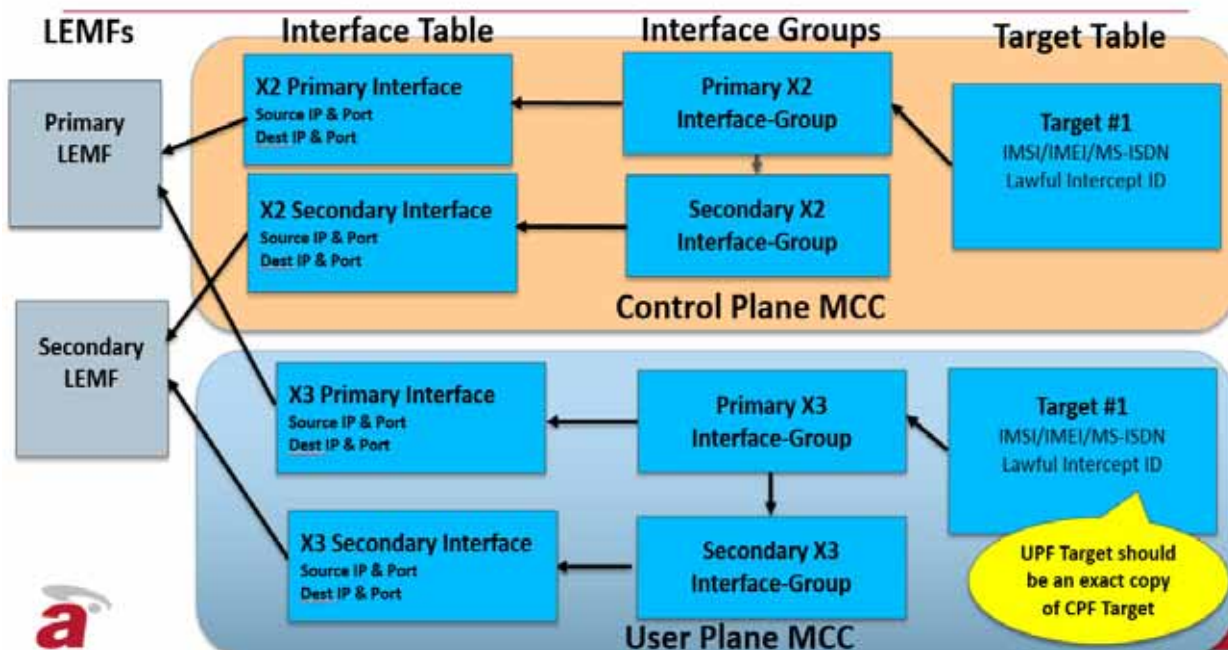
For example:

```
admin% # zone default gateway user-plane-service UPS-1 network-profile
UPS1-LI-PROF
```

Call Content from UPF Static LI Targets

Configure the CPF and UPF MCC gateways as described in this section if you want the UPF to deliver Call Content (X3/HI3) directly to the LEMF using static targets configured by the ADMF on both CPF and UPF MCC.

Figure 2-3 Call Content from UPF Static LI Targets



This option *does not* require configuration of the gateway lawful-intercept-profile or the network-profile of type lawful-intercept.

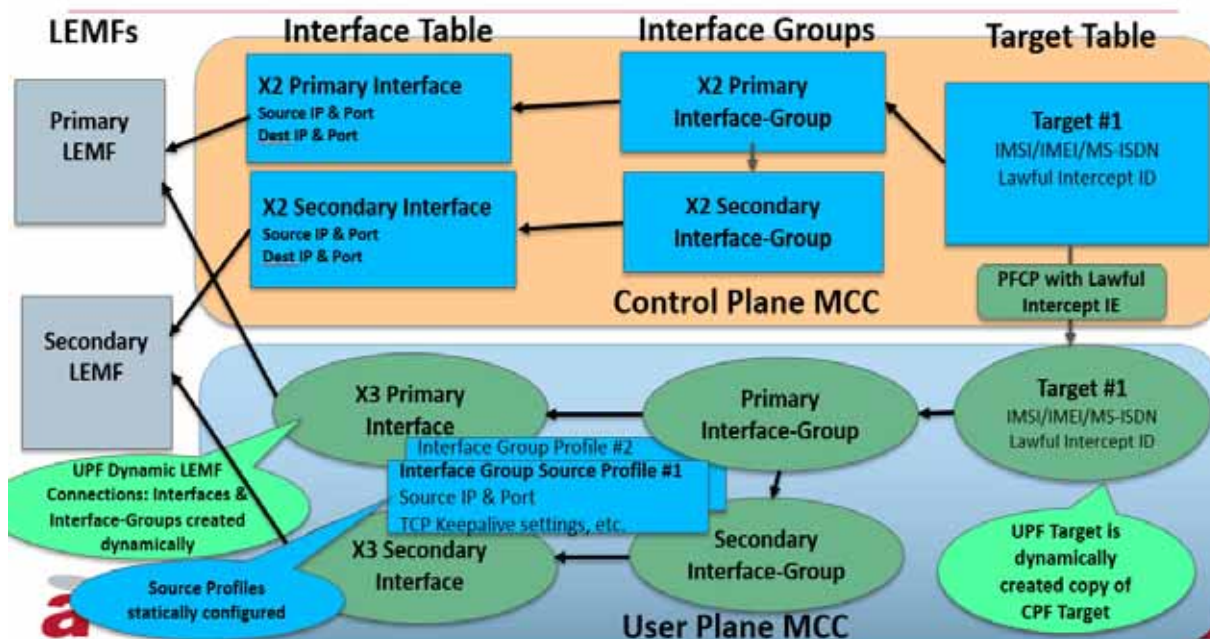
Instead, the CALEA user provisions the following entities:

- CALEA ADMF configures X2 and X3 interfaces and interface-groups on the CPF MCC
- X3 interfaces and interface-groups on the CPF are effectively dummy interfaces (are not used on CPF MCC in this configuration)
- ADMF configures X2 and X3 Interfaces and interface-groups on the UPF MCC
- X2 interfaces and interface-groups on the UPF are effectively dummy interfaces (are not used on UPF MCC in this configuration)
- ADMF provisions calea targets on both the CPF MCC and the UPF MCC

Call Content from UPF Dynamic LI Targets

Configure the CPF and UPF MCC gateways as described in this section if you want the UPF to deliver Call Content (X3/HI3) directly to the LEMF with targets configured by ADMF only on the CPF MCC.

Figure 2-4 Call Content from UPF Dynamic LI Targets



This option uses the following configuration:

To provision UPF Dynamic LI Targets:

- 1 CALEA ADMF configures pfcplawfulinterceptie on both the CPF MCC and UPF MCC. See [calea pfcplawfulinterceptie](#) for details of this configuration.
- 2 CALEA ADMF configures X2 and X3 interfaces and interface-groups on the CPF MCC. See [calea interface-table](#) and [calea interface-group <name>](#) for details of this configuration.
- 3 CALEA ADMF configures X2 and X3 interface-group-source-profiles on the UPF MCC.

The interface-group-source-profile is required on the UPF MCC to allow Dynamic Targets to establish a connection to the LEMF. Specify source IP and source port for the UPF MCC to use when connecting to the LEMF. See [calea interface-group-source-profile](#) for details of this configuration.

Multiple X2/X3 Delivery Destinations

To facilitate provisioning a system with many different X2/X3 delivery destinations, the calea target-table syntax has been enhanced to allow direct specification of the X2 and X3 LEMF IP address and port numbers. When a target is configured with these values, the MCC dynamically creates the corresponding calea interface-groups and calea interfaces:

- X2 interface-group(s) and X2 interface(s) are dynamically created by the MCC when the target is provisioned.
- X3 interface-group(s) and X3 interface(s) are dynamically created when a session is intercepted.

Prior to configuring targets in this manner, configure the required calea interface-group-source-profiles for the X2 and X3 LEMF IP addresses and ports (see [calea interface-group-source-profile](#)). If the required interface-group-source-profiles do not exist when the MCC attempts to dynamically create the target's calea interface-groups and interfaces, the dynamic creation fails. An alarm is raised to indicate that a required calea interface-group-source-profile was not found.

To configure a target in this manner, specify the following options in the calea target-table entry (see [calea target-table <liid>](#)):

- x2-primary-ip-address
- x2-primary-port
- x3-primary-ip-address
- x3-primary-port

Optionally, you can also specify secondary IP address and port numbers using the following keywords:

- x2-secondary-ip-address
- x2-secondary-port
- x3-secondary-ip-address
- x3-secondary-port

The MCC first attempts to establish TCP connections to the LEMF(s) specified by the primary IP address and port numbers. If MCC is unable to establish a TCP connection to a specified primary LEMF, it then attempts to establish a TCP connection to the LEMF specified by secondary IP address and port number.

Multiple X2/X3 Delivery Destinations is also supported with non-persistent targets. See [calea createTarget](#) for details.

Behavior Change

In previous releases, the show calea target-status command displayed the following fields:

Only for persistent targets:

- x2-interface-group-name
- x3-interface-group-name

Only for non-persistent targets:

- x2-interface-name
- x3-interface-name

In this release, the show calea target-status command now displays all of the fields according to the follow rules:

When non-persistent targets are associated directly with an interface and no Multiple HI2/HI3 configuration is present:

- x2-interface-name - populated
- x3-interface-name - populated (for iri-plus-cc targets), otherwise displays empty string ""
- x2-interface-group-name - displays empty string ""
- x3-interface-group-name - displays empty string ""

When non-persistent targets are used with the Multiple HI2/HI3 feature configured:

- x2-interface-name - displays empty string ""
- x3-interface-name - displays empty string ""
- x2-interface-group-name - populated
- x3-interface-group-name - populated (for iri-plus-cc targets with Multiple HI3 LEMF once the dynamic X3 group has been created), otherwise displays empty string ""

Example 1: Target with Specified Primary X2 and X3 LEMF IP Addresses and Port Numbers

```
an3000-mcm-slot7cpu1# show running-config calea target-table
```

```
calea target-table $3$r/zkEp20u736RWBZHQsfXQ==
identityType          imsi
identityVal           $3$JX4Dsz6Twd6dwte5gobZug==
dataCollectionType    iri-plus-cc
gwType                sae-gw
wlanGwType            none
operator-id           att
location-info         true
iri-event-time-stamp-type local-time
cc-data-time-stamp-type local-time
packet-data-header-info-report PDHIR_DISABLE
x2-primary-ip-address 10.33.39.49
x2-primary-port       7453
x3-primary-ip-address 10.33.39.49
x3-primary-port       7463
!
```

Example 2: Target with Specified Primary and Secondary X2 and X3 LEMF IP Addresses and Port Numbers

```
an3000-mcm-slot7cpu1# show running-config calea target-table
```

```
calea target-table $3$r/zkEp20u736RWBZHQsfXQ==
identityType          imsi
identityVal           $3$JX4Dsz6Twd6dwte5gobZug==
dataCollectionType    iri-plus-cc
gwType                sae-gw
wlanGwType            none
operator-id           att
location-info         true
iri-event-time-stamp-type local-time
cc-data-time-stamp-type local-time
packet-data-header-info-report PDHIR_DISABLE
x2-primary-ip-address 10.1.1.1
x2-primary-port       6022
```

```
x2-secondary-ip-address      10.2.2.2
x2-secondary-port            7022
x3-primary-ip-address        10.1.1.1
x3-primary-port              6033
x3-secondary-ip-address      10.2.2.2
x3-secondary-port            7033
!
```

As shown in these examples, the Lawful Intercept Identifier and IdentityVal values are displayed in encrypted form.

When this feature is used on CUPS systems, for targets provisioned with data-collection-type iri-plus-cc:

- The dynamic X2 interface-group and X2 interface are created on the CPF at the time of target provisioning, regardless of whether or not the session is established.
- The dynamic X3 interface-group and X3 interface are created on the UPF at the time of session interception (such as, at the time the session is established).

AN-FORMAT ASN.1 Definition

```
-- Affirmed Networks CALEA Module Definition
AffirmedCaleaModule DEFINITIONS AUTOMATIC TAGS ::=
BEGIN

IMPORTS
    LawfulInterceptionIdentifier
        FROM AffirmedCaleaHIModule
    lawfulInterceptDomainId
        FROM AffirmedCaleaHIModule
    threeGPPSUBDomainId
        FROM AffirmedCaleaHIModule
    TimeStamp
        FROM AffirmedCaleaHIModule
    LocalTimeStamp
        FROM AffirmedCaleaHIModule
    National-Parameters
        FROM AffirmedCaleaHIModule
    Network-Identifier
        FROM AffirmedCaleaHIModule
    Network-Element-Identifier
        FROM AffirmedCaleaHIModule
    DataNodeAddress
        FROM AffirmedCaleaHIModule
    IPAddress
        FROM AffirmedCaleaHIModule
    IP-value
        FROM AffirmedCaleaHIModule
    X25Address
        FROM AffirmedCaleaHIModule
    Version
        FROM AffirmedCaleaHIModule
```

AN-FORMAT ASN.1 Definition

```
National-HI2-ASN1parameters
    FROM AffirmedCaleaHIModule
TPDU-direction
    FROM AffirmedCaleaHIModule
National-HI3-ASN1parameters
    FROM AffirmedCaleaHIModule
ICE-type
    FROM AffirmedCaleaHIModule;

-- Start of Affirmed Networks Defined Types

-- LIID
AnLawfulInterceptionIdentifier ::= OCTET STRING (SIZE (1..25))

-- Sequence number and reason code
AnSequenceNumber ::= INTEGER
AnReasonCode ::= ENUMERATED
{
    success(1),
    operation-not-supported(2),
    asn1-syntax-error(3),
    general-system-error(4),
    ...
}

-- Party identification
AnPartyInformation ::= SEQUENCE
{
    party-Qualifier [0] ENUMERATED
    {
        gPRSorEPS-Target(3),
        ...
    },

    -- All identities are in packed BCD Format.  If length is odd, the last nibble will
    -- contain 0xF.
    partyIdentity [1] SEQUENCE
    {
        imei[1] OCTET STRING (SIZE (8)) OPTIONAL,
        imsi[3] OCTET STRING (SIZE (3..8)) OPTIONAL,
        msISDN[6] OCTET STRING (SIZE (1..9)) OPTIONAL,
        ...
    },
}
```



```

    ...
}

-- Type of bearer/PDP context
AnTypeOfBearer ::= ENUMERATED
{
    defaultBearer          (1),
    dedicatedBearer        (2),
    ...
}
AnTypeOfPdpContext ::= ENUMERATED
{
    primary                (1),
    secondary              (2),
    ...
}
AnBearerActivationType ::= AnTypeOfBearer
AnBearerDeactivationType ::= AnTypeOfBearer

-- Procedure initiator
AnInitiator ::= ENUMERATED
{
    not-Available(0),
    originating-Target(1),
    -- in case of GPRS, this indicates that the PDP context activation, modification
    -- or deactivation is MS requested
    -- in case of EPS, this indicated that the EPS detach, bearer activation,
    -- modification or deactivation is UE requested
    terminating-Target(2),
    -- in case of GPRS, this indicates that the PDP context activation, modification or
    -- deactivation is network initiated
    -- in case of EPS, this indicated that the EPS detach, bearer activation,
    -- modification or deactivation is network initiated
    ...
}

-- Indication of handover
AnHandoverIndication ::= NULL

-- Unique identifier of NE sending event
AnNetworkElementIdentifier ::= OCTET STRING

-- Correlation number used to correlated IRI and CC traffic

```

AN-FORMAT ASN.1 Definition

```
AnCorrelationNumber ::= OCTET STRING (SIZE(8..20))

-- All the following parameters are coded as the corresponding IEs without the octets
-- containing type and length. Unless differently stated, they are coded according to
-- 3GPP TS 29.060 (in the case of GGSN) or 3GPP TS 29.274 (in the case of SGW/PDNGW)
-- The octet containing the instance shall also be not included in the SGW/PDNGW case
-- Start
AnPDNAddressAllocation ::= OCTET STRING
AnAPN ::= OCTET STRING (SIZE (1..100))
AnProtConfigOptions ::= SEQUENCE
{
    ueToNetwork                [1] OCTET STRING (SIZE(1..251))      OPTIONAL,
    networkToUe                [2] OCTET STRING (SIZE(1..251))      OPTIONAL,
    ...
}
AnRATType ::= OCTET STRING (SIZE (1))
AnFailedBearerActivationReason ::= OCTET STRING
AnFailedBearerModificationReason ::= OCTET STRING
AnEPSBearerQos ::= OCTET STRING
AnAPN-AMBR ::= OCTET STRING
AnProcedureTransactionID ::= OCTET STRING
AnTFT ::= OCTET STRING
AnFailedUEReqBearerResModReason ::= OCTET STRING
AnTrafficAggregateDescription ::= OCTET STRING
AnBearerDeactivationCause ::= OCTET STRING
AnEPSBearerIdentity ::= OCTET STRING (SIZE (1))
AnLinkedEPSBearerID ::= OCTET STRING (SIZE (1))
AnEndUserAddress ::= OCTET STRING
AnQos ::= OCTET STRING
AnFailedContextActivationReason ::= OCTET STRING (SIZE (1))
AnContextDeactivationCause ::= OCTET STRING (SIZE (1))
AnNsapi ::= OCTET STRING (SIZE (1))
AnLocationOfTheTarget ::= SEQUENCE
{
    userLocationInfo    [1] OCTET STRING (SIZE (1..34)) OPTIONAL,
    olduserLocationInfo[4] OCTET STRING (SIZE (1..34))OPTIONAL,
    ...
}
-- End

-- Top level Lawful Intercept Message
AN-LI-MESSAGE ::= CHOICE
{
```

```

    adminMessage AN-ADMIN-MESSAGE,
    iriMessage AN-IRI-MESSAGE,
    ccPdu AN-CC-PDU
}

AN-ADMIN-MESSAGE ::= CHOICE
{
    reset AN-RESET,
    resetResponse AN-RESET-RESPONSE,
    echoRequest AN-ECHO-REQUEST,
    echoResponse AN-ECHO-RESPONSE
}

AN-RESET ::= AnSequenceNumber
AN-RESET-RESPONSE ::= SEQUENCE
{
    sequenceNumber AnSequenceNumber,
    reasonCode AnReasonCode
}

AN-ECHO-REQUEST ::= SEQUENCE
{
}

AN-ECHO-RESPONSE ::= SEQUENCE
{
}

AN-IRI-MESSAGE ::= CHOICE
{
    -- 3G Events
    pdpContextActivation AN-PDP-CONTEXT-ACTIVATION,
    startofInterceptionWithActivePdpContext
AN-START-OF-INTERCEPTION-WITH-ACTIVE-PDP-CONTEXT,
    pdpContextModification AN-PDP-CONTEXT-MODIFICATION,
    pdpContextDeactivation AN-PDP-CONTEXT-DEACTIVATION,
    -- 4G Events
    bearerActivation AN-BEARER-ACTIVATION,
    startofInterceptionWithActiveBearer
AN-START-OF-INTERCEPTION-WITH-ACTIVE-BEARER,
    bearerModification AN-BEARER-MODIFICATION,
    bearerDeactivation AN-BEARER-DEACTIVATION,
    ueRequestedBearerResourceModification
AN-UE-REQUESTED-BEARER-RESOURCE-MODIFICATION,
    ...
}

```

AN-FORMAT ASN.1 Definition

```
}

-- 4G Event Definitions
AN-BEARER-ACTIVATION ::= SEQUENCE
{
    partyInformation          AnPartyInformation,
        liid                  AnLawfulInterceptionIdentifier,
        iceType                ICE-type,
    ratType                    AnRATType OPTIONAL,
    pdnAddressAllocation      AnPDNAddressAllocation OPTIONAL,
    timestamp                  TimeStamp,
    correlationNumber          AnCorrelationNumber OPTIONAL,
    apn                        AnAPN,
    bearerActivationType       AnBearerActivationType,
    networkElementIdentifier   AnNetworkElementIdentifier,
    locationOfTheTarget        AnLocationOfTheTarget OPTIONAL,
    failedBearerActivationReason AnFailedBearerActivationReason OPTIONAL,
    epsBearerQOS               AnEPSBearerQos OPTIONAL,
    apnAMBR                    AnAPN-AMBR OPTIONAL,
    epsBearerIdentity          AnEPSBearerIdentity,
    protConfigOptions          AnProtConfigOptions OPTIONAL,
    initiator                  AnInitiator,
    procedureTransactionID     AnProcedureTransactionID OPTIONAL,
    linkedEPSBearerID          AnLinkedEPSBearerID OPTIONAL,
    tft                        AnTFT OPTIONAL,
    handoverIndication         AnHandoverIndication OPTIONAL,
    ...
}

AN-START-OF-INTERCEPTION-WITH-ACTIVE-BEARER ::= SEQUENCE
{
    partyInformation          AnPartyInformation,
        liid                  AnLawfulInterceptionIdentifier,
        iceType                ICE-type,
    ratType                    AnRATType OPTIONAL,
    pdnAddressAllocation      AnPDNAddressAllocation OPTIONAL,
    timestamp                  TimeStamp,
    correlationNumber          AnCorrelationNumber,
    apn                        AnAPN,
    bearerActivationType       AnBearerActivationType,
    networkElementIdentifier   AnNetworkElementIdentifier,
    locationOfTheTarget        AnLocationOfTheTarget OPTIONAL,
    epsBearerQOS               AnEPSBearerQos OPTIONAL,
```

apnAMBR	AnAPN-AMBR OPTIONAL,
epsBearerIdentity	AnEPSBearerIdentity,
protConfigOptions	AnProtConfigOptions OPTIONAL,
initiator	AnInitiator,
procedureTransactionID	AnProcedureTransactionID OPTIONAL,
linkedEPSBearerId	AnLinkedEPSBearerID OPTIONAL,
tft	AnTFT OPTIONAL,
handoverIndication	AnHandoverIndication OPTIONAL,
...	

}

AN-BEARER-MODIFICATION ::= SEQUENCE

{

partyInformation	AnPartyInformation, --targetIdentity
liid	AnLawfulInterceptionIdentifier,
iceType	ICE-type,
timestamp	TimeStamp,
correlationNumber	AnCorrelationNumber,
networkElementIdentifier	AnNetworkElementIdentifier,
locationOfTheTarget	AnLocationOfTheTarget OPTIONAL,
initiator	AnInitiator,
epsBearerQoS	AnEPSBearerQos OPTIONAL,
epsBearerIdentity	AnEPSBearerIdentity,
procedureTransactionID	AnProcedureTransactionID OPTIONAL,
ratType	AnRATType OPTIONAL,
apnAMBR	AnAPN-AMBR OPTIONAL,
tft	AnTFT OPTIONAL,
handoverIndication	AnHandoverIndication OPTIONAL,
failedBearerModificationReason	AnFailedBearerModificationReason OPTIONAL,
protConfigOptions	AnProtConfigOptions OPTIONAL,
...	

}

AN-BEARER-DEACTIVATION ::= SEQUENCE

{

partyInformation	AnPartyInformation, --targetIdentity
liid	AnLawfulInterceptionIdentifier,
iceType	ICE-type,
timestamp	TimeStamp,
correlationNumber	AnCorrelationNumber,
bearerDeactivationType	AnBearerDeactivationType,
networkElementIdentifier	AnNetworkElementIdentifier,
locationOfTheTarget	AnLocationOfTheTarget OPTIONAL,
epsBearerIdentity	AnEPSBearerIdentity,
initiator	AnInitiator,

AN-FORMAT ASN.1 Definition

```

    procedureTransactionID          AnProcedureTransactionID OPTIONAL,
    bearerDeactivationCause         AnBearerDeactivationCause,
    ...
}

AN-UE-REQUESTED-BEARER-RESOURCE-MODIFICATION ::= SEQUENCE
{
    partyInformation                AnPartyInformation, --targetIdentity,
    liid                           AnLawfulInterceptionIdentifier,
    iceType                        ICE-type,
    timestamp                      TimeStamp,
    correlationNumber               AnCorrelationNumber,
    networkElementIdentifier        AnNetworkElementIdentifier,
    locationOfTheTarget             AnLocationOfTheTarget OPTIONAL,
    procedureTransactionID          AnProcedureTransactionID OPTIONAL,
    epsBearerIdentity              AnEPSBearerIdentity,
    linkedEPSBearerId              AnLinkedEPSBearerID OPTIONAL,
    epsBearerQOS                   AnEPSBearerQos OPTIONAL,
    trafficAggregateDescription     AnTrafficAggregateDescription OPTIONAL,
    failedUERequestedBearerResModReason
OPTIONAL,
    protConfigOptions              AnProtConfigOptions OPTIONAL,
    ...
}

-- 3G Event Definitions
AN-PDP-CONTEXT-ACTIVATION ::= SEQUENCE
{
    partyInformation                AnPartyInformation,
    liid                           AnLawfulInterceptionIdentifier,
    iceType                        ICE-type,
    pdpAddress                     AnEndUserAddress OPTIONAL,
    timestamp                      TimeStamp,
    correlationNumber               AnCorrelationNumber,
    apn                            AnAPN,
    networkElementIdentifier        AnNetworkElementIdentifier,
    locationOfTheTarget             AnLocationOfTheTarget OPTIONAL,
    failedCtxtActivationReason      AnFailedContextActivationReason OPTIONAL,
    pdpCtxtQOS                     AnQos OPTIONAL,
    nsapi                          AnNsapi,
    initiator                      AnInitiator,
    pdpType                        AnTypeOfPdpContext,
    ...
}
```

}

AN-START-OF-INTERCEPTION-WITH-ACTIVE-PDP-CONTEXT ::= SEQUENCE

```
{
    partyInformation          AnPartyInformation,
        liid                  AnLawfulInterceptionIdentifier,
        iceType                ICE-type,
    pdpAddress                AnEndUserAddress OPTIONAL,
    timestamp                 TimeStamp,
    correlationNumber          AnCorrelationNumber,
    apn                       AnAPN,
    networkElementIdentifier   AnNetworkElementIdentifier,
    locationOfTheTarget        AnLocationOfTheTarget OPTIONAL,
    pdpCtxtQOS                AnQos OPTIONAL,
    nsapi                     AnNsapi,
    initiator                  AnInitiator,
    pdpType                    AnTypeOfPdpContext,
    ...
}
```

AN-PDP-CONTEXT-MODIFICATION ::= SEQUENCE

```
{
    partyInformation          AnPartyInformation,
        liid                  AnLawfulInterceptionIdentifier,
        iceType                ICE-type,
    pdpAddress                AnEndUserAddress OPTIONAL,
    timestamp                 TimeStamp,
    correlationNumber          AnCorrelationNumber,
    apn                       AnAPN,
    networkElementIdentifier   AnNetworkElementIdentifier,
    locationOfTheTarget        AnLocationOfTheTarget OPTIONAL,
    pdpCtxtQOS                AnQos OPTIONAL,
    initiator                  AnInitiator,
    pdpType                    AnTypeOfPdpContext,
    ...
}
```

AN-PDP-CONTEXT-DEACTIVATION ::= SEQUENCE

```
{
    partyInformation          AnPartyInformation,
        liid                  AnLawfulInterceptionIdentifier,
        iceType                ICE-type,
    pdpAddress                AnEndUserAddress OPTIONAL,
    timestamp                 TimeStamp,
    correlationNumber          AnCorrelationNumber,
```

AN-FORMAT ASN.1 Definition

```

    apn                      AnAPN,
    networkElementIdentifier AnNetworkElementIdentifier,
    locationOfTheTarget      AnLocationOfTheTarget OPTIONAL,
    deactivationCause        AnContextDeactivationCause,
    nsapi                   AnNsapi,
    initiator                AnInitiator,
    ...
}

AN-CC-PDU ::= SEQUENCE
{
    uLIC-header[1] AN-ULIC-HEADER,
    payload      [2] OCTET STRING
}

AN-ULIC-HEADER ::= SEQUENCE
{
    hi3DomainId      [0] OBJECT IDENTIFIER,
    liid             [2] AnLawfulInterceptionIdentifier,
    correlationNumber [3] AnCorrelationNumber,
    timeStamp        [4] TimeStamp OPTIONAL,
    sequence-number  [5] INTEGER (0..65535),
    t-PDU-direction [6] TPDU-direction,
    ...,
    national-HI3-ASN1parameters[7] National-HI3-ASN1parameters OPTIONAL,
    -- encoded per national requirements
    ice-type          [8] ICE-type
    -- The ICE-type indicates the applicable Intercepting Control Element
}

END -- AffirmedCaleaModule
```


EPS HI Operations

```
EpsHI2Operations {itu-t(0) identified-organization(4) etsi(0) securityDomain(2)
lawfulIntercept(2) threeGPP(4) hi2eps(8) r12(12) version-57 (57)}
```

```
DEFINITIONS IMPLICIT TAGS ::=
```

```
BEGIN
```

```
IMPORTS
```

```
LawfulInterceptionIdentifier
```

```
    FROM AffirmedCaleaHIModule
```

```
TimeStamp
```

```
    FROM AffirmedCaleaHIModule
```

```
National-Parameters
```

```
    FROM AffirmedCaleaHIModule
```

```
Network-Identifier
```

```
    FROM AffirmedCaleaHIModule
```

```
DataNodeAddress
```

```
    FROM AffirmedCaleaHIModule
```

```
IPAddress
```

```
    FROM AffirmedCaleaHIModule
```

```
National-HI2-ASN1parameters
```

```
    FROM AffirmedCaleaHIModule
```

```
TPDU-direction
```

```
    FROM AffirmedCaleaHIModule
```

```
National-HI3-ASN1parameters
```

```
    FROM AffirmedCaleaHIModule
```

```
ICE-type
```

```
    FROM AffirmedCaleaHIModule
```

```
GeographicalCoordinates
```

```
    FROM AffirmedCaleaHIModule
```

```
PacketDataHeaderInformation
```

```
    FROM AffirmedCaleaHIModule;
```

```
-----
--IMPORTS
```

```
--OPERATION,
```

```
--    ERROR
```

```
--    FROM Remote-Operations-Information-Objects
```

```
--{joint-iso-itu-t(2) remote-operations(4) informationObjects(5) version1(0)}
```

```
--LawfulInterceptionIdentifier,
```

AN-FORMAT ASN.1 Definition

```

--      TimeStamp,
--      Network-Identifier,
--      National-Parameters,
--National-HI2-ASN1parameters,
--DataNodeAddress,
--IPAddress,
--IP-value,
--X25Address

--FROM HI2Operations
--{itu-t(0) identified-organization(4) etsi(0) securityDomain(2)
--      lawfulIntercept(2) hi2(1) version10(10)};
-- Imported from TS 101 671v3.3.1
-----

-- Object Identifier Definitions

-- Security DomainId

-- Security Subdomains
-- hi2epsDomainId value took from 33.108-12.7 spec, if spec changes update version
-- info and in the calea application.
hi2epsDomainId OBJECT IDENTIFIER ::= {threeGPPSUBDomainId hi2eps(8) r12(12)
version-57 (57)}
hi3DomainId OBJECT IDENTIFIER ::= {threeGPPSUBDomainId hi3eps(9) r12(12)
version-0(0)}

-----

-- User defined definitions.

-----

-- From ETSI spec- ETSI TS 101 671 V3.5.1 (2009-11) HI2Operations.
-----
-----
-----

-- 3gpp 33.108 rel 12.7 G.2.1.3
-- EpsIRIContent includes events that correspond to EPS and UMTS/GPRS.
LawfulIntercept ::= CHOICE

```

```

{
    keep-Alive          [0] NULL,
    envelopedIRIContent [1] EnvelopedIRIContent,
    ...
}
EnvelopedIRIContent ::= SEQUENCE OF EpsIRIContent

EpsIRIContent ::= CHOICE
{
    iRI-Begin-record      [1] IRI-Parameters, -- include at least one optional
    parameter
    iRI-End-record        [2] IRI-Parameters,
    iRI-Continue-record   [3] IRI-Parameters, -- include at least one optional
    parameter
    iRI-Report-record     [4] IRI-Parameters -- include at least one optional
    parameter
}
-- the EpsIRIContent may provide events that correspond to UMTS/GPRS as well.

-- Parameters having the same tag numbers must be identical in Rel-5 and onwards
-- modules.
IRI-Parameters ::= SEQUENCE
{
    hi2epsDomainId        [0] OBJECT IDENTIFIER, -- 3GPP HI2 EPS domain
    lawfulInterceptionIdentifier [1] LawfulInterceptionIdentifier,
    -- This identifier is associated to the target.
    timeStamp             [3] TimeStamp,
    -- date and time of the event triggering the report.)
    initiator              [4] ENUMERATED
    {
        not-Available      (0),
        originating-Target (1),
        -- in case of GPRS, this indicates that the PDP context activation,
        -- modification or deactivation is MS requested
        -- in case of EPS, this indicated that the EPS detach, bearer activation,
        -- modification or deactivation is UE requested
        terminating-Target (2),
        -- in case of GPRS, this indicates that the PDP context activation,
        -- modification or deactivation is network initiated
        -- in case of EPS, this indicated that the EPS detach, bearer activation,
        -- modification or deactivation is network initiated
        ...
    }
}

```

AN-FORMAT ASN.1 Definition

```
    } OPTIONAL,

locationOfTheTarget      [8] Location OPTIONAL,
    -- location of the target
partyInformation         [9] SET SIZE (1..10) OF PartyInformation OPTIONAL,
-- This parameter provides the concerned party, the identiy(ies) of the party
-- )and all the information provided by the party.

    ePSCorrelationNumber [18] EPSCorrelationNumber OPTIONAL,
-- this parameter provides GPRS Correlation number when the event corresponds to
-- UMTS/GPRS.
    ePSevent              [20] EPSevent OPTIONAL,
-- This information is used to provide particular action of the target
-- such as attach/detach
    gPRSOperationErrorCode [22] GPRSOperationErrorCode OPTIONAL,
qoS                      [25] UmtsQos OPTIONAL,
networkIdentifier        [26] Network-Identifier OPTIONAL,
-- Octets are coded according to 3GPP TS 23.003 [25]
    ...,
-- Tag [33] was taken into use by ETSI module in TS 101 671v2.13.1
    correlation           [35] CorrelationValues OPTIONAL,
ePS-GTPV2-specificParameters [36] EPS-GTPV2-SpecificParameters OPTIONAL,
packetDataHeaderInformation [49] PacketDataHeaderInformation OPTIONAL
}
-- Parameters having the same tag numbers must be identical in Rel-5 and onwards modules

-- PARAMETERS FORMATS

PartyInformation ::= SEQUENCE
{
    party-Qualifier [0] ENUMERATED
    {
        gPRSorEPS-Target(3),
        ...
    },
    partyIdentity [1] SEQUENCE
    {
        imei [1] OCTET STRING (SIZE (8)) OPTIONAL,
        -- See MAP format [4]

        imsi [3] OCTET STRING (SIZE (3..8)) OPTIONAL,
        -- See MAP format [4] International Mobile
        -- Station Identity E.212 number beginning with Mobile Country Code
    }
}
```

```

        msISDN                [6] OCTET STRING (SIZE (1..9)) OPTIONAL
        -- MSISDN of the target, encoded in the same format as the AddressString
        -- parameters defined in MAP format document TS 29.002 [4]

    },

    services-Data-Information  [4] Services-Data-Information OPTIONAL,
    -- This parameter is used to transmit all the information concerning the
    -- complementary information associated to the basic data call

    ...

}

Location ::= SEQUENCE
{
    globalCellID               [2] GlobalCellID    OPTIONAL,
    --see MAP format (see [4])
    rAI                        [4] Rai             OPTIONAL,
    -- the Routeing Area Identifier in the current SGSN is coded in
    -- accordance with the 10.5.5.15 of document [9] without the
    -- Routing Area Identification IEI
    -- (only the last 6 octets are used)
    gsmLocation                [5] GSMLocation OPTIONAL,
    umtsLocation                [6] UMTSLocation OPTIONAL,
    sai                         [7] Sai            OPTIONAL,
    -- format: PLMN-ID 3 octets (no. 1 3)
    --          LAC      2 octets (no. 4 5)
    --          SAC      2 octets (no. 6 7)
    --          (according to 3GPP TS 25.413 [62])
    ...,
    oldRAI                     [8] Rai            OPTIONAL
    -- the Routeing Area Identifier in the old SGSN is coded in
    -- accordance with the
    -- 10.5.5.15 of document [9] without the Routing Area
    -- Identification IEI
    -- (only the last 6 octets are used).
}

GlobalCellID ::= OCTET STRING (SIZE (5..7))
Rai          ::= OCTET STRING (SIZE (6))
Sai          ::= OCTET STRING (SIZE (7))

```

AN-FORMAT ASN.1 Definition

```
GSMLocation ::= CHOICE
{
    geoCoordinates [1] SEQUENCE
    {
        latitude      [1] PrintableString (SIZE(7..10)),
        -- format :      XDDMMSS.SS
        longitude      [2] PrintableString (SIZE(8..11)),
        -- format :      XDDMMSS.SS
        mapDatum       [3] MapDatum DEFAULT wGS84,
        ...,
        azimuth        [4] INTEGER (0..359) OPTIONAL
        -- The azimuth is the bearing, relative to true north.
    },
    -- format :      XDDMMSS.SS
    --              X          : N(orth), S(outh), E(ast), W(est)
    --              DD or DDD   : degrees (numeric characters)
    --              MM         : minutes (numeric characters)
    --              SS.SS      : seconds, the second part (.SS) is optionnal
    -- Example :
    --          latitude short form      N502312
    --          longitude long form      E1122312.18

    utmCoordinates [2] SEQUENCE
    {
        utm-East      [1] PrintableString (SIZE(10)),
        utm-North      [2] PrintableString (SIZE(7)),
        -- example utm-East      32U0439955
        --          utm-North      5540736
        mapDatum       [3] MapDatum DEFAULT wGS84,
        ...,
        azimuth        [4] INTEGER (0..359) OPTIONAL
        -- The azimuth is the bearing, relative to true north.
    },

    utmRefCoordinates [3] SEQUENCE
    {
        utmref-string  PrintableString (SIZE(13)),
        mapDatum       MapDatum DEFAULT wGS84,
        ...
    },
    -- example 32UPU91294045
```

```

        wGS84Coordinates    [4] OCTET STRING
        -- format is as defined in [37].
    }

MapDatum ::= ENUMERATED
{
    wGS84,
    wGS72,
    eD50,    -- European Datum 50
    ...
}

UMTSLocation ::= CHOICE {
    point                [1] GA-Point,
    pointWithUncertainty [2] GA-PointWithUncertainty
    -- polygon           [3] GA-Polygon
}

GA-Point ::= SEQUENCE {
    geographicalCoordinates    GeographicalCoordinates,
    ...
}

GA-PointWithUncertainty ::=SEQUENCE {
    geographicalCoordinates    GeographicalCoordinates,
    uncertaintyCode            INTEGER (0..127)
}

maxNrOfPoints            INTEGER ::= 15

GA-Polygon ::= SEQUENCE (SIZE (1..maxNrOfPoints)) OF
SEQUENCE {
    geographicalCoordinates    GeographicalCoordinates,
    ...
}

EPSCorrelationNumber ::= OCTET STRING
-- In case of PS interception, the size will be in the range (8..20)
CorrelationValues ::= CHOICE {

    iri-to-CC    [0]    IRI-to-CC-Correlation, -- correlates IRI to Content(s)
    iri-to-iri   [1]    IRI-to-IRI-Correlation, -- correlates IRI to IRI

```

AN-FORMAT ASN.1 Definition

```
both-IRI-CC [2]    SEQUENCE { -- correlates IRI to IRI and IRI to Content(s)
    iri-CC  [0] IRI-to-CC-Correlation,
    iri-IRI [1] IRI-to-IRI-Correlation}
}

IRI-to-CC-Correlation ::= SEQUENCE { -- correlates IRI to Content
    cc      [0] SET OF OCTET STRING, -- correlates IRI to multiple CCs
    iri     [1] OCTET STRING OPTIONAL
    -- correlates IRI to CC with signaling
}

IRI-to-IRI-Correlation ::= OCTET STRING -- correlates IRI to IRI

EPSEvent ::= ENUMERATED
{
    pDPContextActivation          (1),
    startOfInterceptionWithPDPCContextActive (2),
    pDPContextDeactivation        (4),
    pDPContextModification        (13),
    ... ,
    bearerActivation              (18),
    startOfInterceptionWithActiveBearer (19),
    bearerModification            (20),
    bearerDeactivation            (21),
    uERequestedBearerResourceModification (22),
    packetDataHeaderInformation    (43)
}

-- see [19]

Services-Data-Information ::= SEQUENCE
{
    gPRS-parameters [1] GPRS-parameters OPTIONAL,
    ...
}

GPRS-parameters ::= SEQUENCE
{
    pDP-address-allocated-to-the-target [1] DataNodeAddress OPTIONAL,
    aPN [2] OCTET STRING (SIZE(1..100)) OPTIONAL,
    pDP-type [3] OCTET STRING (SIZE(2)) OPTIONAL,
    -- Include either Octets 3 and 4 of the Packet Data
```



```

        -- Protocol Address information element
        -- of 3GPP TS 24.008 [9] or Octets 4 and 5 of the End
        -- User Address IE of 3GPP TS 29.060 [17].
        -- when PDP-type is IPv4 or IPv6, the IP address is carried
        -- by parameter pDP-address-allocated-to-the-target
        -- when PDP-type is IPv4v6, the additional IP address is
        -- carried by parameter additionalIPAddress

    ...,
    nSAPI [4] OCTET STRING (SIZE (1)) OPTIONAL,
    -- Include either Octet 2 of the NSAPI IE of 3GPP TS 24.008 [9]
    -- or Octet 2 of the NSAPI IE of 3GPP TS 29.060 [17].
    additionalIPAddress [5] DataNodeAddress OPTIONAL
}

```

GPRSOperationErrorCode ::= OCTET STRING

-- The parameter shall carry the GMM cause value or the SM cause value, as defined in
 -- the standard [9], without the IEI.

UmtsQos ::= CHOICE

```

{
    qosMobileRadio [1] OCTET STRING,
        -- The qosMobileRadio parameter shall be coded in accordance with the
        -- 10.5.6.5 of document [9] without the Quality of service IEI and
        -- Length of quality of service IE (. That is, first
        -- two octets carrying 'Quality of service IEI' and 'Length of
        -- quality of service IE' shall be excluded).
        qosGn [2] OCTET STRING
        -- qosGn parameter shall be coded in accordance with 7.7.34 of
document [17]
}

```

EPS-GTPV2-SpecificParameters ::= SEQUENCE

```

{
    pDNAddressAllocation [1] OCTET STRING OPTIONAL,
    aPN [2] OCTET
STRING (SIZE (1..100)) OPTIONAL,
    protConfigOptions [3]
ProtConfigOptions OPTIONAL,
    ePSEBearerIdentity [5] OCTET
STRING OPTIONAL,

```

AN-FORMAT ASN.1 Definition

```

                                rATType                                [7]   OCTET
STRING (SIZE (1))            OPTIONAL,
                                failedBearerActivationReason          [8]   OCTET
STRING (SIZE (1))            OPTIONAL,
                                ePSBearerQoS                            [9]   OCTET
STRING                        OPTIONAL,
                                bearerActivationType                    [10]
TypeOfBearer                  OPTIONAL,
                                aPN-AMBR                                [11]  OCTET
STRING                        OPTIONAL,
                                -- Only octets 5 onwards of AMBR IE from 3GPP
                                -- TS 29.274 [46] shall be included.
                                procedureTransactionId                  [12]
OCTET STRING                  OPTIONAL,
                                linkedEPSBearerId                      [13]  OCTET
STRING                        OPTIONAL,
                                --The Linked EPS Bearer Identity shall be included
                                -- and coded according to 3GPP TS 29.274 [46].
                                tFT                                      [14]
OCTET STRING                  OPTIONAL,
                                -- Only octets 3 onwards of TFT IE from 3GPP TS
                                -- 24.008 [9] shall be included.
                                handoverIndication                      [15]
NULL                          OPTIONAL,
                                failedBearerModReason                  [16]  OCTET
STRING (SIZE (1))            OPTIONAL,
                                trafficAggregateDescription              [17]  OCTET
STRING                        OPTIONAL,
                                bearerDeactivationType                  [21]
TypeOfBearer                  OPTIONAL,
                                bearerDeactivationCause                  [22]  OCTET
STRING (SIZE (1))            OPTIONAL,
                                ePSlocationOfTheTarget                  [23]
EPSLocation                   OPTIONAL,
                                -- the use of ePSLocationOfTheTarget is mutually
                                -- exclusive with the use of locationOfTheTarget
                                -- ePSlocationOfTheTarget allows using the
                                -- coding of the paramater according to
                                -- SAE stage 3.
                                ...,
                                pDNType                                [24]   OCTET
STRING (SIZE (1))            OPTIONAL,
                                -- coded according to TS 24.301 [47]
                                extendedHandoverIndication              [27]
OCTET STRING (SIZE (1))      OPTIONAL,
                                -- This parameter with value 1 indicates handover

```

```

-- based on the flags in the TS 29.274 [46].
-- Otherwise set to the value 0.
-- The use of extendedHandoverIndication
-- and handoverIndication parameters is
-- mutually exclusive and depends on the
-- actual ASN.1 encoding method.

                                ulITimestamp                                [28]
OCTET STRING (SIZE (8))        OPTIONAL

}

-- All the parameters within EPS-GTPV2-SpecificParameters are coded as the
-- corresponding IEs without the octets containing type and length. Unless differently
-- stated, they are coded according to 3GPP TS 29.274 [46]; in this case the octet
-- containing the instance shall also be not included.

TypeOfBearer ::= ENUMERATED
{
    defaultBearer                (1),
                                dedicatedBearer        (2),
                                ...
}

EPSLocation ::= SEQUENCE
{
    userLocationInfo             [1] OCTET STRING (SIZE (1..39)) OPTIONAL,
                                -- coded according to 3GPP TS 29.274 [46]; the type IE is not
                                -- included
                                gsmLocation              [2] GSMLocation OPTIONAL,
                                umtsLocation             [3] UMTSLocation OPTIONAL,
                                olduserLocationInfo      [4] OCTET STRING (SIZE (1..39)) OPTIONAL,
                                -- coded in the same way as userLocationInfo
                                lastVisitedTAI           [5] OCTET STRING (SIZE (1..5)) OPTIONAL,
                                -- the Tracking Area Identity is coded in accordance with the
                                -- TAI field in 3GPP TS 29.274 [46].
}

```

AN-FORMAT ASN.1 Definition

```
        tAIlList          [6] OCTET STRING (SIZE (7..97)) OPTIONAL,
-- the TAI List is coded according to 3GPP TS 24.301 [47], without
-- the TAI list IEI
        ...,
        threeGPP2Bsid     [7] OCTET STRING (SIZE (1..12)) OPTIONAL
-- contains only the payload from the 3GPP2-BSID AVP described
-- in the 3GPP TS 29.212 [56].

}

ProtConfigOptions ::= SEQUENCE

{
    ueToNetwork           [1] OCTET STRING (SIZE(1..251))          OPTIONAL,
-- This shall be coded with octet 3 onwards of
-- the Protocol Configuration Options IE in
-- accordance with 3GPP TS 24.008 [9].
    networkToUe           [2] OCTET STRING (SIZE(1..251))          OPTIONAL,
-- This shall be coded with octet 3 onwards of
-- the Protocol Configuration Options IE in
-- accordance with 3GPP TS 24.008 [9].
    ...
}

-----
-----
--   HI3 Format ASN
-----
-----

-- Object Identifier Definitions
ConnectionStatus ::= CHOICE
{
    keep-Alive            [0] NULL,
    lawfulInterceptionIdentifier [1] LawfulInterceptionIdentifier,
    ...
}

CC-PDU ::= SEQUENCE
{
    uLIC-header           [1] ULIC-header,
```

```

        payload          [2] OCTET STRING
    }

    ULIC-header ::= SEQUENCE
    {
        hi3DomainId        [0] OBJECT IDENTIFIER, -- 3GPP HI3 Domain
        lIID                [2] LawfulInterceptionIdentifier OPTIONAL,
        correlation-Number  [3] EPSCorrelationNumber,
        timeStamp           [4] TimeStamp OPTIONAL,
        sequence-number     [5] INTEGER (0..65535),
        t-PDU-direction    [6] TPDU-direction,
        ...,
        national-HI3-ASN1parameters [7]
    National-HI3-ASN1parameters OPTIONAL,
        -- encoded per national requirements
        ice-type            [8] ICE-type OPTIONAL
        -- The ICE-type indicates the applicable Intercepting
        -- Control Element(see ref [19]) in which the T-PDU
        -- is intercepted.
    }

    -----
    -----

    END -- OF EpsHI20operations

```

HI Common

```
-- Affirmed Networks CALEA HI Module Definition
AffirmedCaleaHIModule
DEFINITIONS IMPLICIT TAGS ::=
BEGIN

-----

-- User defined definitions.

-----

-----

-- From ETSI spec- ETSI TS 101 671 V3.5.1 (2009-11) HI20operations.

-----

LawfulInterceptionIdentifier ::= OCTET STRING (SIZE (1..25))
-- It is recommended to use ASCII characters in "a"<85>"z", "A"<85>"Z", "-", "_",
-- ".", and "0"<85>"9".
-- For subaddress option only "0"..."9" shall be used.

-- Security DomainId
lawfulInterceptDomainId OBJECT IDENTIFIER ::= {itu-t(0) identified-organization(4)
etsi(0) securityDomain(2) lawfulIntercept(2)}

-- Security Subdomains
threeGPPSUBDomainId OBJECT IDENTIFIER ::= {lawfulInterceptDomainId threeGPP(4)}

-----

-- From ETSI spec- ETSI TS 101 671 V3.5.1 (2009-11) HI20operations.

-----

ICE-type ::= ENUMERATED
{
    ggsn                (2),
    s-GW                (3),
    pDN-GW              (4),
    colocated-SAE-GWs   (5),
    ePDG                (6),
    iWLAN                (7)  -- not yet specified - setting to 7 for now
}
```

National-HI3-ASN1parameters ::= SEQUENCE

```
{
    countryCode[1] PrintableString (SIZE (2)),
    -- Country Code according to ISO 3166-1 [39],
    -- the country to which the parameters inserted after the extension marker
    -- apply
    ...
    -- In case a given country wants to use additional national parameters
    -- according to its law, these national parameters should be defined using
    -- the ASN.1 syntax and added after the extension marker (...).
    -- It is recommended that "version parameter" and "vendor identification
    -- parameter" are
    -- included in the national parameters definition. Vendor identifications can
    -- be
    -- retrieved from IANA web site. It is recommended to avoid
    -- using tags from 240 to 255 in a formal type definition.
}
```

TimeStamp ::= CHOICE

```
{
    -- The minimum resolution required is one second.
    -- "Resolution" is the smallest incremental change that can be measured for time and
    -- is expressed with a definite number of decimal digits or bits.
    localTime [0] LocalTimeStamp,
    utcTime [1] UTCTime
}
```

LocalTimeStamp ::= SEQUENCE

```
{
    generalizedTime [0] GeneralizedTime,
    -- The minimum resolution required is one second.
    -- "Resolution" is the smallest incremental change that can be measured for time and
    -- is expressed with a definite number of decimal digits or bits.
    winterSummerIndication [1] ENUMERATED
    {
        notProvided(0),
        winterTime(1),
        summerTime(2),
        ...
    }
}
```

AN-FORMAT ASN.1 Definition

```
National-Parameters ::= SET SIZE (1..40) OF OCTET STRING (SIZE (1..256))
    -- Content defined by national law.
```

```
Network-Identifier ::= SEQUENCE
{
    operator-Identifier [0] OCTET STRING (SIZE (1..5)),
    -- It is a notification of the NW0/AP/SvP in ASCII- characters.
    -- For subaddress option only "0"..."9" shall be used.
    -- The parameter is mandatory.
    network-Element-Identifier [1] Network-Element-Identifier OPTIONAL,
    ...
}
```

```
Network-Element-Identifier ::= CHOICE
{
    e164-Format [1] OCTET STRING (SIZE (1..25)),
    -- E164 address of the node in international format. Coded in the same format as the
    -- calling party number parameter of the ISUP (parameter part: EN 300 356 [5]).
    x25-Format [2] OCTET STRING (SIZE (1..25)),
    -- X25 address
    iP-Format [3] OCTET STRING (SIZE (1..25)),
    -- IP address
    dNS-Format [4] OCTET STRING (SIZE (1..25)),
    -- DNS address
    ...,
    iP-Address [5] IPAddress,
    -- IP address
    generic-String [6] OCTET STRING (SIZE (1..256))
    -- Arbitrary unique string that identifies the network element.
}
```

```
DataNodeAddress ::= CHOICE
{
    ipAddress [1] IPAddress,
    x25Address [2] X25Address,
    ...
}
```

```
IPAddress ::= SEQUENCE
{
    iP-type [1] ENUMERATED
    {
```



```

        iPV4(0),
        iPV6(1),
        ...
    },
    iP-value [2] IP-value,
    iP-assignment [3] ENUMERATED
    {
        static(1),
        -- The static coding shall be used to report a static address.
        dynamic(2),
        -- The dynamic coding shall be used to report a dynamically allocated
        -- address.
        notKnown(3),
        -- The notKnown coding shall be used to report other than static or
        -- dynamically allocated IP addresses.
        ...
    } OPTIONAL,
    ...
}

IP-value ::= CHOICE
{
    iPBinaryAddress [1] OCTET STRING (SIZE(4..16)),
    iPTextAddress [2] IA5String (SIZE(7..45)),
    ...
}

X25Address ::= OCTET STRING (SIZE(1..25))

National-HI2-ASN1parameters ::= SEQUENCE
{
    countryCode [1] PrintableString (SIZE (2)),
        -- Country Code according to ISO 3166-1 [67],
        -- the country to which the parameters inserted after the extension
        -- marker apply.
        ...
        -- In case a given country wants to use additional national
        -- parameters according to its law, these national parameters
        -- should be defined using the ASN.1 syntax and added after the
        -- extension marker (...).
        -- It is recommended that "version parameter" and "vendor
        -- identification parameter" are included in the national
        -- parameters definition. Vendor identifications can be

```

AN-FORMAT ASN.1 Definition

```
-- retrieved from the IANA web site (see annex J). Besides, it is
-- recommended to avoid using tags from 240 to 255 in a formal type
-- definition.
}

Version ::= ENUMERATED
{
    version1(1),
    ...,
    version3(3) ,
    -- versions 4-7 were omitted to align with UmtsHI2Operations.
    lastVersion(8)
    -- Mandatory parameter "version" (tag 1) was always redundant in 33.108,
    -- because the object identifier "hi3DomainId" was introduced into
    -- "ULIC-headerV in the initial version of 33.108v5.0.0 In order to keep
    -- backward compatibility, even when the version of the "hi3DomainId"
    -- parameter will be incremented it is recommended to always send to LEMF the
    -- same: enumeration value "lastVersion(8)".
}

TPDU-direction ::= ENUMERATED
{
    from-target (1),
    to-target (2),
    unknown (3)
}

GPSCorrelationNumber ::= OCTET STRING (SIZE(8..20))
CorrelationNumber ::= OCTET STRING (SIZE(8..20))

GeographicalCoordinates ::= SEQUENCE {
    latitudeSign      ENUMERATED { north, south },
    latitude          INTEGER (0..8388607),
    longitude         INTEGER (-8388608..8388607),
    ...
}

PacketDataHeaderInformation ::= CHOICE
{
    packetDataHeader      [1] PacketDataHeader,
    packetDataHeaderSummary [2] PacketDataHeaderSummary,
    ...
}
```

}

PacketDataHeader ::= CHOICE

{

packetDataHeaderMapped [1] PacketDataHeaderMapped,

packetDataHeaderCopy [2] PacketDataHeaderCopy,

...

}

PacketDataHeaderMapped ::= SEQUENCE

{

sourceIPAddress [1] IPAddress OPTIONAL,

sourcePortNumber [2] INTEGER (0..65535) OPTIONAL,

destinationIPAddress [3] IPAddress OPTIONAL,

destinationPortNumber [4] INTEGER (0..65535) OPTIONAL,

transportProtocol [5] INTEGER OPTIONAL,

-- For IPv4, report the ?Protocol? field and for IPv6 report ?Next Header? field.

-- Assigned Internet Protocol Numbers can be found at

-- <http://www.iana.org/assignments/protocol-numbers/protocol-numbers.xml>

packetSize [6] INTEGER OPTIONAL,

flowLabel [7] INTEGER OPTIONAL,

packetCount [8] INTEGER OPTIONAL,

direction [9] TPDU-direction,

...

}

PacketDataHeaderCopy ::= SEQUENCE

{

direction [1] TPDU-direction,

headerCopy [2] OCTET STRING, -- includes a copy of the packet header
at the IP

-- network layer and above including extension headers, but excluding

-- contents.

...

}

PacketDataHeaderSummary ::= SEQUENCE OF PacketFlowSummary

PacketFlowSummary ::= SEQUENCE

{

AN-FORMAT ASN.1 Definition

```
sourceIPAddress      [1] IPAddress,
sourcePortNumber     [2] INTEGER (0..65535) OPTIONAL,
destinationIPAddress [3] IPAddress,
destinationPortNumber [4] INTEGER (0..65535) OPTIONAL,
transportProtocol    [5] INTEGER,
-- For IPv4, report the ?Protocol? field and for IPv6 report ?Next Header? field.
-- Assigned Internet Protocol Numbers can be found at
-- http://www.iana.org/assignments/protocol-numbers/protocol-numbers.xml
flowLabel            [6] INTEGER OPTIONAL,
summaryPeriod        [7] ReportInterval,
packetCount          [8] INTEGER,
sumOfPacketSizes     [9] INTEGER,
packetDataSummaryReason [10] ReportReason,
...
}

ReportReason ::= ENUMERATED
{
    timerExpired          (0),
    countThresholdHit     (1),
    pDPContextDeactivated (2),
    pDPContextModification (3),
    otherOrUnknown        (4),
    ...
}

ReportInterval ::= SEQUENCE
{
    firstPacketTimeStamp [0] TimeStamp,
    lastPacketTimeStamp  [1] TimeStamp,
    ...
}

END
```

I-WLAN UMTS HI2 Operations

Note: In release 9.0 and later, Lawful Intercept for ePDG and TWAG is no longer supported.

```
IWLANUmtsHI2Operations {itu-t(0) identified-organization(4) etsi(0) securityDomain(2)
lawfulintercept(2) threeGPP(4) hi2wlan(6) r12(12) version-3(3)}
```

```
DEFINITIONS IMPLICIT TAGS ::=
```

```
BEGIN
```

```
IMPORTS
```

```
LawfulInterceptionIdentifier,
threeGPPSUBDomainId,
TimeStamp,
National-Parameters,
Network-Identifier,
DataNodeAddress,
IPAddress,
National-HI2-ASN1parameters,
CorrelationNumber,
GeographicalCoordinates,
PacketDataHeaderInformation
    FROM AffirmedCaleaHIModule;
```

```
-- Object Identifier Definitions
```

```
-- Security Subdomains
```

```
hi2wlanDomainId OBJECT IDENTIFIER ::= {threeGPPSUBDomainId hi2wlan(6) r12 (12)
version-3(3)}
```

```
-- Note: The prefix AN-WLAN has been added to some constructs below to prevent overlap
-- with defined OIDs in the context of EPS in calea_Eps_HI_Operations.asn
```

```
-- Note: As per TS 33.107, the following OIDs are used for both TWAG and ePDG.
```

```
-- 3gpp 33.108 rel 11.1 G.2.1.3
```

```
AN-WLAN-LawfulIntercept ::= CHOICE
```

```
{
    keep-Alive [0] NULL,
    envelopedIRIContent[1] AN-WLAN-EnvelopedIRIContent,
```

AN-FORMAT ASN.1 Definition

```
...
}

AN-WLAN-EnvelopedIRIContent ::= SEQUENCE OF IWLANUmtsIRIContent

IWLANUmtsIRIContent ::= CHOICE
{
    iRI-Begin-record[1] AN-WLAN-IRI-Parameters,
    iRI-End-record[2] AN-WLAN-IRI-Parameters,
    iRI-Report-record[3] AN-WLAN-IRI-Parameters,
    ...
}

AN-WLAN-IRI-Parameters ::= SEQUENCE
{
    hi2iwlanDomainId[0] OBJECT IDENTIFIER, -- 3GPP HI2 WLAN domain
    lawfulInterceptionIdentifier[2] LawfulInterceptionIdentifier,
    -- This identifier is associated to the target.
    timeStamp [3] TimeStamp,
    -- date and time of the event triggering the report.
    initiator [4] ENUMERATED
    {
        not-Available(0),
        originating-Target(1),
        -- in case of I-WLAN, this indicates that the I-WLAN tunnel disconnect is
        -- WLAN UE requested.
        terminating-Target(2),
        -- in case of I-WLAN, this indicates that the I-WLAN tunnel disconnect is
        -- network initiated.
        ...
    } OPTIONAL,

    partyInformation [5] SET SIZE (1..10) OF AN-WLAN-PartyInformation OPTIONAL,
    -- This parameter provides the concerned party, the identiy(ies) of the party
    -- and all the information provided by the party.

    national-Parameters[6] National-Parameters OPTIONAL,
    networkIdentifier[7] Network-Identifier OPTIONAL,
    i-WLANevent[8] I-WLANevent OPTIONAL,
    correlationNumber[9] CorrelationNumber OPTIONAL,
    i-WLANOperationErrorCode[10] I-WLANOperationErrorCode OPTIONAL,

    i-WLANinformation[11] I-WLANinformation OPTIONAL,
```

```

visitedPLMNID[12] VisitedPLMNID OPTIONAL,
national-HI2-ASN1parameters[255]National-HI2-ASN1parametersOPTIONAL,
...,
nSAPI      [13] OCTET STRING (SIZE (1)) OPTIONAL,
packetDataHeaderInformation[14] PacketDataHeaderInformationOPTIONAL
}

-- PARAMETERS FORMATS

AN-WLAN-PartyInformation ::= SEQUENCE
{
    party-Qualifier [0]  ENUMERATED
    {
        iWLAN-Target(1),
        ...
    },
    partyIdentity [1] SEQUENCE
    {
        imsi [2] OCTET STRING (SIZE (3..8)) OPTIONAL,
            -- See MAP format [4] International Mobile
            -- Station Identity E.212 number beginning with Mobile Country Code

        msISDN [3] OCTET STRING (SIZE (1..9)) OPTIONAL,
            -- MSISDN of the target, encoded in the same format as the AddressString
            -- parameters defined in MAP format document TS 29.002 [4]

        nai [7] OCTET STRING OPTIONAL,
            -- NAI of the target, encoded in the same format as
            -- defined in 3GPP TS 29.234 [41].
        ...
    },

    services-Data-Information [2] AN-WLAN-Services-Data-Information OPTIONAL,
        -- This parameter is used to transmit all the information concerning the
        -- complementary information associated to the basic data call
    ...
}

I-WLANEvent ::= ENUMERATED
{
    --Events (1) and (2) might not be applicable for ePDG/WAG

```

AN-FORMAT ASN.1 Definition

```
i-WLANAccessInitiation (1),
i-WLANAccessTermination(2),
i-WLANTunnelEstablishment(3),
i-WLANTunnelDisconnect(4),
startOfInterceptionCommunicationActive(5),
...,
packetDataHeaderInformation(6)
}
```

```
AN-WLAN-Services-Data-Information ::= SEQUENCE
{
    i-WLAN-parameters [1] I-WLAN-parameters OPTIONAL,
    ...
}
```

```
I-WLAN-parameters ::= SEQUENCE
{
    wlan-local-IP-address-of-the-target [1] DataNodeAddress OPTIONAL,
    -- This represents the endpoint of the remote side of the tunnel
    -- used by the UE. For WAG, this is the far end of the GRE tunnel
    w-APN [2] OCTET STRING (SIZE(1..100))OPTIONAL,
    -- The Access Point Name (APN) is coded in accordance with
    -- 3GPP TS 24.008 [9] without the APN IEI (only the last 100 octets are used).
    -- Octets are coded according to 3GPP TS 23.003 [25].
    wlan-remote-IP-address-of-the-target [3] DataNodeAddress OPTIONAL,
    -- This represents the IP address used by the UE inside the tunnel

    wlan-pdg-tunnel-endpoint-address [10] DataNodeAddress OPTIONAL,
    -- Affirmed addition. This is the IP address of the PGW when S2a/S2b is used.
    ...
}
```

```
I-WLANOperationErrorCode ::= OCTET STRING
-- The parameter shall carry the I-WLAN failed tunnel establishment reason,
-- the I-WLAN Failed Access Initiation reason or the I-WLAN session termination reason.
```

```
I-WLANinformation ::= SEQUENCE
{
    wlanOperatorName [1] OCTET STRING OPTIONAL,
```



```
wLANLocationData[2] OCTET STRINGOPTIONAL,  
wLANLocationInformation [3] OCTET STRINGOPTIONAL,  
nASIPv6Address[4]IPAddressOPTIONAL,  
wLANMACAddress[5]OCTET STRINGOPTIONAL,  
sessionAliveTimer[6]SessionAliveTimeOPTIONAL,  
...,  
--These parameters are defined in 3GPP TS 29.234.  
    geographicalCoordinates[7] GeographicalCoordinates OPTIONAL  
    --civicAddress[8] CivicAddress OPTIONAL  
}  
  
VisitedPLMNID ::= OCTET STRING  
-- The parameter shall carry the VisitedPLMNID as defined in 3GPP TS 29.234.  
  
SessionAliveTime ::= OCTET STRING  
--The parameter shall carry the SessionAliveTime as defined in 3GPP TS 29.234.  
  
END -- OF IWLANUmtsHI2Operations
```

UMTS HI3 PS

```
Umts-HI3-PS {itu-t(0) identified-organization(4) etsi(0) securityDomain(2)
lawfulintercept(2) threeGPP(4) hi3(2) r7(7) version-0(0)}
```

```
DEFINITIONS IMPLICIT TAGS ::=
```

```
BEGIN
```

```
IMPORTS
```

```
GPRSCorrelationNumber
```

```
    FROM AffirmedCaleaHIModule
```

```
LawfulInterceptionIdentifier
```

```
    FROM AffirmedCaleaHIModule
```

```
threeGPPSUBDomainId
```

```
    FROM AffirmedCaleaHIModule
```

```
TimeStamp
```

```
    FROM AffirmedCaleaHIModule
```

```
Version
```

```
    FROM AffirmedCaleaHIModule
```

```
National-HI3-ASN1parameters
```

```
    FROM AffirmedCaleaHIModule
```

```
TPDU-direction
```

```
    FROM AffirmedCaleaHIModule
```

```
ICE-type
```

```
    FROM AffirmedCaleaHIModule
```

```
CC-PDU
```

```
    FROM AffirmedCaleaHIModule;
```

```
-- Object Identifier Definitions
```

```
-- Security Subdomains
```

```
--hi3DomainId OBJECT IDENTIFIER ::= {threeGPPSUBDomainId hi3(2) r7(7) version-0(0)}
```

```
AN-WLAN-CC-PDU ::= SEQUENCE
```

```
{
```

```
    uLIC-header[1] AN-WLAN-ULIC-header,
```

```
    payload    [2] OCTET STRING
```

```
}
```

```
AN-WLAN-ULIC-header ::= SEQUENCE
```

```
{
```

```
    hi3DomainId  [0]OBJECT IDENTIFIER, -- 3GPP HI3 Domain
```

```
    version      [1]Version,
```

```

    lIID          [2] LawfulInterceptionIdentifier OPTIONAL,
    correlation-Number[3]GPRSCorrelationNumber,
    timeStamp     [4]TimeStamp OPTIONAL,
    sequence-number[5]INTEGER (0..65535),
    t-PDU-direction[6] TPDU-direction,
    ...,
    national-HI3-ASN1parameters[7] National-HI3-ASN1parameters OPTIONAL,
    -- encoded per national requirements
    ice-type      [8] ICE-type OPTIONAL
    -- The ICE-type indicates the applicable Intercepting Control Element(see ref [19])
    -- in which the T-PDU is intercepted.
}

END-- OF Umts-HI3-PS

```


AN-FORMAT IRI Event Details

This chapter lists the AN-FORMAT Intercept Related Information (IRI) Events.

Table 4-1 AN-BEARER-ACTIVATION

Parameter	Availability
partyInformation	Always
IMSI	Always
IMEI	If Available
MSISDN	If Available
liid	Always
iceType	Always
ratType	If Available
pdnAddressAllocation	Available if Activation Successful
timestamp	Always
correlationNumber	Available if Activation Successful
apn	Always
bearerActivationType	Always
networkElementIdentifier	Always
locationOfTheTarget	If Available
failedBearerActivationReason	Available if activation failed
epsBearerQOS	Available for GBR bearer
apnAMBR	Available for non-GBR bearers
epsBearerIdentity	Always

Table 4-1 AN-BEARER-ACTIVATION (Continued)

Parameter	Availability
protConfigOptions	If Available
initiator	Always
procedureTransactionID	If Available
linkedEPSBearerId	Available if Dedicated Bearer
tft	Available if Dedicated Bearer
handoverIndication	Present if activation due to Handover

Table 4-2 AN-START-OF-INTERCEPTION-WITH-ACTIVE-BEARER

Parameter	Availability
partyInformation	Always
IMSI	Always
IMEI	If Available
MSISDN	If Available
liid	Always
iceType	Always
ratType	If Available
pdnAddressAllocation	Available if Activation Successful
timestamp	Always
correlationNumber	Always
apn	Always
bearerActivationType	Always
networkElementIdentifier	Always
locationOfTheTarget	If Available
epsBearerQOS	Available for GBR bearer
apnAMBR	Available for non-GBR bearers
epsBearerIdentity	Always
protConfigOptions	If Available
initiator	Always
procedureTransactionID	If Available
linkedEPSBearerId	Available if Dedicated Bearer
tft	Available if Dedicated Bearer
handoverIndication	Present if activation due to Handover

Table 4-3 AN-BEARER-MODIFICATION

Parameter	Availability
partyInformation	Always
IMSI	Always
IMEI	If Available
MSISDN	If Available
liid	Always
iceType	Always
timestamp	Always
correlationNumber	Always
networkElementIdentifier	Always
locationOfTheTarget	If Available
initiator	Always
epsBearerQOS	If available for GBR bearer
epsBearerIdentity	Always
procedureTransactionID	If Available
ratType	If Available
apnAMBR	If available for non-GBR bearers
tft	If available if Dedicated Bearer
handoverIndication	Present if activation due to Handover
failedBearerModificationReason	Available if modification failed
protConfigOptions	If Available

Table 4-4 AN-BEARER-DEACTIVATION

Parameter	Availability
partyInformation	Always
IMSI	Always
IMEI	If Available
MSISDN	If Available
liid	Always
iceType	Always
timestamp	Always
correlationNumber	Always
bearerDeactivationType	Always
networkElementIdentifier	Always
locationOfTheTarget	If Available
epsBearerIdentity	Always
initiator	Always
procedureTransactionID	If Available
bearerDeactivationCause	Always

Table 4-5 AN-UE-REQUESTED-BEARER-RESOURCE-MODIFICATION

Parameter	Availability
partyInformation	Always
IMSI	Always
IMEI	If Available
MSISDN	If Available
liid	Always
iceType	Always
timestamp	Always
correlationNumber	Always
networkElementIdentifier	Always
locationOfTheTarget	If Available
procedureTransactionID	If Available
epsBearerIdentity	Always
linkedEPSBearerId	Available if Dedicated Bearer
epsBearerQOS	If available for GBR bearer
trafficAggregateDescription	If Available
failedUERequestedBearerResModReason	Available if modification failed
protConfigOptions	If Available

Table 4-6 AN-PDP-CONTEXT-ACTIVATION

Parameter	Availability
partyInformation	Always
IMSI	Always
IMEI	If Available
MSISDN	If Available
liid	Always
iceType	Always
pdpAddress	Available if Activation Successful
timestamp	Always
correlationNumber	Always
apn	Always
networkElementIdentifier	Always
locationOfTheTarget	If Available
failedCtxtActivationReason	Available if activation failed
pdpCtxtQOS	If Available
nsapi	Always
initiator	Always
pdpType	Always

Table 4-7 AN-START-OF-INTERCEPTION-WITH-ACTIVE-PDP-CONTEXT

Parameter	Availability
partyInformation	Always
IMSI	Always
IMEI	If Available
MSISDN	If Available
liid	Always
iceType	Always
pdpAddress	If Available
timestamp	Always
correlationNumber	Always
apn	Always
networkElementIdentifier	Always
locationOfTheTarget	If Available
pdpCtxtQOS	If Available
nsapi	Always
initiator	Always
pdpType	Always

Table 4-8 AN-PDP-CONTEXT-MODIFICATION

Parameter	Availability
partyInformation	Always
IMSI	Always
IMEI	If Available
MSISDN	If Available
liid	Always
iceType	Always
pdpAddress	If Available
timestamp	Always
correlationNumber	Always
apn	Always
networkElementIdentifier	Always
locationOfTheTarget	If Available
pdpCtxtQOS	If Available
initiator	Always
pdpType	Always

Table 4-9 AN-PDP-CONTEXT-DEACTIVATION

Parameter	Availability
partyInformation	Always
IMSI	Always
IMEI	If Available
MSISDN	If Available
liid	Always
iceType	Always
pdpAddress	If Available
timestamp	Always
correlationNumber	Always
apn	Always
networkElementIdentifier	Always
locationOfTheTarget	If Available
deactivationCause	Always
nsapi	Always
initiator	Always

HI-FORMAT ASN.1 Definition

ASN.1 for HI2 Mediation Function Messages

```
DEFINITIONS IMPLICIT TAGS ::=
```

```
LawfulIntercept ::= CHOICE
```

```
{
    keep-Alive [0] NULL,
    envelopedIRIContent[1] EnvelopedIRIContent,
    ...
}
```

```
EnvelopedIRIContent ::= SEQUENCE OF UmtsIRIContent //only for UMTS case.
```

```
EnvelopedIRIContent ::= SEQUENCE OF EpsIRIContent // for both EPC and UMTS. //AN3k use this.
```

Intercept Related Information (HI2 SAE/EPS)

This section contains excerpts from the 3GPP 33.108 rel 12.7 specification.

ASN1 description of IRI (HI2 interface)

```
EpsHI2Operations {itu-t(0) identified-organization(4) etsi(0) securityDomain(2)
lawfulIntercept(2) threeGPP(4) hi2eps(8) r12(12) version-57(57)}
```

```
DEFINITIONS IMPLICIT TAGS ::=
```

```
BEGIN
```

```
IMPORTS
```

```
    OPERATION,
    ERROR
```

HI-FORMAT ASN.1 Definition

```
FROM Remote-Operations-Information-Objects
{joint-iso-itu-t(2) remote-operations(4) informationObjects(5)
version1(0)}

    LawfulInterceptionIdentifier,
    TimeStamp,
    Network-Identifier,
    National-Parameters,
    National-HI2-ASN1parameters,
    DataNodeAddress,
    IPAddress,
    IP-value,
    X25Address

FROM HI2Operations
{itu-t(0) identified-organization(4) etsi(0) securityDomain(2)
    lawfulIntercept(2) hi2(1) version10(10)}; -- Imported from TS 101
671v3.3.1

-- Object Identifier Definitions

-- Security DomainId
lawfulInterceptDomainId OBJECT IDENTIFIER ::= {itu-t(0) identified-organization(4)
etsi(0)
securityDomain(2) lawfulIntercept(2)}

-- Security Subdomains
threeGPPSUBDomainId OBJECT IDENTIFIER ::= {lawfulInterceptDomainId threeGPP(4)}
hi2epsDomainId OBJECT IDENTIFIER ::= {threeGPPSUBDomainId hi2eps(8) r12(12)
version-57(57)}

EpsIRIContent ::= CHOICE
{
    iRI-Begin-record[1] IRI-Parameters, -- include at least one optional parameter
    iRI-End-record[2] IRI-Parameters,
    iRI-Continue-record[3] IRI-Parameters, -- include at least one optional parameter
    iRI-Report-record[4] IRI-Parameters -- include at least one optional parameter
}

-- Parameters having the same tag numbers must be identical in Rel-5 and onwards
modules.
IRI-Parameters ::= SEQUENCE
{
```



```

hi2epsDomainId[0] OBJECT IDENTIFIER, -- 3GPP HI2 EPS domain
lawfulInterceptionIdentifier[1] LawfulInterceptionIdentifier,
    -- This identifier is associated to the target.
timeStamp [3] TimeStamp,
    -- date and time of the event triggering the report.)
initiator [4] ENUMERATED
{
    not-Available(0),
    originating-Target(1),
        -- in case of GPRS, this indicates that the PDP context activation,
modification
        -- or deactivation is MS requested
        -- in case of EPS, this indicated that the EPS detach, bearer activation,
modification
        -- or deactivation is UE requested
    terminating-Target(2),
        -- in case of GPRS, this indicates that the PDP context activation,
modification or
        -- deactivation is network initiated
        -- in case of EPS, this indicated that the EPS detach, bearer activation,
modification
        -- or deactivation is network initiated
    ...
} OPTIONAL,

locationOfTheTarget[8] Location OPTIONAL,
    -- location of the target subscriber
partyInformation [9] SET SIZE (1..10) OF PartyInformation OPTIONAL,
    -- This parameter provides the concerned party, the identiy(ies) of the party
    --)and all the information provided by the party.

national-Parameters[16] National-Parameters OPTIONAL,
ePSCorrelationNumber[18] EPSCorrelationNumber OPTIONAL,
ePSevent [20] EPSEvent OPTIONAL,
    -- This information is used to provide particular action of the target
    -- such as attach/detach
gPRSOperationErrorCode [22] GPRSOperationErrorCode OPTIONAL,
ggsnAddress [24] DataNodeAddress OPTIONAL,
qoS [25] UmtsQos OPTIONAL,
networkIdentifier[26] Network-Identifier OPTIONAL,
...,
    -- Tag[33] was taken into use by ETSI module in TS 101 671v2.13.1

correlation [35] CorrelationValues OPTIONAL,

```

HI-FORMAT ASN.1 Definition

```
ePS-GTPV2-specificParameters [36] EPS-GTPV2-SpecificParameters OPTIONAL,
    -- contains parameters to be used in case of GTPV2 based intercepted messages
    packetDataHeaderInformation [49] PacketDataHeaderInformation OPTIONAL
}
-- Parameters having the same tag numbers must be identical in Rel-5 and onwards modules

-- PARAMETERS FORMATS

PartyInformation ::= SEQUENCE
{
    party-Qualifier [0] ENUMERATED
    {
        gPRSorEPS-Target(3),
        ...
    },
    partyIdentity [1] SEQUENCE
    {
        imei [1] OCTET STRING (SIZE (8)) OPTIONAL,
            -- See MAP format [4]

        imsi [3] OCTET STRING (SIZE (3..8)) OPTIONAL,
            -- See MAP format [4] International Mobile
            -- Station Identity E.212 number beginning with Mobile Country Code

        msISDN [6] OCTET STRING (SIZE (1..9)) OPTIONAL,
            -- MSISDN of the target, encoded in the same format as the AddressString
            -- parameters defined in MAP format document TS 29.002 [4]

    },

    services-Data-Information [4] Services-Data-Information OPTIONAL,
        -- This parameter is used to transmit all the information concerning the
        -- complementary information associated to the basic data call
    ...
}

Location ::= SEQUENCE
{
    e164-Number [1] OCTET STRING (SIZE (1..25)) OPTIONAL,
        -- Coded in the same format as the ISUP location number (parameter
        -- field) of the ISUP (see EN 300 356 [30]).
    globalCellID[2] GlobalCellIDOPTIONAL,
```

```

--see MAP format (see [4])
rAI      [4] RaiOPTIONAL,
-- the Routeing Area Identifier in the current SGSN is coded in accordance
with the
-- § 10.5.5.15 of document [9] without the Routing Area Identification IEI
-- (only the last 6 octets are used)
gsmLocation[5] GSMLocation OPTIONAL,
umtsLocation[6] UMTSLocation OPTIONAL,
sAI      [7] SaiOPTIONAL,
-- format:PLMN-ID3 octets (no. 1 - 3)
-- LAC2 octets (no. 4 - 5)
-- SAC2 octets (no. 6 - 7)
-- (according to 3GPP TS 25.413)
...,
oldRAI   [8] RaiOPTIONAL
-- the Routeing Area Identifier in the old SGSN is coded in accordance with
the
-- § 10.5.5.15 of document [9] without the Routing Area Identification IEI
-- (only the last 6 octets are used).
}
GlobalCellID ::= OCTET STRING (SIZE (5..7))
Rai          ::= OCTET STRING (SIZE (6))
Sai          ::=OCTET STRING (SIZE (7))

```

```

GSMLocation ::= CHOICE
{
  geoCoordinates [1] SEQUENCE
  {
    latitude[1]PrintableString (SIZE(7..10)),
      -- format : XDDMMSS.SS
    longitude[2]PrintableString (SIZE(8..11)),
      -- format : XDDMMSS.SS
    mapDatum[3]MapDatum DEFAULT wGS84,
    ...,
    azimuth[4] INTEGER (0..359) OPTIONAL
      -- The azimuth is the bearing, relative to true north.
  },
  -- format : XDDMMSS.SS
  -- X : N(orth), S(outh), E(ast), W(est)
  -- DD or DDD : degrees (numeric characters)
  -- MM: minutes (numeric characters)
  -- SS.SS : seconds, the second part (.SS) is optionnal

```

HI-FORMAT ASN.1 Definition

```
-- Example :
--   latitude short formN502312
--   longitude long formE1122312.18

utmCoordinates[2] SEQUENCE
{
    utm-East [1] PrintableString (SIZE(10)),
    utm-North [2] PrintableString (SIZE(7)),
    -- exampleutm-East32U0439955
    -- utm-North5540736
    mapDatum[3]MapDatum DEFAULT wGS84,
    ...,
    azimuth[4] INTEGER (0..359) OPTIONAL
    -- The azimuth is the bearing, relative to true north.
},

utmRefCoordinates [3] SEQUENCE
{
    utmref-stringPrintableString (SIZE(13)),
    mapDatumMapDatum DEFAULT wGS84,
    ...
},
-- example32UPU91294045

wGS84Coordinates [4] OCTET STRING
-- format is as defined in [37].
}

MapDatum ::= ENUMERATED
{
    wGS84,
    wGS72,
    eD50, -- European Datum 50
    ...
}

UMTSLocation ::= CHOICE {
    point [1]GA-Point,
    pointWithUncertainty[2]GA-PointWithUncertainty,
    polygon [3]GA-Polygon
}

GeographicalCoordinates ::= SEQUENCE {
```

```

    latitudeSignENUMERATED { north, south },
    latitude  INTEGER (0..8388607),
    longitude INTEGER (-8388608..8388607),
    ...
}

GA-Point ::= SEQUENCE {
    geographicalCoordinatesGeographicalCoordinates,
    ...
}

GA-PointWithUncertainty ::=SEQUENCE {
    geographicalCoordinatesGeographicalCoordinates,
    l    uncertaintyCodeINTEGER (0..127)
}

maxNrOfPoints    INTEGER ::= 15

GA-Polygon ::= SEQUENCE (SIZE (1..maxNrOfPoints)) OF
    SEQUENCE {
        geographicalCoordinatesGeographicalCoordinates,
        ...
    }

EPSCorrelationNumber ::= OCTET STRING
    -- In case of PS interception, the size will be in the range (8..20)
CorrelationValues ::= CHOICE {

    iri-to-CC [0]IRI-to-CC-Correlation, -- correlates IRI to Content(s)
    iri-to-iri[1]IRI-to-IRI-Correlation, -- correlates IRI to IRI
    both-IRI-CC[2]SEQUENCE { -- correlates IRI to IRI and IRI to Content(s)
        iri-CC[0]IRI-to-CC-Correlation,
        iri-IRI[1]IRI-to-IRI-Correlation}
}

IRI-to-CC-Correlation ::= SEQUENCE { -- correlates IRI to Content
    cc [0] SET OF OCTET STRING,-- correlates IRI to multiple CCs
    iri [1] OCTET STRING OPTIONAL
    -- correlates IRI to CC with signaling
}

IRI-to-IRI-Correlation ::= OCTET STRING -- correlates IRI to IRI

```

EPSEvent ::= ENUMERATED

```
{
    pDPContextActivation (1),
    startOfInterceptionWithPDPCContextActive(2),
    pDPContextDeactivation(4),

    pDPContextModification(13),

    ... ,

    bearerActivation (18),
    startOfInterceptionWithActiveBearer (19),
    bearerModification (20),
    bearerDeactivation (21),
    uERequestedBearerResourceModification (22)
    packetDataHeaderInformation (43)
}
-- see [19]
```

Services-Data-Information ::= SEQUENCE

```
{
    gPRS-parameters [1] GPRS-parameters OPTIONAL,
    ...
}
```

GPRS-parameters ::= SEQUENCE

```
{
    pDP-address-allocated-to-the-target [1] DataNodeAddress OPTIONAL,
    aPN [2] OCTET STRING (SIZE(1..100)) OPTIONAL,
    pDP-type [3] OCTET STRING (SIZE(2)) OPTIONAL,
    -- when PDP-type is IPv4 or IPv6, the IP address is carried by parameter
    -- pDP-address-allocated-to-the-target
    -- when PDP-type is IPv4v6, the additional IP address is carried by parameter
    -- additionalIPAddress
    ...,
    nSAPI [4] OCTET STRING (SIZE (1)) OPTIONAL,
    additionalIPAddress[5] DataNodeAddress OPTIONAL
}
```

GPRSOperationErrorCode ::= OCTET STRING

-- The parameter shall carry the GMM cause value or the SM cause value, as defined in the

-- standard [9], without the IEI.

LDIevent ::= ENUMERATED

```
{
    targetEntersIA (1),
    targetLeavesIA(2),
    ...
}
```

UmtsQos ::= CHOICE

```
{
    qosMobileRadio [1] OCTET STRING,
    -- The qosMobileRadio parameter shall be coded in accordance with the §
10.5.6.5 of
    -- document [9] without the Quality of service IEI and Length of
    -- quality of service IE (. That is, first
    -- two octets carrying 'Quality of service IEI' and 'Length of quality of
service
    -- IE' shall be excluded).
    qosGn [2] OCTET STRING
    -- qosGn parameter shall be coded in accordance with § 7.7.34 of document [17]
}
```

EPS-GTPV2-SpecificParameters ::= SEQUENCE

```
{
    pDNAddressAllocation          [1]  OCTET STRING          OPTIONAL,
    aPN                          [2]  OCTET STRING (SIZE (1..100))  OPTIONAL,
    protConfigOptions            [3]  ProtConfigOptions      OPTIONAL,
    ePSBearerIdentity            [5]  OCTET STRING          OPTIONAL,
    rATType                      [7]  OCTET STRING (SIZE (1))    OPTIONAL,
    failedBearerActivationReason [8]  OCTET STRING (SIZE (1))    OPTIONAL,
    ePSBearerQoS                 [9]  OCTET STRING          OPTIONAL,
    bearerActivationType         [10] TypeOfBearer          OPTIONAL,
    aPN-AMBR                     [11] OCTET STRING          OPTIONAL,
    procedureTransactionId       [12] OCTET STRING          OPTIONAL,
    linkedEPSBearerId            [13] OCTET STRING          OPTIONAL,
    tFT                          [14] OCTET STRING          OPTIONAL,
```

HI-FORMAT ASN.1 Definition

```
    handoverIndication          [15]  NULL                                OPTIONAL,
    failedBearerModReason       [16]  OCTET STRING (SIZE (1))
OPTIONAL,
    trafficAggregateDescription [17]  OCTET STRING
OPTIONAL,
    bearerDeactivationType      [21]  TypeOfBearer                      OPTIONAL,
    bearerDeactivationCause     [22]  OCTET STRING (SIZE (1))
OPTIONAL,
    ePSlocationOfTheTarget      [23]  EPSLocation                      OPTIONAL,
    -- the use of ePSlocationOfTheTarget is mutually exclusive with the use of
locationOfTheTarget
    -- ePSlocationOfTheTarget allows using the coding of the paramater according to
SAE stage 3.
    ...,
    pDNType                    [24]  OCTET STRING (SIZE (1))
OPTIONAL,
    -- coded according to TS 24.301 [47]

    requestType                [25]  OCTET STRING (SIZE (1))            OPTIONAL,
    -- coded according to TS 24.301 [47]
    uEReqPDNConnFailReason     [26]  OCTET STRING (SIZE (1))            OPTIONAL
    -- coded according to TS 24.301 [47]

    -- All the parameters are coded as the corresponding IEs without the octets
containing type and
    -- length. Unless differently stated, they are coded according to 3GPP TS 29.274
[46]; in this
    -- case the octet containing the instance shall also be not included.
}
```

```
TypeOfBearer ::= ENUMERATED
{
    defaultBearer      (1),
    dedicatedBearer    (2),
    ...
}
```

```
EPSLocation ::= SEQUENCE
{
```



```

userLocationInfo    [1] OCTET STRING (SIZE (1..34)) OPTIONAL,
  -- coded according to 3GPP TS 29.274 [46]; the type IE is not included
umtsLocation[3] UMTSLocation OPTIONAL,
olduserLocationInfo[4] OCTET STRING (SIZE (1..34))OPTIONAL,
  -- coded in the same way as userLocationInfo
lastVisitedTAI      [5] OCTET STRING (SIZE (1..5))OPTIONAL,
  -- the Tracking Area Identity is coded in accordance with the TAI field in 3GPP
TS 29.274
  -- [46].
tailist             [6] OCTET STRING (SIZE (7..97))OPTIONAL,
  -- the TAI List is coded according to 3GPP TS 24.301 [47], without the TAI list IEI
...,
}

```

ProtConfigOptions ::= SEQUENCE

```

{
  ueToNetwork          [1] OCTET STRING (SIZE(1..251))      OPTIONAL,
  networkToUe          [2] OCTET STRING (SIZE(1..251))      OPTIONAL,
  ...
}

```

PacketDataHeaderInformation ::= CHOICE

```

{
  packetDataHeader      [1] PacketDataHeader,
  ...
}

```

PacketDataHeader ::= CHOICE

```

{
  packetDataHeaderMapped [1] PacketDataHeaderMapped,
  packetDataHeaderCopy   [2] PacketDataHeaderCopy,
  ...
}

```

PacketDataHeaderMapped ::= SEQUENCE

```

{
  sourceIPAddress      [1] IPAddress  OPTIONAL,
  sourcePortNumber     [2] INTEGER (0..65535) OPTIONAL,
  destinationIPAddress [3] IPAddress  OPTIONAL,

```

HI-FORMAT ASN.1 Definition

```
destinationPortNumber  [4] INTEGER (0..65535) OPTIONAL,
transportProtocol      [5] INTEGER OPTIONAL,
-- For IPv4, report the ?Protocol? field and for IPv6 report ?Next Header? field.
-- Assigned Internet Protocol Numbers can be found at
-- http://www.iana.org/assignments/protocol-numbers/protocol-numbers.xml
packetSize            [6] INTEGER OPTIONAL,
flowLabel             [7] INTEGER OPTIONAL,
direction             [9] TPDU-direction,
...
}

PacketDataHeaderCopy ::= SEQUENCE
{
    direction          [1] TPDU-direction,
    headerCopy         [2] OCTET STRING, -- includes a copy of the packet header
at the IP              -- network layer and above including extension headers, but excluding
contents.
    ...
}

END -- OF EpsHI2Operations
```

Content of Communication (HI3 EPS)

ASN.1 for HI3 Mediation Function Messages

```
DEFINITIONS IMPLICIT TAGS ::=
```

```
ConnectionStatus ::= CHOICE
```

```
{
    keep-Alive [0] Null,
    lawfulInterceptionIdentifier [1] LawfulInterceptionIdentifier,
    ...
}
```

```
Eps-HI3-PS {itu-t(0) identified-organization(4) etsi(0) securityDomain(2)
lawfulIntercept(2) threeGPP(4) hi3eps(9) r12(12) version-0(0)}
```

```
DEFINITIONS IMPLICIT TAGS ::=
```

```
BEGIN
```

```
IMPORTS
```

```
EPSCorrelationNumber
```

```
    FROM EpsHI2Operations
```

```
    {itu-t(0) identified-organization(4) etsi(0) securityDomain(2)
lawfulIntercept(2) threeGPP(4) hi2eps(8) r8(8) version-0(0)}    -- Imported from TS
33.108 v.8.6.0
```

```
LawfulInterceptionIdentifier,
```

```
TimeStamp
```

```
    FROM HI2Operations
```

```
    {itu-t(0) identified-organization(4) etsi(0) securityDomain(2)
lawfulIntercept(2) hi2(1) version10(10)}; -- from ETSI HI2Operations TS 101 671 v3.3.1
```

```
-- Object Identifier Definitions
```

```
-- Security DomainId
```

```
lawfulInterceptDomainId OBJECT IDENTIFIER ::= {itu-t(0) identified-organization(4)
etsi(0)
```

```
securityDomain(2) lawfulIntercept(2)}
```

```
-- Security Subdomains
```

HI-FORMAT ASN.1 Definition

```
threeGPPSUBDomainId OBJECT IDENTIFIER ::= {lawfulInterceptDomainId threeGPP(4)}
hi3DomainId OBJECT IDENTIFIER ::= {threeGPPSUBDomainId hi3eps(9) r8(8) r12(12)
version-0(0)}

CC-PDU ::= SEQUENCE
{
    uLIC-header[1] ULIC-header,
    payload [2] OCTET STRING
}

ULIC-header ::= SEQUENCE
{
    hi3DomainId[0]OBJECT IDENTIFIER, -- 3GPP HI3 Domain
    lIID [2] LawfulInterceptionIdentifier OPTIONAL,
    correlation-Number[3]EPSCorrelationNumber,
    timeStamp [4]TimeStamp OPTIONAL,
    sequence-number[5]INTEGER (0..65535),
    t-PDU-direction[6] TPDU-direction,
    ...,
    national-HI3-ASN1parameters[7] National-HI3-ASN1parameters OPTIONAL,
    -- encoded per national requirements
    ice-type [8] ICE-type OPTIONAL
    -- The ICE-type indicates the applicable Intercepting Control Element(see ref
[19]) in which
    -- the T-PDU is intercepted.
}

TPDU-direction ::= ENUMERATED
{
    from-target (1),
    to-target (2),
    unknown (3)
}

National-HI3-ASN1parameters ::= SEQUENCE
{
    countryCode[1] PrintableString (SIZE (2)),
    -- Country Code according to ISO 3166-1 [39],
    -- the country to which the parameters inserted after the extension marker
apply
    ...
    -- In case a given country wants to use additional national parameters
according to its law,
```

```
-- these national parameters should be defined using the ASN.1 syntax and
added after the
-- extension marker (...).
-- It is recommended that "version parameter" and "vendor identification
parameter" are
-- included in the national parameters definition. Vendor identifications can
be
-- retrieved from IANA web site. It is recommended to avoid
-- using tags from 240 to 255 in a formal type definition.
}
```

```
ICE-type ::= ENUMERATED
{
    sgsn          (1),
    ggsn          (2),
    ...,
    S-GW          (3),
    PDN-GW        (4),
    colocated-SAE-GWs (5)
}
```

```
END-- OF Eps-HI3-PS
```


HI-FORMAT IRI Event Details

This chapter lists the HI-FORMAT Intercept Related Information (IRI) Events.

Evolved Packet System

This section specifies requirements for the handover interfaces (HI2 and HI3) in the Evolved Packet System.

When a PDN-GW provides a Gn/Gp interface for interworking with a SGSN, from the LI perspective, the PDN-GW acts as a GGSN towards the involved SGSN. In this case, in addition to the requirements specified in this chapter, all the requirements specified in this document for PS interception that apply to the GGSN also apply to the PDN-GW. PDP contexts/EPS bearer modification signaling detected by the PDN-GW during a handover between different accesses involving a Gn/Gp interface (that is, from E-UTRAN to 2G/3G and vice versa) is reported inside the IRI BEGIN-END transaction. The same correlation number shall be used before and after the handover during the same IRI transaction. After the handover, the events sent by the PDN-GW shall be mapped into IRIs according to the requirements for the new access.

Identifiers

Specific identifiers are necessary to identify a target for interception uniquely and to correlate between the data, which is conveyed over the different handover interfaces (HI2 and HI3). The identifiers are defined in the following sections.

For the delivery of CC and IRI, the S-GW or PDN-GW provide correlation numbers and target identities to the HI2 and HI3. The correlation number is unique per EPS bearer/tunnel and is used to correlate CC with IRI and the different IRI's of one EPS bearer/tunnel.

Note: When different protocols (that is, GTP and PMIP) are used in the networks, different values for the correlation number can be generated by different nodes for the same communication.

Lawful Interception Identifier

For each target identity related to an interception measure, the authorized Operator (NO/AN/SP) shall assign a special Lawful Interception Identifier (LIID), which has been agreed between the LEA and the Operator (NO/AN/SP).

Using an indirect identification pointing to a target identity makes it easier to keep the knowledge about a specific interception target limited within the authorized Operator (NO/AN/SP) and the handling agents at the LEA.

The LIID is a component of the CC delivery procedure and of the IRI records. It shall be used within any information exchanged at the handover interfaces HI2 and HI3 for identification and correlation purposes.

The LIID format shall consist of alphanumeric characters. It might, for example, among other information, contain a lawful authorization reference number and the date when the lawful authorization was issued.

The authorized Operator (NO/AN/SP) shall either enter a unique LIID for each target identity of the interception subject or a single LIID for multiple target identities all pertaining to the same interception subject.

If more than one LEA intercepts the same target identity, there shall be unique LIIDs assigned relating to each LEA.

Network Identifier

The network identifier (NID) is a mandatory parameter; it should be internationally unique. It consists of the following two identifiers.

- Operator- (NO/AN/SP) identifier (mandatory):
Unique identification of network Operator, access network provider or service provider.
- Network element identifier NEID (optional):
The purpose of the network element identifier is to uniquely identify the relevant network element carrying out the LI operations, such as LI activation, IRI record sending, and so on.

A network element identifier may be an IP address or other identifier. National regulations may mandate the sending of the NEID.

For WAG and ePDG events, the NEID can be configured through CLI using the following command:

```
zone <default> gateway [epdg|wag] <name> network-element-identifier <string>
```

The **network-element-identifier** parameter specifies a unique network element identifier that associates the service with Lawful Intercept support. If the parameter is not configured, the S2a/S2b IP address is used (if available). Otherwise the gateway name is used as the NEID.

Correlation Number

The Correlation Number is unique per EPS bearer/tunnel and is used for the following purposes:

- Correlate CC with IRI
- Correlate different IRI records within one EPS bearer/tunnel.

Note: *The Correlation Number is at a minimum unique for each concurrent communication (for example, EPS bearer/tunnel) of a subject within a lawful authorization. However when different protocols (that is, GTP and PMIP) are used in the networks, different values for the correlation number can be generated by different nodes for the same communication.*

In case of handover between different accesses involving a Gn/Gp interface that is, from E-UTRAN to 2G/3G and vice versa), the same correlation number for the PDP context/bearer shall be used before and after the handover during the same IRI transaction.

EPC IRI Events

Bearer Activation

When a bearer activation is generated, a bearer activation-event is generated by the S-GW/PDN-GW. The following elements are delivered to the DF2, if available.

Table 6-1 Bearer Activation (successful) and Start of Interception with Active Bearer BEGIN Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed ME Id		
event type	C	Provide, depending on the reported event, Bearer activation or Start of interception with active bearer event type.
event date	M	Provide the date and time the event is detected.
event time		
access point name	C	Provide to identify the packet data network to which the connection is made (valid only for default bearer activation).
PDN address allocation	C	Provides the PDN type and PDN address(es) used by the network.
initiator	C	Provide to indicate whether the EPS bearer activation is network-initiated, intercept-subject-initiated, or not available.
network identifier	M	Shall be provided.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's UE.
EPS bearer QOS	C	Provide to identify the QOS parameters. It carries the EPS bearer QOS associated to the established bearer.
Bearer activation type	C	Provides information on default or dedicated bearer activation.

Table 6-1 Bearer Activation (successful) and Start of Interception with Active Bearer BEGIN Record (Continued)

Parameter	MOC	Description/Conditions
APN-AMBR	C	The Aggregate Maximum Bit Rate foreseen for the APN. The parameter carries the APN-AMBR used for the established bearer
Protocol configuration options	C	Provide information about the protocol configuration options requested by the UE.
Procedure transaction identifier	C	Used to associate the EPS bearer activation to other messages triggering the procedure.
EPS bearer id	C	Provides the EPS bearer ID allocated by the network.
Linked EPS bearer identity	C	Provides, in case of dedicated bearer activation, the EPS bearer ID of the associated default bearer; not applicable in case of default bearer activation.
Traffic Flow Template(s) TFT	C	The TFT associated to the dedicated bearer activation; not applicable in case of default bearer activation.
Handover indication	C	Provide information that the procedure is triggered as part of a handover.
RAT type	C	The Radio Access Type used by the target subscriber (only applicable to default bearer activation).
Correlation number	C	Provide to uniquely identify the EPS bearer delivered to the LEMF and to correlate IRI records with CC.

Table 6-2 Bearer Activation (unsuccessful) REPORT Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed ME Id		
PDN address allocation	C	Provides the PDN type and PDN address(es) used by the network.
event type	C	Provide EPS Bearer Activation event type.

Table 6-2 Bearer Activation (unsuccessful) REPORT Record (Continued)

Parameter	MOC	Description/Conditions
event date	M	Provide the date and time the event is detected.
event time		
access point name	C	Provide to identify the packet data network to which the attempt to connect was made; this information may be provided by the UE; the parameter is applicable only for default bearer activation.
RAT type	C	Provide the Radio Access Type used by the target subscriber.
initiator	C	Provide to indicate whether the EPS bearer activation is network-initiated, intercept-subject-initiated, or not available.
network identifier	M	Shall be provided.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's UE.
failed bearer activation reason	C	Provide information about the reason for failed bearer activation attempts of the target subscriber.
EPS bearer QOS	C	Provide to identify the QOS parameters. The parameter carries the requested EPS bearer QOS.
Bearer activation type	C	Provides information on default or dedicated bearer failed activation
APN-AMBR	C	The Aggregate Maximum Bit Rate foreseen for the APN. The parameter carries the subscribed APN-AMBR.
Protocol configuration options	C	Provide information about the protocol configuration options requested by the UE
Procedure transaction identifier	C	Used to associate the EPS bearer activation attempt to other messages triggering the procedure.
Linked EPS bearer identity	C	Provides, in case of failed dedicated bearer activation attempt, the EPS bearer ID of the associated default bearer; not applicable in case of default bearer activation attempt.
Traffic Flow Template TFT	C	The TFT associated to the dedicated bearer activation attempt; not applicable in case of default bearer activation attempt
Handover indication	C	Provide information that the procedure is triggered as part of a handover

Bearer Modification

When a bearer modification is detected, a bearer modification-event is generated. The following elements are delivered by the S-GW/PDN-GW to the DF2, if available.

Table 6-3 Bearer Modification CONTINUE Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed ME Id		
event type	C	Provide Bearer modification event type.
event date	M	Provide the date and time the event is detected.
event time		
initiator	C	Provide to indicate whether the EPS bearer modification is network-initiated, intercept-subject-initiated, or not available.
network identifier	M	Shall be provided.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's UE.
EPS bearer QOS	C	Provide to identify the QOS parameters.
APN-AMBR	C	The Aggregate Maximum Bit Rate for the APN.
Procedure transaction identifier	C	Used to associate the EPS bearer modification to other messages triggering the procedure.
EPS bearer id	C	Provides the EPS bearer ID allocated by the network.
Traffic Flow Template(s) TFT	C	The TFT associated to the EPS bearer modification;
RAT type	C	The Radio Access Type used by the target subscriber.
APN-AMBR	C	The Aggregate Maximum Bit Rate foreseen for the APN.
Handover indication	C	Provide information that the procedure is triggered as part of a handover.
Correlation number	C	Provide to uniquely identify the EPS bearer delivered to the LEMF and to correlate IRI records with CC.

Table 6-3 Bearer Modification CONTINUE Record (Continued)

Parameter	MOC	Description/Conditions
Failed bearer modification reason	C	Provide information about the reason for failed bearer modification

Start of Interception with Active Bearer

This event is generated if interception for a target is started and if the target has at least the default bearer active. If more than one bearer is active, an event record is generated for each. The parameters defined for bearer activation (see [GGSN IRI Events](#)) are sent, if available, by the S-GW/PDN-GW to the DF2

As an option, in case the event is sent due to a change of the involved S-GW, the new S-GW may provide as additional parameter, (*old location information*). However, the absence of this information does not imply that interception has not started in the old location S-GW for an active bearer.

Table 6-4 Start of Interception with Active Bearer CONTINUE Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed ME Id		
event type	C	Provide Start of interception with active bearer event type.
event date	M	Provide the date and time the event is detected.
event time		
access point name	C	Provide to identify the packet data network to which the connection is made (valid only for default bearer).
PDN address allocation	C	Provides the PDN type and PDN address(es) used by the network.
network identifier	M	Shall be provided.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's UE.

Table 6-4 Start of Interception with Active Bearer CONTINUE Record (Continued)

Parameter	MOC	Description/Conditions
EPS bearer QOS	C	Provide to identify the QOS parameters.
Bearer activation type	C	Provides information on default or dedicated bearer.
APN-AMBR	C	The Aggregate Maximum Bit Rate foreseen for the APN.
Protocol configuration options	C	Provide, if available, information about the protocol configuration options requested by the UE (valid only for default bearer).
Procedure transaction identifier	C	Used, if available, to associate the EPS bearer to other messages triggering the previous bearer activation.
EPS bearer id	C	Provides the EPS bearer ID allocated by the network.
Linked EPS bearer identity	C	Provides, in case of dedicated bearer, the EPS bearer ID of the associated default bearer; not applicable in case of default bearer.
Traffic Flow Template(s) TFT	C	The TFT associated to the dedicated bearer; not applicable in case of default bearer.
Handover indication	C	Provide information that the procedure is triggered as part of a handover.
RAT type	C	The Radio Access Type used by the target subscriber (only applicable to default bearer).
Correlation number	C	Provide to uniquely identify the EPS bearer delivered to the LEMF and to correlate IRI records with CC.

UE Requested Bearer Resource Modification

When UE requested Bearer Resource Modification [22] is detected at the S-GW/PDN-GW, an UE requested Bearer Resource Modification event is generated. The following elements are delivered to the DF2, if available.

Table 6-5 UE Requested Bearer Resource Modification REPORT Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed ME Id		
event type	C	Provide UE requested bearer resource modification event type.
event date	M	Provide the date and time the event is detected.
event time		
network identifier	M	Shall be provided.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's UE.
failed bearer modification reason	C	Provide information about the reason for failed UE requested bearer resource modification.
EPS bearer QOS	C	Provide to identify the QOS parameters.
Procedure transaction identifier	C	Used to associate the UE requested bearer resource modification to other messages related to the procedure.
Linked EPS bearer identity	C	Provides the EPS bearer ID of the associated default bearer.
EPS Bearer identity	C	Provides the EPS bearer ID of the bearer which the request refers to.
Traffic Aggregate Description	C	Description of the packet filter(s) for the traffic flow aggregate
Protocol Configuration Options	C	Provide information about the protocol configuration options requested by the UE.

Packet Data Header Information

Packet Data Header Information Reporting can be configured for either Raw or Mapped header reporting format.

The Raw format delivers the raw hex bytes in a single parameter:

- Packet dataheader copy

The Mapped mode maps the various header fields into the following parameters:

- Source IP address
- Source port number
- Destination IP address
- Destination port number
- Transport protocol
- Flow label
- Direction
- Packet size

This event is used to provide packet header reports on either:

- Per-packet basis (non-summarized reporting), triggered by each packet sent or received by the target
- Packet flow summary basis (when configured for summary reports), triggered by expiration of the packet-data-header-summary report-interval timer or traffic exceeding the packet-data-header-summary byte-count-threshold.

These elements are delivered by the S-GW/PDN-GW either directly to DF2 or via another network entity, if available.

Table 6-6 Packet Data Heater Information REPORT Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed ME Id		
event type	C	Provide Packet Data Header Information event type.

Table 6-6 Packet Data Header Information REPORT Record (Continued)

Parameter	MOC	Description/Conditions
event date	M	Provide the date and time the event is detected.
event time		
network identifier	M	Shall be provided.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's UE.
EPS bearer id	C	Provides the EPS bearer ID allocated by the network.
Correlation number	M	Provide to uniquely identify the EPS bearer/tunnel delivered to the LEMF and to correlate IRI records with CC.
APN	C	Provides the Access Point Name
packet data header information	M	Provides the packet header information to be reported on a per-packet basis as defined in Table 6-7 .

Table 6-7 Contents of a Per-packet Packet Data Header Information Parameter

Parameter	MOC	Description/Conditions
source IP address <i>Is populated in Mapped Mode only.</i>	C	Provide when mapping packet header information to identify the source IP address for a particular packet flow.
source port number <i>Is populated in Mapped Mode only.</i>	C	Provide when mapping packet header information to report the source port number for a particular packet flow when the transport protocol supports port numbers.
destination IP address <i>Is populated in Mapped Mode only.</i>	C	Provide when mapping packet header information to identify the destination IP address for a particular packet flow.
destination port number <i>Is populated in Mapped Mode only.</i>	C	Provide when mapping packet header information to report the destination port number for a particular packet flow when the transport protocol supports port numbers.
transport protocol <i>Is populated in Mapped Mode only.</i>	C	Provide when mapping packet header information to identify the transport protocol (for example, TCP) for a particular packet flow.

Table 6-7 Contents of a Per-packet Packet Data Header Information Parameter (Continued)

Parameter	MOC	Description/Conditions
flow label <i>Is populated in Mapped Mode only.</i>	C	Provide when mapping packet header information for IPv6 only for a particular packet flow.
direction <i>Is populated in Mapped Mode only.</i>	M	Shall be provided. Identifies the direction of the packet (from subject or to subject).
packet size <i>Is populated in Mapped Mode only.</i>	C	Provide when mapping packet header information to convey the value contained in Total Length Fields of the IPv4 packets or the value contained in the Payload Length fields of the IPv6 packets.
packet data header copy <i>Is populated in Raw Mode only.</i>	C	Provide when reporting a copy of the entire packet header information rather than mapping individual information.

When configured for Packet Data Summary Reports (PDHIR_SUMMARY), the system reports summary information for the packet flows in the session using Packet Data Header Summary records defined in 3GPP TS 33.108.

Table 6-8 Contents of Packet Flow Summary Parameter

Parameter	MOC	Description/Conditions
source IP address	M	Shall be provided. Identifies the source IP address for a particular packet flow.
source port number	C	Provide to report the source port number for a particular packet flow when the transport protocol supports port numbers.
destination IP address	M	Shall be provided. Identifies the destination IP address for a particular packet flow.
destination port number	C	Provide to report the destination port number for a particular packet flow when the transport protocol supports port numbers.
transport protocol	M	Identifies the transport protocol (e.g. TCP) for a particular packet flow.
flow label	C	Provide for IPv6 only for a particular packet flow.

Table 6-8 Contents of Packet Flow Summary Parameter (Continued)

Parameter	MOC	Description/Conditions
summary period	M	Provides the period of time during which the packets of a particular packet flow of the summary report were sent or received by the target and defined by specifying the time when the first packet and the last packet of the reporting period were detected.
packet count	M	Provides the number of packets detected for a particular packet flow.
sum of packet sizes	M	Provides the sum of values contained in Total Length Fields of the IPv4 packets or the sum of the values contained in the Payload Length fields of the IPv6 packets.
packet data summary reason	M	Provides the reason for the report being delivered to the LEMF (i.e. timeout, count limit, end of session).

Note: One or more Packet Flow Summary parameters is contained in a single Packet Data Header Summary Report event.

Bearer Deactivation

When a bearer deactivation is generated, a bearer deactivation-event is generated by the S-GW/PDN-GW. The following elements are delivered to the DF2, if available.

In case all the bearers belonging to the same PDN connection are released at the same time, one event is sent for each bearer.

Note: Cause can be present, for example, in the case of inter S-GW TAU, when the new S-GW sends a bearer deactivation request to the old S-GW.

Table 6-9 Bearer Deactivation END Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed ME Id		
event type	C	Provide Bearer Deactivation event type.
event date	M	Provide the date and time the event is detected.
event time		
initiator	C	Provide to indicate whether the EPS deactivation is network-initiated, intercept-subject-initiated, or not available.
network identifier	M	Shall be provided.
correlation number	C	Provide to uniquely identify the PDP context delivered to the LEM and to correlate IRI records with CC.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's MS.
Bearer deactivation type	C	Provides information on default or dedicated bearer deactivation.
Bearer deactivation cause	C	Provide to indicate reason for deactivation.
EPS bearer id	O	Provides the identity of the deactivated bearer.
Procedure Transaction Identifier	C	Used to associate the EPS bearer deactivation to other messages triggering the procedure.

Mapping Between EPC (S/PGW) Events and HI2 Records Type from 3GPP 33.108

Table 6-10 Mapping Between EPC (S/PGW) Events and HI2 Records Type

SLNO	S/PGW Events	IRI Record Type
1	Bearer activation (successful)	BEGIN
2	Bearer modification	CONTINUE
3	UE Requested bearer resource modification	REPORT
4	Bearer activation (unsuccessful)	REPORT
5	Start of interception with active bearer	BEGIN or optionally CONTINUE
6	Bearer deactivation	END
7	Packet Data Header Information	REPORT

EPC IRI Events Mapping Parameters

Mapping between Events information and IRI information (**Note:** LIID parameter must be present in each record sent to the LEMF).

Table 6-11 EPC IRI Events Mapping Parameters

SLNO	Parameter	HI2 ASN.1 parameter	Description
1	access point name	aPN	When provided by the MME, the parameter carries the Access Point Name provided by the UE. When provided by the S-GW/PDN-GW, it is the APN used for the PDN connection
2	APN-AMBR	aPN-AMBR	Contains the Aggregate Maximum Bit Rate for the APN
3	Bearer activation/ deactivation type	bearerActivationType, bearerDeactivationType	Indicates the type of bearer being activated/deactivated, that is, default or dedicated.
4	bearer deactivation reason	bearerDeactivationCause	This field gives information about the reason for bearer deactivation of the target subscriber.
5	correlation number	ePSCorrelationNumber	Unique number for each target connection delivered to the LEMF, to help the LEA, to have a correlation between each target connection and the IRI.

Table 6-11 EPC IRI Events Mapping Parameters (Continued)

SLNO	Parameter	HI2 ASN.1 parameter	Description
6	EPS bearer identity	ePSBearerIdentity	Identifies an EPS bearer for one UE accessing via E-UTRAN. It is allocated by the MME.
7	EPS bearer QOS	ePSBearerqOS	This field indicates the Quality of Service associated with the EPS bearer procedure.
8	event date	Timestamp	Date of the event generation in the node
9	event time	Timestamp	Time of the event generation in the node
10	event type	ePSevent	Description which type of event is delivered
11	failed bearer activation reason	failedBearerActivationReason	This field gives information about the reason for a failed bearer activation of the target subscriber.
12	Failed Bearer Modification reason	failedBearerModReason	The reason for failure of Bearer Modification
13	Handover indication	handoverIndication, extendedHandoverIndication	Provides information that the procedure is triggered as part of a handover
14	initiator	initiator	This field indicates whether the procedure is UE or network initiated.
15	lawful interception identifier	lawfulInterceptionIdentifier	Unique number for each lawful authorization.
16	Linked EPS bearer identity	linkedEPSBearerId	Indicates, in case of dedicated bearer, the EPS bearer identity of the default bearer.
17	location information	ePSlocationOfTheTarget	When authorized, this field provides the location information of the target that is present at the node at the time of event record production.
18	network identifier	networkIdentifier	Operator ID plus node address.
19	observed IMSI	partyInformation (party-identity)	Target Identifier with the IMSI of the target subscriber (monitored subscriber).
20	observed ME Id	partyInformation (party-identity)	Target Identifier with the ME Id of the target subscriber (monitored subscriber)
21	observed MSISDN	partyInformation (party-identity)	Target Identifier with the MSISDN of the target subscriber (monitored subscriber).
22	Old location information	ePSlocationOfTheTarget	Location information of the target subscriber before Tracking Area Update.
23	PDN address allocation	pDNAddressAllocation	Provides the IP version (IPv4, IPv6, IPv4/IPv6) and the IP address(es) allocated for the UE.

Table 6-11 EPC IRI Events Mapping Parameters (Continued)

SLNO	Parameter	HI2 ASN.1 parameter	Description
24	Procedure Transaction Identifier	procedureTransactionId	Identifies a set of messages belonging to the same procedure; the parameter is dynamically allocated by the UE
25	Protocol Configuration Options	protConfigOptions	Are used to transfer parameters between the UE and the PDN-GW (for example, address allocation preference by DHCP)
26	RAT type	rATType	Radio Access Type
27	Traffic Aggregate Description (TAD)	trafficAggregateDescription	Consists of the description of the packet filter(s) for the traffic flow aggregate.
28	Traffic Flow Template (TFT)	tFT	Collection of all packet filters associated with the EPS bearer.
29	destination IP address	destinationIPAddress	Identifies the destination IP address of a packet.
30	destination port number	destinationPortNumber	Identifies the destination port number of a packet.
31	source IP address	sourceIPAddress	Identifies the source IP address of a packet.
32	source port number	sourcePortNumber	Identifies the source port number of a packet.
33	transport protocol	transportProtocol	Identifies the transport protocol (that is, Protocol Field in IPv4 or Next Header Field in IPv6).
34	flow label	flowLabel	The field in the IPv6 header that is used by a source to label packets of a flow (see RFC 3697 [c])
35	packet size	packetsize	The size of a packet (that is, Total Length Field in IPv4 [a] or Payload Length Field in IPv6 [b])
36	packet direction	packetDirection	Identifies the direction of the intercepted packet (from subject or to subject)
37	packet data header copy	packetHeaderCopy	Provides a copy of the packet headers including IP layer and next layer, and extensions, but excluding content.

Parameter to Event Mapping

Table 6-12 Parameter to Event Mapping

SLNO	Parameter	Description	HI2 ASN.1 Parameter	Bearer Activation	Bearer Deactivation	Bearer Modification	Start of Intercept with Active Bearer	UE Requested Bearer Resource Modification	Packet Data Header Information
1	access point name	When provided by the MME, the parameter carries the Access Point Name provided by the UE. When provided by the S-GW/PDN-GW, it is the APN used for the PDN connection	aPN	C			C		C
2	APN-AMBR	Contains the Aggregate Maximum Bit Rate for the APN	aPN-AMBR	C		C	C		
3	Bearer activation/deactivation type	Indicates the type of bearer being activated/deactivated, that is, default or dedicated.	bearerActivationType, bearerDeactivationType	C	C		C		
4	bearer deactivation reason	This field gives information about the reason for bearer deactivation of the target subscriber.	bearerDeactivationCause		C				

Table 6-12 Parameter to Event Mapping (Continued)

SLNO	Parameter	Description	HI2 ASN.1 Parameter	Bearer Activation	Bearer Deactivation	Bearer Modification	Start of Intercept with Active Bearer	UE Requested Bearer Resource Modification	Packet Data Header Information
5	correlation number	Unique number for each target connection delivered to the LEMF, to help the LEA, to have a correlation between each target connection and the IRI.	ePSCorrelationNumber		C	C	C		M
6	EPS bearer identity	Identifies an EPS bearer for one UE accessing via E-UTRAN. It is allocated by the MME.	ePSBearerId entity	C	O	C	C	C	C
7	EPS bearer QOS	This field indicates the Quality of Service associated with the EPS bearer procedure.	ePSBearerqOS	C		C	C	C	
8	event date	Date of the event generation in the node	Timestamp	M	M	M	M	M	M
9	event time	Time of the event generation in the node		M	M	M	M	M	M
10	event type	Description which type of event is delivered	ePSevent	M	C	C	C	C	C

Table 6-12 Parameter to Event Mapping (Continued)

SLNO	Parameter	Description	Hi2 ASN.1 Parameter	Bearer Activation	Bearer Deactivation	Bearer Modification	Start of Interception with Active Bearer	UE Requested Bearer Resource Modification	Packet Data Header Information
11	failed bearer activation reason	This field gives information about the reason for a failed bearer activation of the target subscriber.	failedBearerActivationReason	C					
12	Failed Bearer Modification reason	The reason for failure of Bearer Modification	failedBearerModReason			C		C	
13	Handover indication	Provides information that the procedure is triggered as part of a handover	handoverIndication, extendedHandoverIndication	C		C	C		C
14	initiator	This field indicates whether the procedure is UE or network initiated.	initiator	C	C	C	C		C
15	lawful interception identifier	Unique number for each lawful authorization.	lawfulInterceptionIdentifier	M	M	M	M	M	M
16	Linked EPS bearer identity	Indicates, in case of dedicated bearer, the EPS bearer identity of the default bearer.	linkedEPSBearerId	C			C	C	

Table 6-12 Parameter to Event Mapping (Continued)

SLNO	Parameter	Description	HI2 ASN.1 Parameter	Bearer Activation	Bearer Deactivation	Bearer Modification	Start of Intercept with Active Bearer	UE Requested Bearer Resource Modification	Packet Data Header Information
17	location information	When authorized, this field provides the location information of the target that is present at the node at the time of event record production.	ePSlocationOfTheTarget	C	C	C	C	C	C
18	network identifier	Operator ID plus node address.	networkIdentifier	M	M	M	M		M
19	observed IMSI	Target Identifier with the IMSI of the target subscriber (monitored subscriber).	partyInformation (party-identity)	C	C	C	C	C	C
20	observed ME Id	Target Identifier with the ME Id of the target subscriber (monitored subscriber)	partyInformation (party-identity)	C	C	C	C	C	C
21	observed MSISDN	Target Identifier with the MSISDN of the target subscriber (monitored subscriber).	partyInformation (party-identity)	C	C	C	C	C	C
22	Old location information	Location information of the target subscriber before Tracking Area Update.	ePSlocationOfTheTarget						

Table 6-12 Parameter to Event Mapping (Continued)

SLNO	Parameter	Description	HI2 ASN.1 Parameter	Bearer Activation	Bearer Deactivation	Bearer Modification	Start of Intercept with Active Bearer	UE Requested Bearer Resource Modification	Packet Data Header Information
23	PDN address allocation	Provides the IP version (IPv4, IPv6, IPv4/IPv6) and the IP address(es) allocated for the UE.	pDNAddressAllocation	C			C		
24	Procedure Transaction Identifier	Identifies a set of messages belonging to the same procedure; the parameter is dynamically allocated by the UE	procedureTransactionId	C	C	C	C	C	
25	Protocol Configuration Options	Are used to transfer parameters between the UE and the PDN-GW (for example, address allocation preference by DHCP)	protConfigOptions	C			C	C	
26	RAT type	Radio Access Type	rATType	C		C	C		
27	Traffic Aggregate Description (TAD)	Consists of the description of the packet filter(s) for the traffic flow aggregate.	trafficAggregateDescription					C	
28	Traffic Flow Template (TFT)	Collection of all packet filters associated with the EPS bearer.	tFT	C		C	C		

Table 6-12 Parameter to Event Mapping (Continued)

SLNO	Parameter	Description	HI2 ASN.1 Parameter	Bearer Activation	Bearer Deactivation	Bearer Modification	Start of Intercept with Active Bearer	UE Requested Bearer Resource Modification	Packet Data Header Information
29	ULI Timestamp	Indicates the time when the User Location Information was acquired.	uLITimestamp		O				
30	destination IP address	Identifies the destination IP address of a packet.	destinationIPAddress						C
31	destination port number	Identifies the destination port number of a packet.	destinationPortNumber						C
32	source IP address	Identifies the source IP address of a packet.	sourceIPAddress						C
33	source port number	Identifies the source port number of a packet.	sourcePortNumber						C
34	transport protocol	Identifies the transport protocol (that is, Protocol Field in IPv4 or Next Header Field in IPv6).	transportProtocol						C
35	flow label	The field in the IPv6 header that is used by a source to label packets of a flow (see RFC 3697 [c])	flowLabel						C

Table 6-12 Parameter to Event Mapping (Continued)

SLNO	Parameter	Description	HI2 ASN.1 Parameter	Bearer Activation	Bearer Deactivation	Bearer Modification	Start of Intercept with Active Bearer	UE Requested Bearer Resource Modification	Packet Data Header Information
36	packet size	The size of a packet (that is, Total Length Field in IPv4 [a] or Payload Length Field in IPv6 [b])							C
37	packet direction	Identifies the direction of the intercepted packet (from subject or to subject)							M
38	packet header copy	Provides a copy of the packet headers including IP layer and next layer, and extensions, but excluding content.							C

GGSN IRI Events

This section contains excerpts from the 3GPP 33.107 specification.

The following events are applicable to the 3G GGSN:

- PDP context activation
- PDP context modification
- PDP context deactivation
- Start of interception with PDP context active
- Packet Data Header Information

Packet Data PDP Context Activation

When a PDP context activation is generated, a PDP context activation-event is generated. The following elements are delivered to the DF2, if available.

Table 6-13 PDP Context Activation (successful) BEGIN Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed IMEI		
observed PDP address	C	Provide to identify one of the following: ■ static address requested by the intercept subject's MS, and allocated by the Network for a successful PDP context activation; ■ address allocated dynamically by the network to the intercept subject MS in association with a PDP context activation (that is, address is sent by the Network in an Activate PDP Context Accept) for a successful PDP context activation procedure when the PDP Context activation request does not contain a static PDP address; or ■ address offered by the network in association with a network-initiated PDP context activation request when the intercept subject's MS accepts the network-initiated PDP context activation request.
iP assignment	C	Provide to indicate observed PDP address is statically or dynamically assigned.

Table 6-13 PDP Context Activation (successful) BEGIN Record (Continued)

Parameter	MOC	Description/Conditions
event type	C	Provide PDP Context Activation event type.
event date	M	Provide the date and time the event is detected.
event time		
access point name	C	Provide to identify the: ■ packet data network to which the intercept subject requested to be connected when the intercept subject's MS is successful at performing a PDP context activation procedure (MS to Network). ■ access point of the packet data network that requested to be connected to the MS when the intercept subject's MS accepts a network-initiated PDP context activation (Network to MS).
PDP type	C	Provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS.
initiator	C	Provide to indicate whether the PDP context activation is network-initiated, intercept-subject-initiated, or not available.
network identifier	M	Shall be provided.
correlation number	C	Provide to uniquely identify the PDP context delivered to the LEMF and to correlate IRI records with CC.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's MS.
umts QOS	C	Provide to identify the QOS parameters.
NSAPI	O	Provided for additional information.

Table 6-14 PDP Context Activation (unsuccessful) REPORT Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed IMEI		
observed PDP address	C	Provide to identify either the: ■ static address requested by the intercept subject's MS in association with a subject-initiated PDP context activation request for unsuccessful PDP context activation requests; or ■ address offered by the network in association with a network-initiated PDP context activation request when the intercept subject's MS rejects the network-initiated PDP context activation.
iP assignment	C	Provide to indicate observed PDP address is statically or dynamically assigned.
event type	C	Provide PDP Context Activation event type.
event date	M	Provide the date and time the event is detected.
event time		
access point name	C	Provide to identify either the: ■ packet data network to which the intercept subject requested to be connected when the intercept subject's mobile station is unsuccessful at performing a PDP context activation procedure (MS to Network) ■ access point of the packet data network that requested to be connected to the MS when the intercept subject's mobile station rejects a network-initiated PDP context activation (Network to MS).
PDP type	C	Provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS.
initiator	C	Provide to indicate whether the PDP context activation is network-initiated, intercept-subject-initiated, or not available.
network identifier	M	Shall be provided.
lawful intercept identifier	M	Shall be provided.

Table 6-14 PDP Context Activation (unsuccessful) REPORT Record (Continued)

Parameter	MOC	Description/Conditions
location information	C	Provide, when authorized, to identify location information for the intercept subject's MS.
failed context activation reason	C	Provide information about the reason for failed context activation attempts of the target subscriber.
umts QOS	C	Provide to identify the QOS parameters.

Packet Data PDP Context Modification

This event is generated if an active PDP context for the target subscriber is modified. The following elements are delivered to the DF2, if available.

Table 6-15 PDP Context Modification CONTINUE Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed IMEI		
observed PDP address	C	The observed address after modification Provide to identify the: ■ static address requested by the intercept subject's MS, and allocated by the Network for a successful PDP context activation. ■ address allocated dynamically by the network to the intercept subject MS in association with a PDP context activation (that is, address is sent by the Network in an Activate PDP Context Accept) for a successful PDP context activation procedure when the PDP Context activation request does not contain a static PDP address. ■ address offered by the network in association with a network-initiated PDP context activation request when the intercept subject's MS accepts the network-initiated PDP context activation request.
event type	C	Provide the PDP Context Modification event type.

Table 6-15 PDP Context Modification CONTINUE Record (Continued)

Parameter	MOC	Description/Conditions
event date	M	Provide the date and time the event is detected.
event time		
access point name	C	Provide to identify the: ■ packet data network to which the intercept subject requested to be connected when the intercept subject's MS is successful at performing a PDP context activation procedure (MS to Network). ■ access point of the packet data network that requested to be connected to the MS when the intercept subject's MS accepts a network-initiated PDP context activation (Network to MS).
PDP type	C	Provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS.
initiator	C	Provide to indicate whether the PDP context modification is network-initiated, intercept-subject-initiated, or not available.
network identifier	M	Shall be provided.
correlation number	C	Provide to uniquely identify the PDP context delivered to the LEMF used to correlate IRI records with CC.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's MS.
umts QOS	C	Provide to identify the QOS parameters.
NSAPI	O	Provided for additional information.

Start of Interception with PDP Context Active

This event is generated if interception for a target is started and if the target has at least one PDP context active. If more than one PDP context is open, an event record is generated for each. The following elements are delivered to the DF2, if available.

Table 6-16 Start Of Interception (with PDP Context Active) BEGIN Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed IMEI		
observed PDP address	C	Provide to identify the: ■ static address requested by the intercept subject's MS, and allocated by the Network for a successful PDP context activation. ■ address allocated dynamically by the network to the intercept subject MS in association with a PDP context activation (that is, address is sent by the Network in an Activate PDP Context Accept) for a successful PDP context activation procedure when the PDP Context activation request does not contain a static PDP address. ■ address offered by the network in association with a network-initiated PDP context activation request when the intercept subject's MS accepts the network-initiated PDP context activation request.
event type	C	Provide Start Of Interception With PDP Context Active event type.
event date	M	Provide the date and time the event is detected.
event time		
access point name	C	Provide to identify the: ■ packet data network to which the intercept subject requested to be connected when the intercept subject's MS is successful at performing a PDP context activation procedure (MS to Network). ■ access point of the packet data network that requested to be connected to the MS when the intercept subject's MS accepts a network-initiated PDP context activation (Network to MS).

Table 6-16 Start Of Interception (with PDP Context Active) BEGIN Record (Continued)

Parameter	MOC	Description/Conditions
PDP type	C	Provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS.
initiator	C	Provide to indicate whether the PDP context activation is network-initiated, intercept-subject-initiated, or not available.
network identifier	M	Shall be provided.
correlation number	C	Provide to uniquely identify the PDP context delivered to the LEMF and to correlate IRI records with CC.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's MS.
umts QOS	C	Provide to identify the QOS parameters.
NSAPI	O	Provided for additional information.

Packet Data Header Information

Packet Header Information Reporting can be configured for either Raw header format or Mapped header format. This event is used to provide packet header reports on a per packet basis and is triggered by each packet sent or received by the target. These elements are delivered either directly to DF2 or via another network entity, if available.

Table 6-17 Packet Data Heater Information REPORT Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed ME Id		

Table 6-17 Packet Data Header Information REPORT Record (Continued)

Parameter	MOC	Description/Conditions
observed PDP address	M	Provide to identify the: ■ static address requested by the intercept subject's MS, and allocated by the Network for a successful PDP context activation. ■ address allocated dynamically by the network to the intercept subject MS in association with a PDP context activation (that is, address is sent by the Network in an Activate PDP Context Accept) for a successful PDP context activation procedure when the PDP Context activation request does not contain a static PDP address. ■ address offered by the network in association with a network-initiated PDP context activation request when the intercept subject's MS accepts the network-initiated PDP context activation request.
event type	M	Provide the Packet Data Header Information event type.
event date	M	Provide the date and time the event is detected.
event time		
access point name	M	Provide to identify the packet data network to which the intercept subject is connected.
PDP type	M	Provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS.
network identifier	M	Shall be provided.
correlation number	C	Provide to uniquely identify the PDP context delivered to the LEMF used to correlate IRI records with CC.
lawful intercept identifier	M	Shall be provided.
packet data header information	M	Provides the packet header information to be reported on a per-packet basis as defined in Table 6-18 .
NSAPI	O	Provided for additional information.

Table 6-18 Contents of a Per-packet Packet Data Header Information Parameter

Parameter	MOC	Description/Conditions
source IP address	C	Provide when mapping packet header information to identify the source IP address for a particular packet flow.
source port number	C	Provide when mapping packet header information to report the source port number for a particular packet flow when the transport protocol supports port numbers.
destination IP address	C	Provide when mapping packet header information to Identify the destination IP address for a particular packet flow.
destination port number	C	Provide when mapping packet header information to report the destination port number for a particular packet flow when the transport protocol supports port numbers.
transport protocol	C	Provide when mapping packet header information to identify the transport protocol (for example, TCP) for a particular packet flow.
flow label	C	Provide when mapping packet header information for IPv6 only for a particular packet flow.
direction	M	Shall be provided. Identifies the direction of the packet (from subject or to subject).
packet size	O	Provide when mapping packet header information to convey the value contained in Total Length Fields of the IPv4 packets or the value contained in the Payload Length fields of the IPv6 packets.
packet data header copy	C	Provide when reporting a copy of the entire packet header information rather than mapping individual information.

Packet Data PDP Context Deactivation

At PDP context deactivation, a PDP context deactivation-event is generated. The following elements are delivered to the DF2, if available.

Table 6-19 PDP Context Deactivation END Record

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed IMEI		
observed PDP address	C	Provide to identify the PDP address assigned to the intercept subject, if available.
event type	C	Provide PDP Context Deactivation event type.
event date	M	Provide the date and time the event is detected.
event time		
access point name	C	Provide to identify the packet data network to which the intercept subject is connected.
PDP type	C	Provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS.
initiator	C	Provide to indicate whether the PDP context deactivation is network-initiated, intercept-subject-initiated, or not available.
network identifier	M	Shall be provided.
correlation number	C	Provide to uniquely identify the PDP context delivered to the LEM and to correlate IRI records with CC.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's MS.
context deactivation reason	C	Provide to indicate reason for deactivation.
NSAPI	O	Provided for additional information.

Table 6-20 Start Of Interception (with PDP Context Active) CONTINUE Record (optional)

Parameter	MOC	Description/Conditions
observed MSISDN	C	Provide at least one and others when available.
observed IMSI		
observed IMEI		
observed PDP address	C	Provide to identify the: ■ static address requested by the intercept subject's MS, and allocated by the Network for a successful PDP context activation. ■ address allocated dynamically by the network to the intercept subject MS in association with a PDP context activation (that is, address is sent by the Network in an Activate PDP Context Accept) for a successful PDP context activation procedure when the PDP Context activation request does not contain a static PDP address. ■ address offered by the network in association with a network-initiated PDP context activation request when the intercept subject's MS accepts the network-initiated PDP context activation request.
event type	C	Provide the Continue interception with active PDP event type.
event date	M	Provide the date and time the event is detected.
event time		
access point name	C	Provide to identify the: ■ packet data network to which the intercept subject requested to be connected when the intercept subject's MS is successful at performing a PDP context activation procedure (MS to Network). ■ access point of the packet data network that requested to be connected to the MS when the intercept subject's MS accepts a network-initiated PDP context activation (Network to MS).
PDP type	C	Provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS.
network identifier	M	Shall be provided.

Table 6-20 Start Of Interception (with PDP Context Active) CONTINUE Record (optional) (Continued)

Parameter	MOC	Description/Conditions
correlation number	C	Provide to uniquely identify the PDP context delivered to the LEMF used to correlate IRI records with CC.
lawful intercept identifier	M	Shall be provided.
location information	C	Provide, when authorized, to identify location information for the intercept subject's MS.
umts QOS	C	Provide to identify the QOS parameters.
NSAPI	O	Provided for additional information.

Presence of the optional Old Location Information field indicates that PDP context was already active, and being intercepted. However, the absence of this information does not imply that interception has not started in the old location SGSN for an active PDP context.

Start of interception with PDP context active is sent regardless of whether a Start of interception with mobile station attached has already been sent.

Mapping Between GGSN Events and HI2 Records Type from 3GPP 33.108

Table 6-21 Mapping Between GGSN Events and HI2 Records Type

SLNO	GGSN Event	IRI Record Type
1	PDP context activation (successful)	BEGIN
2	PDP context modification	CONTINUE
3	PDP context activation (unsuccessful)	REPORT
4	Start of interception with PDP context active	BEGIN or optionally CONTINUE
5	PDP context deactivation	END
6	Packet Data Header Information	REPORT

3G IRI Events Mapping Parameters

Table 6-22 3G IRI Events Mapping Parameters

SLNO	Parameter	HI2 ASN.1 Parameter	Description
1	access point name	partyInformation (services-data-information)	The Access Point Name contains a logical name (see 3GPP TS 23.060 [---TBD---])
2	context deactivation reason	gPRSOperationErrorCode	This field gives information about the reason for context deactivation of the target subscriber.
3	correlation number	gPRSCorrelationNumber	Unique number for each PDP context delivered to the LEMF, to help the LEA, to have a correlation between each PDP Context and the IRI.
4	event date	Timestamp	Date of the event generation in the xGSN
5	event time		Time of the event generation in the xGSN
6	event type	gPRSevent (when using Annex B.3) or ePSevent (when using Annex B.9)	Description which type of event is delivered: PDP Context Activation, PDP Context Deactivation, GPRS Attach, etc.
7	initiator	initiator	This field indicates whether the PDP context activation, deactivation, or modification is MS directed or network initiated.
8	iP assignment	iP-assignment	Observed PDP address is statically or dynamically assigned.
9	failed context activation reason	gPRSOperationErrorCode	This field gives information about the reason for a failed context activation of the target subscriber.
10	lawful interception identifier	lawfulInterceptionIdentifier	Unique number for each lawful authorization.
11	location information	locationOfTheTarget	When authorized, this field provides the location information of the target that is present at the SGSN at the time of event record production.
12	network identifier	networkIdentifier	Operator ID plus SGSN, GGSN, or HLR address.

Table 6-22 3G IRI Events Mapping Parameters (Continued)

SLNO	Parameter	HI2 ASN.1 Parameter	Description
13	NSAPI	nSAPI	Network layer Service Access Point Identifier information element contains an NSAPI identifying a PDP Context in a mobility management context specified by the Tunnel Endpoint Identifier Control Plane This is an optional parameter to help DF/MF and LEA's to distinguish between the sending mobile access networks when the GGSN is used as element of the PDG according TS 23.234 [43].
14	observed IMEI	partyInformation (party-identity)	Target Identifier with the IMEI of the target subscriber (monitored subscriber)
15	observed IMSI		Target Identifier with the IMSI of the target subscriber (monitored subscriber).
16	observed MSISDN		Target Identifier with the MSISDN of the target subscriber (monitored subscriber).
17	observed PDP address	partyInformation (services-data-information)	PDP address(es) used by the target. In case of IPv4v6 two addresses may be carried.
18	PDP type	partyInformation (services-data-information)	This field describes the PDP type as defined in 3GPP TS 29.060 [17], TS 24.008 [9], TS 29.002 [4]
19	umts QOS	qOS	This field indicates the Quality of Service associated with the PDP Context procedure.
20	destination IP address	destinationIPAddress	Identifies the destination IP address of a packet.
21	destination port number	destinationPortNumber	Identifies the destination port number of a packet.
22	source IP address	sourceIPAddress	Identifies the source IP address of a packet.
23	source port number	sourcePortNumber	Identifies the source port number of a packet.
24	transport protocol	transportProtocol	Identifies the transport protocol (that is, Protocol Field in IPv4 or Next Header Field in IPv6.
25	flow label	flowLabel	The field in the IPv6 header that is used by a source to label packets of a flow (see RFC 3697 [c])
26	packet size	packetsize	The size of a packet (that is, Total Length Field in IPv4 [a] or Payload Length Field in IPv6 [b])

Table 6-22 3G IRI Events Mapping Parameters (Continued)

SLNO	Parameter	HI2 ASN.1 Parameter	Description
27	packet direction	packetDirection	Identifies the direction of the intercepted packet (from subject or to subject)
28	packet data header copy	packetHeaderCopy	Provides a copy of the packet headers including IP layer and next layer, and extensions, but excluding content.

Parameter to Event Mapping

Table 6-23 Parameter to Event Mapping

SLNO	Parameter	Description	HI2 ASN.1 Parameter	PDP context activation	PDP context modification	PDP context deactivation	Start of interception with PDP context active	Packet Data Header Information
1	access point name	The Access Point Name contains a logical name (see 3GPP TS 23.060 [---TBD---])	partyInformation					
			(services-data-information)	C		C	C	M
2	context deactivation reason	This field gives information about the reason for context deactivation of the target subscriber.	gPRSOperationErrorCode			C		
3	correlation number	Unique number for each PDP context delivered to the LEMF, to help the LEA, to have a correlation between each PDP Context and the IRI.	gPRSCorrelationNumber		C	C	C	M
4	event date	Date of the event generation in the xGSN	Timestamp	M	M	M	M	M
5	event time	Time of the event generation in the xGSN		M	M	M	M	M
6	event type	Description which type of event is delivered: PDP Context Activation, PDP Context Deactivation, GPRS Attach, etc.	gPRSevent (when using Annex B.3) or ePSevent (when using Annex B.9)	C	C	C	C	M
7	initiator	This field indicates whether the PDP context activation, deactivation, or modification is MS directed or network initiated.	initiator	C	C	C	C	

Table 6-23 Parameter to Event Mapping (Continued)

SLNO	Parameter	Description	HI2 ASN.1 Parameter	PDP context activation	PDP context modification	PDP context deactivation	Start of interception with PDP context active	Packet Data Header Information
8	lawful interception identifier	Unique number for each lawful authorization.	lawfulInterceptionIdentifier	M	M	M	M	M
9	location information	When authorized, this field provides the location information of the target that is present at the SGSN at the time of event record production.	locationOfTheTarget	C	C	C	C	
10	network identifier	Operator ID plus SGSN, GGSN, or HLR address.	networkIdentifier	M	M	M	M	M
11	NSAPI	Network layer Service Access Point Identifier	nSAPI					
		information element contains an NSAPI identifying a PDP Context in a mobility management context specified by the Tunnel Endpoint Identifier Control Plane		O	O	O	O	O
		This is an optional parameter to help DF/MF and LEA's to distinguish between the sending mobile access networks when the GGSN is used as element of the PDG according TS 23.234 [43].		O	O	O	O	O
12	observed IMEI	Target Identifier with the IMEI of the target subscriber (monitored subscriber)	partyInformation (party-identity)	C	C	C	C	C

Table 6-23 Parameter to Event Mapping (Continued)

SLNO	Parameter	Description	HI2 ASN.1 Parameter	PDP context activation	PDP context modification	PDP context deactivation	Start of interception with PDP context active	Packet Data Header Information
13	observed IMSI	Target Identifier with the IMSI of the target subscriber (monitored subscriber).	partyInformation (party-identity)	C	C	C	C	C
14	observed MSISDN	Target Identifier with the MSISDN of the target subscriber (monitored subscriber).	partyInformation (party-identity)	C	C	C		C
15	observed PDP address	PDP address(es) used by the target. In case of IPv4v6 two addresses may be carried.	partyInformation					
			(services-data-information)	C	C	C	C	M
16	PDP type	This field describes the PDP type as defined in 3GPP TS 29.060 [17], TS 24.008 [9], TS 29.002 [4]	partyInformation					
			(services-data-information)	C	C	C	C	M
17	ULI Timestamp	Indicates the time when the User Location Information was acquired.	uLITimestamp			O		
	umts QOS	This field indicates the Quality of Service associated with the PDP Context procedure.	qOS	C	C		C	
18	iP assignment	Observed PDP address is statically or dynamically assigned.	iP-assignment	C				
19	failed context activation reason	This field gives information about the reason for a failed context activation of the target subscriber.	gPRSOperationErrorcode	C				

Table 6-23 Parameter to Event Mapping (Continued)

SLNO	Parameter	Description	HI2 ASN.1 Parameter	PDP context activation	PDP context modification	PDP context deactivation	Start of interception with PDP context active	Packet Data Header Information
20	destination IP address	Identifies the destination IP address of a packet.	destinationIPAddress					C
21	destination port number	Identifies the destination port number of a packet.	destinationPortNumber					C
22	source IP address	Identifies the source IP address of a packet.	sourceIPAddress					C
23	source port number	Identifies the source port number of a packet.	sourcePortNumber					C
24	transport protocol	Identifies the transport protocol (that is, Protocol Field in IPv4 or Next Header Field in IPv6).	transportProtocol					C
25	flow label	The field in the IPv6 header that is used by a source to label packets of a flow (see RFC 3697 [c])	flowLabel					C
26	packet size	The size of a packet (that is, Total Length Field in IPv4 [a] or Payload Length Field in IPv6 [b])	packetSize					C
27	packet direction	Identifies the direction of the intercepted packet (from subject or to subject)	packetDirection					M
28	packet header copy	Provides a copy of the packet headers including IP layer and next layer, and extensions, but excluding content.	packetHeaderCopy					C

WAG and ePDG IRI Events

Note: In release 9.0 and later, Lawful Intercept for ePDG and TWAG is no longer supported.

Tunnel Activation Event

Table 6-24 Tunnel Activation Event

Parameter	Availability WAG	Availability ePDG
Domain ID	Always	Always
LIID	Always	Always
Party Identity	-	-
IMSI	If Available	If Available
MSISDN	If Available	If Available
NAI	If Available	If Available
Correlation Number	Always	Always
Timestamp	Always	Always
NE Identifier	Always	Always
APN	Always	Always
MAC Address	If Available	N/A
PGW Address	If Available	If Available
UE IP Address	If Available	If Available

Start of Interception with Communication Active Event

Table 6-25 Start of Interception with Communication Active Event

Parameter	Availability WAG	Availability ePDG
Domain ID	Always	Always
LIID	Always	Always
Party Identity	-	-
IMSI	If Available	If Available
MSISDN	If Available	If Available
NAI	If Available	If Available
Correlation Number	Always	Always
Timestamp	Always	Always
NE Identifier	Always	Always
APN	Always	Always
MAC Address	If Available	N/A
PGW Address	If Available	If Available
UE IP Address	If Available	If Available

Tunnel Deactivation Event

Table 6-26 Tunnel Deactivation Event

Parameter	Availability WAG	Availability ePDG
Domain ID	Always	Always
LIID	Always	Always
Party Identity	-	-
IMSI	If Available	If Available
MSISDN	If Available	If Available
NAI	If Available	If Available
Correlation Number	Always	Always
Timestamp	Always	Always
NE Identifier	Always	Always
APN	Always	Always
MAC Address	If Available	N/A
PGW Address	If Available	If Available
UE IP Address	If Available	If Available

CALEA Alarms

This chapter lists the CALEA alarms.

Alarm Detail

The alarm details are listed in the following table.

Table 7-1 Alarm Detail

anCaleaPeerLinkDown	Description:	The alarm is generated when the transport connection to the CALEA peer is down. It is cleared when connectivity to the CALEA peer is restored.
	Trap ID:	140
	Status:	Current
	Severity:	Major
	Auto Clearing:	No
	Potential Impact:	If the link to the CALEA peer is down, it can lead to packets being sent over the link to be rerouted over other links, buffered or lost.
	Corrective Action:	Ensure that connectivity to the peer is restored. Loss of connectivity could be due to network issues, problems on the peer side etc.

Table 7-1 Alarm Detail (Continued)

anCaleaPeerLinkUp	Description:	The alarm is generated when the transport connection to the CALEA peer comes up. It has now been deprecated and will be removed in future releases, as the anCaleaPeerLinkDown alarm delivers all link status change notifications. It is not used by the existing code.
	Trap ID:	141
	Status:	Deprecated
	Severity:	Major
	Auto Clearing:	No
	Potential Impact:	This alarm is no longer used.
	Corrective Action:	This alarm is no longer used.
anCaleaTargetsCleared	Description:	The trap is generated when all CALEA targets have been removed and the size of the CALEA target data base becomes zero. It is an informational trap and therefore never cleared.
	Trap ID:	162
	Status:	Current
	Severity:	Info
	Auto Clearing:	Yes
	Potential Impact:	None; this just indicates that there are no CALEA targets configured.
	Corrective Action:	None needed.
anCaleaInterfacesCleared	Description:	This alarm is generated when all Lawful Intercept Mediation device Interfaces have been deleted. It is an informational alarm and is never cleared.
	Trap ID:	163
	Status:	Current
	Severity:	Info
	Auto Clearing:	Yes
	Potential Impact:	None; this just indicates that all configured CALEA interfaces have been removed.
	Corrective Action:	None needed.

Table 7-1 Alarm Detail (Continued)

anCaleaUtilizationLow	Description:	The alarm is generated when the Configured Lawful Intercept Targets Utilization is below 70% of the maximum capacity.
	Trap ID:	164
	Status:	Current
	Severity:	Info
	Auto Clearing:	Yes
	Potential Impact:	None. This just indicates that the CALEA target database is less than 70% full. It is an informational trap only and never cleared.
	Corrective Action:	None needed.
anCaleaUtilizationMedium	Description:	This Alarm is generated when Configured Lawful Intercept Targets are above 70% but below 80% of maximum allowed configurable targets.
	Trap ID:	165
	Status:	Current
	Severity:	Info
	Auto Clearing:	Yes
	Potential Impact:	If additional targets continue to be added, it is possible that the maximum allowed number of targets might be reached or exceeded, preventing configuration of additional targets.
	Corrective Action:	Delete some LI targets to bring down the utilization to acceptable levels.
anCaleaUtilizationHigh	Description:	This Alarm is generated when Configured Lawful Intercept Targets are above 80% but below 90% of maximum allowed targets. This is an informational alarm and is never cleared.
	Trap ID:	166
	Status:	Current
	Severity:	Info
	Auto Clearing:	Yes
	Potential Impact:	If additional targets continue to be added, it is possible that the maximum allowed number of targets might be reached or exceeded, preventing configuration of additional targets.
	Corrective Action:	Delete some LI targets to bring down the utilization to acceptable levels.

Table 7-1 Alarm Detail (Continued)

anCaleaUtilizationCritical	Description:	This Alarm is generated when Configured Lawful Intercept Targets are above 90% but below 100% of maximum allowed targets. This is an informational alarm and is never cleared.
	Trap ID:	167
	Status:	Current
	Severity:	Info
	Auto Clearing:	Yes
	Potential Impact:	If additional targets continue to be added, it is possible that the maximum allowed number of targets might be reached or exceeded, preventing configuration of additional targets.
	Corrective Action:	Delete some LI targets to bring down the utilization to acceptable levels.
anCaleaCCBufferConfiguredSizeBelowThreshold	Description:	The alarm is generated when the configured size for the CC buffer on the interface group is less than the minimum threshold value (10MB).
	Trap ID:	189
	Status:	Current
	Severity:	Major
	Auto Clearing:	No
	Potential Impact:	If the size of the CC buffer on the interface group is too small, this can lead to packets getting lost when connectivity is lost and packets are being buffered.
	Corrective Action:	A larger value for the CC buffer size should be configured.
anCaleaCCBufferUtilizationThresholdReached	Description:	This alarm is generated when the buffer utilization threshold is reached for the CC buffer on the interface group.
	Trap ID:	190
	Status:	Current
	Severity:	Critical
	Auto Clearing:	No
	Potential Impact:	If the CALEA CC buffer utilization continues to grow, it could cause the buffer to fill up completely and potentially overflow, leading to loss of data from the buffer.
	Corrective Action:	Ensure that connectivity is restored, which would cause the packets stored in the buffer to be sent out on the available CALEA interfaces that are part of this interface group.

Table 7-1 Alarm Detail (Continued)

anCalealRIBufferUtilizationThresholdReached	Description:	This alarm is generated when the buffer utilization threshold is reached for the CC buffer on the interface group.
	Trap ID:	191
	Status:	Current
	Severity:	Critical
	Auto Clearing:	No
	Potential Impact:	If the CALEA IRI buffer utilization continues to grow, it could cause the buffer to fill up completely and potentially overflow, leading to loss of data from the buffer.
	Corrective Action:	Ensure that connectivity is restored, which would cause the packets stored in the buffer to be sent out on the available CALEA interfaces that are part of this interface group.
anCaleaTargetDBMaxSubsLimitReached	Description:	The alarm is generated when the number of entries in the target DB reaches the maximum allowed value of 32,768.
	Trap ID:	192
	Status:	Current
	Severity:	Critical
	Auto Clearing:	No
	Potential Impact:	Any attempts to add LI targets after this target will fail as the limit for the maximum number of allowed subscribers has been reached.
	Corrective Action:	Delete some targets to ensure that additional targets can be configured. The alarm will be cleared when the count goes down to 99% of the maximum allowed targets.
anCalealRIBufferConfiguredSizeBelowThreshold	Description:	The alarm is generated when the configured size for the IRI buffer on the interface group is less than the minimum threshold value (100 KB).
	Trap ID:	193
	Status:	Current
	Severity:	Major
	Auto Clearing:	No
	Potential Impact:	If the size of the IRI buffer on the interface group is too small, this can lead to packets getting lost when connectivity is lost and packets are being buffered.
	Corrective Action:	A larger value for the IRI buffer size should be configured.

Table 7-1 Alarm Detail (Continued)

anCaleaIfOrIfGroupConfig Change	Description:	The alarm is raised when the configuration for a CALEA interface/interface group is modified in any way (added, changed or deleted).
	Trap ID:	194
	Status:	Current
	Severity:	Info
	Auto Clearing:	No
	Potential Impact:	None; configuration changes will be applied.
	Corrective Action:	None needed.
anCaleaLITargetCreatedOr Deleted	Description:	The alarm is generated whenever a CALEA Lawful Intercept target is created or deleted.
	Trap ID:	195
	Status:	Current
	Severity:	Minor
	Auto Clearing:	No
	Potential Impact:	No adverse impact.
	Corrective Action:	None needed.
anCaleaInterfaceGroup StatusDown	Description:	The alarm is generated whenever the interface group status changes from up to down. It is cleared when the status changes back to up.
	Trap ID:	196
	Status:	Current
	Severity:	Major
	Auto Clearing:	No
	Potential Impact:	If this is a primary group and no secondary group is configured, this may lead to packets being buffered, and eventually being lost if the buffer size is exceeded.
	Corrective Action:	Restore connectivity to the CALEA server so that data can be sent via some interfaces in the group and group status changes to up again.

Table 7-1 Alarm Detail (Continued)

anCaleaCCBufferDataLost	Description:	The alarm is generated whenever CC buffer data is lost because the buffer size was exceeded and inserting additional data in the buffer caused earlier packets to be lost.
	Trap ID:	197
	Status:	Current
	Severity:	Info
	Auto Clearing:	No
	Potential Impact:	Buffered data is lost, so not all data will be sent to the remote server.
	Corrective Action:	Increase the CC buffer size or flush the buffers by restoring connectivity to CALEA server.
anCaleaIRIBufferDataLost	Description:	The alarm is generated whenever CC buffer data is lost because the buffer size was exceeded and inserting additional data in the buffer caused earlier packets to be lost.
	Trap ID:	198
	Status:	Current
	Severity:	Minor
	Auto Clearing:	No
	Potential Impact:	Buffered data is lost, so not all data will be sent to the remote server.
	Corrective Action:	Increase the IRI buffer size or flush the buffers by restoring connectivity to CALEA server.
anCaleaUserBanFeature ConfigChange	Description:	The alarm is generated when admin state of Calea User Ban Feature is changed.
	Trap ID:	299
	Status:	Current
	Severity:	Major
	Auto Clearing:	No
	Potential Impact:	The admin state of the calea user-ban feature has been enabled or disabled
	Corrective Action:	Make sure the admin state of calea user-ban feature is set to the desired setting.

Table 7-1 Alarm Detail (Continued)

anCaleaCCDataLost	Description:	The alarm is generated when CC data is first lost on an interface which is down (and no interface-group is provisioned for target)
	Trap ID:	300
	Status:	Current
	Severity:	Info
	Auto Clearing:	No
	Potential Impact:	CC data is lost, so not all data will be sent to the remote server.
	Corrective Action:	Use persistent targets associated with calea interface-groups. Interface-groups provide buffering when interfaces are down/congested
anCaleaIRIDataLost	Description:	The alarm is generated when IRI data is first lost on an interface which is down (and no interface-group is provisioned for target)
	Trap ID:	301
	Status:	Current
	Severity:	Info
	Auto Clearing:	No
	Potential Impact:	IRI data is lost, so not all data will be sent to the remote server.
	Corrective Action:	Use persistent targets associated with calea interface-groups. Interface-groups provide buffering when interfaces are down/congested.
anCaleaGrpSourceProfile ConfigChange	Description:	The alarm is raised when the configuration for a CALEA interface-group-source-profile is modified in any way (added, changed or deleted)
	Trap ID:	323
	Status:	Current
	Severity:	Info
	Auto Clearing:	No
	Potential Impact:	None; configuration changes will be applied
	Corrective Action:	None needed

Table 7-1 Alarm Detail (Continued)

anCaleaIfGrpSourceProfile NotFound	Description:	The alarm is raised when the CALEA subsystem is unable to find an interface-group-source-profile for a dynamic target interface/group
	Trap ID:	324
	Status:	Current
	Severity:	Info
	Auto Clearing:	No
	Potential Impact:	Dynamic calea interface and/or interface-group is not created. LI data is not delivered to the LEMF.
	Corrective Action:	Configure the required interface-group-source-profile(s) for use by dynamic targets
anCaleaPdhIrNotSupported ForSx3lIfTunnelToCPF	Description:	This alarm is generated to indicate control plane will not deliver PDHIR events when CALEA target config is having PDHIR enabled in SX3LIF mode. The Packet Data Header Information reporting (PDHIR) features <i>are not</i> supported on CUPS systems configured for Call Content from CPF LI Targets. This alarm is raised when a session targeted for PDHIR is attempted in the unsupported configuration.
	Trap ID:	342
	Status:	Current
	Severity:	Major
	Auto Clearing:	Yes
	Potential Impact:	Control plane does not deliver the PDHIR events.
	Corrective Action:	Remove Sx3lIf config

Table 7-1 Alarm Detail (Continued)

anCaleaDecryptFail	Description:	This alarm is generated when encrypted CALEA Lawful Intercept data cannot be decrypted (may be due to security key mismatch between CPF and UPF).
	Trap ID:	343
	Status:	Current
	Severity:	Major
	Auto Clearing:	No
	Potential Impact:	Traffic on Session/Bearer which should be intercepted by Lawful Intercept target is not delivered to the Lawful Intercept Mediation Server.
	Corrective Action:	Configure the CALEA PKI security keys and/or calea pfcpl-lawful-intercept-ie encryption and secret-key to identical values on both CPF MCC and UPF MCC.

CALEA Examples

This Appendix provides sample commands for configuring, monitoring, and deleting CALEA information:

- [Example 1: Configuring a CALEA Interface](#)
- [Example 2: Configuring IPsec to be used with a CALEA Interface](#)
- [Example 3: Configuring a CALEA Interface Group](#)
- [Example 4: Configuring Non-Persistent CALEA Targets](#)
- [Example 5: Configuring Persistent CALEA Targets](#)
- [Example 6: Monitoring CALEA Targets](#)
- [Example 7: Monitoring Missing CALEA Targets](#)
- [Example 8: Monitoring Extra CALEA Targets](#)
- [Example 9: Configuring a CALEA interface-group-source-profile](#)
- [Example 10: Configuring a Target with Explicit LEMFs Using the Multiple HI2/HI3 Delivery Destinations Feature](#)
- [Example 11: Configuring a Target for packet-data-header-info-report Features](#)

Example 1: Configuring a CALEA Interface

The following example shows how to configure a CALEA interface with an IPv4 ip-address and source-ip-address name:

```
~/echo $ ssh calea@10.32.1.199
calea@10.32.1.199's password:
calea connected from 10.1.1.60 using ssh on an3000-mcm-slot7cpu1

an3000-mcm-slot7cpu1# config
an3000-mcm-slot7cpu1(config)# calea interface-table X1-1-BER-HI
interface-type x1-1 transport-type tcp ip-address 10.6.1.205 port 1111
output-type ber interfaceFormat-type AN_format echo-state enabled
echo-interval 10 echo-attempt-count 6 tcp-send-sock-buffer-size 128
tcp-user-timeout 900
an3000-mcm-slot7cpu1(config)# commit
Commit complete.

an3000-mcm-slot7cpu1# top
an3000-mcm-slot7cpu1(config)# calea interface-table X2-BER-HI
interface-type x2 transport-type tcp network-context LI
source-ip-address-name x2_lp_v4 source-port 6161 ip-address
10.71.32.14 port 7171 output-type ber interfaceFormat-type HI_format
dscp-value dscp_11 tcp-connection-retry-interval 10 tcp-keep-alive
enabled tcp-keep-alive-interval 75 tcp-keep-alive-probes 9
tcp-keep-alive-time 7200 tpkt-keep-alive enabled
tpkt-keep-alive-time-interval 5 tcp-send-sock-buffer-size 128
tcp-user-timeout 900
an3000-mcm-slot7cpu1(config)# commit
Commit complete.

an3000-mcm-slot7cpu1(config)# top
an3000-mcm-slot7cpu1(config)# calea interface-table X3-BER-HI
interface-type x3 transport-type tcp network-context LI
source-ip-address-name x3_lp_v4 source-port 8181 ip-address
10.71.32.14 port 9191 output-type ber interfaceFormat-type HI_format
dscp-value dscp_11 tcp-connection-retry-interval 10 tcp-keep-alive
enabled tcp-keep-alive-interval 75 tcp-keep-alive-probes 9
tcp-keep-alive-time 7200 tpkt-keep-alive enabled
tpkt-keep-alive-time-interval 5 tcp-send-sock-buffer-size 128
tcp-user-timeout 900
an3000-mcm-slot7cpu1(config)# commit
Commit complete.

an3000-mcm-slot7cpu1(config)# top
an3000-mcm-slot7cpu1(config)# exit
```

```

an3000-mcm-slot7cpu1# show calea interface-status
calea interface-status X1-1-BER
  transport-type          tcp
  interface-type          x1-1
  operational-status      up
  number-of-agent-instances 1
  number-of-x2-events-offered 0
  number-of-x3-events-offered 0
  number-of-x3-event-encode-failures-no-buffer 0
  number-of-x3-event-xer-encode-failures 0
  number-of-x3-event-ber-encode-failures 0
  number-of-transmit-ok 0
  number-of-transmit-fail-bad-interface 0
  number-of-transmit-fail-interface-down 0
  number-of-transmit-fail-no-space 0
  number-of-x2-encode-fail-no-buffer 0
  number-of-x2-ber-encode-fail 0
  number-of-x2-xer-encode-fail 0
  number-of-x3-encode-fail-no-buffer 0
  number-of-x3-ber-encode-fail 0
  number-of-x3-xer-encode-fail 0
  number-of-connection-attempts 0
  number-of-connection-attempt-fail 0
  number-of-connection-congested-high 0
  number-of-connection-congested-low 0
  number-of-connection-congested-medium 0
  number-of-connection-disconnect 0
  number-of-connection-success 0
  number-of-echo-requests-received 127
  number-of-echo-requests-sent 0
  number-of-echo-responses-received 0
  number-of-echo-responses-sent 127
  number-of-echo-timeouts 0
  number-of-targets 1
  number-of-tpkt-keep-alive-messages-sent 90
  number-of-tpkt-keep-alive-ber-encode-fail-count 0
  number-of-tpkt-keep-alive-xer-encode-fail-count 0
  total-cumulative-LI-Sessions 0
  currently-active-li-calls 0
  total-number-of-intercepted-packets-for-current-active-calls 48
  number-of-x2-events-for-current-active-calls 0
  number-of-x3-events-for-current-active-calls 48
  total-number-of-all-terminated-LI-sessions 0

```

total-number-of-terminated-LI-sessions-due-to-call-disconnect	0
total-number-of-terminated-LI-sessions-due-to-deprovision	0
calea interface-status X2-BER-HI	
transport-type	tcp
interface-type	x2
operational-status	up
admin-state	enabled
number-of-agent-instances	1
number-of-x2-events-offered	0
number-of-x3-events-offered	0
number-of-x3-event-encode-failures-no-buffer	0
number-of-x3-event-xer-encode-failures	0
number-of-x3-event-ber-encode-failures	0
number-of-transmit-ok	0
number-of-transmit-fail-bad-interface	0
number-of-transmit-fail-interface-down	0
number-of-transmit-fail-no-space	0
number-of-x2-encode-fail-no-buffer	0
number-of-x2-ber-encode-fail	0
number-of-x2-xer-encode-fail	6
number-of-x3-encode-fail-no-buffer	0
number-of-x3-ber-encode-fail	0
number-of-x3-xer-encode-fail	0
number-of-echo-requests-received	0
number-of-echo-requests-sent	0
number-of-echo-responses-received	0
number-of-echo-responses-sent	0
number-of-echo-timeouts	0
number-of-targets	1
number-of-tpkt-keep-alive-messages-sent	90
number-of-tpkt-keep-alive-ber-encode-fail-count	0
number-of-tpkt-keep-alive-xer-encode-fail-count	0
total-cumulative-LI-Sessions	1
currently-active-li-calls	1
total-number-of-intercepted-packets-for-current-active-calls	6
number-of-x2-events-for-current-active-calls	6
number-of-x3-events-for-current-active-calls	0
total-number-of-all-terminated-LI-sessions	0
total-number-of-terminated-LI-sessions-due-to-call-disconnect	0
total-number-of-terminated-LI-sessions-due-to-deprovision	0
calea interface-status X3-BER-HI	
transport-type	tcp
interface-type	x3

Example 1: Configuring a CALEA Interface

operational-status	up
number-of-agent-instances	1
number-of-x2-events-offered	0
number-of-x3-events-offered	0
number-of-x3-event-encode-failures-no-buffer	0
number-of-x3-event-xer-encode-failures	0
number-of-x3-event-ber-encode-failures	0
number-of-transmit-ok	0
number-of-transmit-fail-bad-interface	0
number-of-transmit-fail-interface-down	0
number-of-transmit-fail-no-space	0
number-of-x2-encode-fail-no-buffer	0
number-of-x2-ber-encode-fail	0
number-of-x2-xer-encode-fail	6
number-of-x3-encode-fail-no-buffer	0
number-of-x3-ber-encode-fail	0
number-of-x3-xer-encode-fail	0
number-of-echo-requests-received	0
number-of-echo-requests-sent	0
number-of-echo-responses-received	0
number-of-echo-responses-sent	0
number-of-echo-timeouts	0
number-of-targets	1
number-of-tpkt-keep-alive-messages-sent	90
number-of-tpkt-keep-alive-ber-encode-fail-count	0
number-of-tpkt-keep-alive-xer-encode-fail-count	0
total-cumulative-LI-Sessions	1
currently-active-li-calls	1
total-number-of-intercepted-packets-for-current-active-calls	6
number-of-x2-events-for-current-active-calls	6
number-of-x3-events-for-current-active-calls	0
total-number-of-all-terminated-LI-sessions	0
total-number-of-terminated-LI-sessions-due-to-call-disconnect	0
total-number-of-terminated-LI-sessions-due-to-deprovision	0

an3000-mcm-slot7cpu1# **show running-config calea interface-table**

calea interface-table X1-1-BER

ip-address	10.6.1.205
port	1111
interface-type	x1-1
interfaceFormat-type	AN_format
transport-type	tcp
output-type	ber

```

echo-state          disabled
dscp-value          dscp_11
tcp-connection-retry-interval  10
tcp-keep-alive      disabled
tcp-keep-alive-interval        75
tcp-keep-alive-probes          9
tcp-keep-alive-time           7200
tpkt-keep-alive      disabled
tcp-send-sock-buffer-size 128
tcp-user-timeout 900
!
calea interface-table X2-BER-HI
network-context      LI
source-ip-address-name  x2_lp_v4
source-port          6161
ip-address            10.71.32.14
port                  7171
interface-type        x2
interfaceFormat-type  HI_format
transport-type         tcp
output-type            ber
echo-state            disabled
dscp-value             dscp_11
tcp-connection-retry-interval  10
tcp-keep-alive         disabled
tcp-keep-alive-interval        75
tcp-keep-alive-probes          9
tcp-keep-alive-time           7200
tpkt-keep-alive         enabled
tpkt-keep-alive-time-interval  5
tcp-send-sock-buffer-size 128
tcp-user-timeout        900
!
calea interface-table X3-BER-HI
network-context      LI
source-ip-address-name  x3_lp_v4
source-port          8181
ip-address            10.71.32.14
port                  9191
interface-type        x3
interfaceFormat-type  HI_format
transport-type         tcp
output-type            ber

```

```
echo-state disabled
dscp-value dscp_11
tcp-connection-retry-interval 10
tcp-keep-alive disabled
tcp-keep-alive-interval 75
tcp-keep-alive-probes 9
tcp-keep-alive-time 7200
tpkt-keep-alive enabled
tpkt-keep-alive-time-interval 5
tcp-send-sock-buffer-size 128
tcp-user-timeout 900
!
```

You can also configure a CALEA interface with an IPv6 ip-address and source-ip-address name. For example:

```
an3000-mcm-slot7cpu1(config)# calea interface-table x2-ber-hi
interface-type x2 source-ip-address-name sgwv6 source-port 7453
ip-address 2001:470:8865:2020::22:31 port 7453
```

Example 2: Configuring IPsec to be used with a CALEA Interface

The following example shows how to configure IPsec to be used with a CALEA Interface. The administrator account (admin) should perform the configuration. Configure an IPsec tunnel, where:

- **loopback-network-context** *<network-context-name>* is the network context in which the CALEA interface loopback-ip exists.
- **loopback-ip** *<loopback-ip-address-name>* is the loopback-ip name of the CALEA interface.
- **remote-ip** *<remote-ip-address>* is the remote IP address of the CALEA LEMF.

```
config# security policy tunnel <network-context-name> <tunnel-name>
transform esp-3des-192-sha1-160
loopback-network-context <network-context-name>
loopback-ip <loopback-ip-address-name>
remote-ip <remote-ip-address>
options "transport-mode auto-initiate long-sequence-numbers
anti-replay"
ike-life-time 86400
ipsec-life-time 3600
ike-version v2-fallback-v1
```

```
ike-groups group5
pfs-groups group5
pre-shared-key <secret-key>
local-identity email <email-address>
remote-identity email <email-address>
```

- 1 Configure the security policy rule:

```
config# security policy rule <network-context-name> <rule-name>
priority <integer>
action pass
to-tunnel <tunnel-name>
traffic-selector-type packet-filter
```

```
config# security policy rule <network-context-name> <rule-name>
source-packet-filter <source-pf-name>
protocol 6
ip-start <loopback-ip-address> ip-prefix-len 32
```

```
config# security policy rule <network-context-name> <rule-name>
destination-packet-filter <destination-pf-name>
protocol 6
ip-start <remote-ip-address> ip-prefix-len 32
port-start <mf-df-tcp-port-number>
```


Example 3: Configuring a CALEA Interface Group

```
an3000-mcm-slot7cpu1# calea interface-group X2-BER-HI-GRP
interface-type x2 interfaceFormat-type HI_format output-type ber
interface X2-BER-HI roll ; top
an3000-mcm-slot7cpu1(config)# calea interface-group X3-BER-HI-GRP
interface-type x3 interfaceFormat-type HI_format output-type ber
interface X3-BER-HI; top
an3000-mcm-slot7cpu1# commit
Commit complete.
```

```
an3000-mcm-slot7cpu1# show running-config calea interface-group
calea interface-group X2-BER-HI-GRP
role                primary
interface-type      x2
output-type         ber
interfaceFormat-type HI_format
interface           X2-BER-HI
!
!
calea interface-group X3-BER-HI-GRP
role                primary
interface-type      x3
output-type         ber
interfaceFormat-type HI_format
interface           X3-BER-HI
```

```
an3000-mcm-slot7cpu1# show calea interface-group-status
calea interface-group-status X2-BER-HI-GRP
role                primary
interface-group-type x2
output-type         ber
interfaceFormat-type HI_format
operational-status  up
iri-buffer-size     0
iri-buffer-current-size 0
iri-buffer-num-packets 0
iri-buffer-high-water-mark 0
iri-buffer-num-buffer-size-exceeded 0
iri-buffer-num-dropped-packets 0
number-of-agent-instances 1
number-of-x2-events-offered 0
```

number-of-transmit-ok	0
number-of-transmit-fail-bad-interface-group	0
number-of-transmit-fail-interface-group-down	0
number-of-transmit-fail-no-space	0
number-of-x2-encode-fail-no-buffer	0
number-of-x2-ber-encode-fail	0
number-of-x2-xer-encode-fail	0
number-of-targets	0
total-cumulative-LI-Sessions	0
currently-active-li-calls	0
total-number-of-intercepted-packets-for-current-active-calls	0
number-of-x2-events-for-current-active-calls	0
total-number-of-all-terminated-LI-sessions	0
total-number-of-terminated-LI-sessions-due-to-call-disconnect	0
total-number-of-terminated-LI-sessions-due-to-deprovision	0
calea interface-group-status X2-BER-HI-GRP-SEC	
role	secondary
interface-group-type	x2
output-type	ber
interfaceFormat-type	HI_format
operational-status	up
number-of-agent-instances	1
number-of-transmit-ok	0
number-of-transmit-fail-bad-interface-group	0
number-of-transmit-fail-interface-group-down	0
number-of-transmit-fail-no-space	0
calea interface-group-status X3-BER-HI-GRP	
role	primary
interface-group-type	x3
output-type	ber
interfaceFormat-type	HI_format
operational-status	up
cc-buffer-size	0
cc-buffer-current-size	0
cc-buffer-num-packets	0
cc-buffer-high-water-mark	0
cc-buffer-num-buffer-size-exceeded	0
cc-buffer-num-dropped-packets	0
number-of-agent-instances	1
number-of-x3-events-offered	0
number-of-x3-event-encode-failures-no-buffer	0

Example 3: Configuring a CALEA Interface Group

number-of-x3-event-xer-encode-failures	0
number-of-x3-event-ber-encode-failures	0
number-of-transmit-ok	0
number-of-transmit-fail-bad-interface-group	0
number-of-transmit-fail-interface-group-down	0
number-of-transmit-fail-no-space	0
number-of-x3-encode-fail-no-buffer	0
number-of-x3-ber-encode-fail	0
number-of-x3-xer-encode-fail	0
number-of-targets	0
total-cumulative-LI-Sessions	0
currently-active-li-calls	0
total-number-of-intercepted-packets-for-current-active-calls	0
number-of-x3-events-for-current-active-calls	0
calea interface-group-status X3-BER-HI-GRP-SEC	
role	secondary
interface-group-type	x3
output-type	ber
interfaceFormat-type	HI_format
operational-status	up
number-of-agent-instances	1
number-of-transmit-ok	0
number-of-transmit-fail-bad-interface-group	0
number-of-transmit-fail-interface-group-down	0
number-of-transmit-fail-no-space	0

Example 4: Configuring Non-Persistent CALEA Targets

The following example shows how to configure targets that will not persist following a chassis reboot.

```
~/echo $ ssh calea@10.32.1.199
calea@10.32.1.199's password:
calea connected from 10.1.1.60 using ssh on an3000-mcm-slot7cpu1
```

```
an3000-mcm-slot7cpu1# calea createTarget liid imsi-target-1
dataCollectionType iri-plus-cc identityType imsi identityVal
17567547898922 x2InterfaceName test-iri x3InterfaceName test-cc
result success
reason CALEA Director: target added
```

```
an3000-mcm-slot7cpu1# calea createTarget liid imsi-target-2
dataCollectionType iri-only identityType imsi identityVal
17567547898922 x2InterfaceName test-iri
result success
reason CALEA Director: target added
```

```
an3000-mcm-slot7cpu1# calea createTarget liid imei-target
dataCollectionType iri-plus-cc identityType imei identityVal
17567547898922 x2InterfaceName test-iri x3InterfaceName test-cc
result success
reason CALEA Director: target added
```

```
an3000-mcm-slot7cpu1# calea createTarget liid msisdn-target
dataCollectionType iri-plus-cc identityType msisdn identityVal
17567547898922 x2InterfaceName test-iri x3InterfaceName test-cc
result success
reason CALEA Director: target added
```

```
an3000-mcm-slot7cpu1# show calea target-status
calea target-status imei-target
identity-type                imei
identity-value                17567547898922
identity-value-mac
x2InterfaceName              test-iri
x3InterfaceName              test-cc
dataCollectionType           iri-plus-cc
gwType                       all-gw
number-of-agent-instances    1
number-of-x3-events-dropped-no-interface 0
```

Example 4: Configuring Non-Persistent CALEA Targets

```
number-of-x3-events-sent 0
number-of-x3-event-encode-failures-no-buffer 0
number-of-x3-event-xer-encode-failures 0
number-of-x3-event-ber-encode-failures 0
number-of-x2-events-dropped-no-interface 0
number-of-x2-events-sent 0
number-of-x2-pdhir-per-packet-raw-events-sent 0
number-of-x2-pdhir-per-packet-mapped-events-sent 0
number-of-x2-packet-data-summary-report-events-sent 11
number-of-x2-encode-fail-no-buffer 0
number-of-x2-ber-encode-fail 0
number-of-x2-xer-encode-fail 0
number-of-x3-ber-encode-fail 0
number-of-x3-xer-encode-fail 0
calea target-status imsi-target-1
  identity-type imsi
  identity-value 17567547898922
  identity-value-mac
  x2InterfaceName test-iri
  x3InterfaceName test-cc
  dataCollectionType iri-plus-cc
  gwType all-gw
  number-of-agent-instances 1
  number-of-x3-events-dropped-no-interface 0
  number-of-x3-events-sent 0
  number-of-x3-event-encode-failures-no-buffer 0
  number-of-x3-event-xer-encode-failures 0
  number-of-x3-event-ber-encode-failures 0
  number-of-x2-events-dropped-no-interface 0
  number-of-x2-events-sent 0
  number-of-x2-pdhir-per-packet-raw-events-sent 0
  number-of-x2-pdhir-per-packet-mapped-events-sent 0
  number-of-x2-packet-data-summary-report-events-sent 11
  number-of-x2-encode-fail-no-buffer 0
  number-of-x2-ber-encode-fail 0
  number-of-x2-xer-encode-fail 0
  number-of-x3-ber-encode-fail 0
  number-of-x3-xer-encode-fail 0
calea target-status imsi-target-2
  identity-type imsi
  identity-value 17567547898922
  identity-value-mac
  x2InterfaceName test-iri
```

x3InterfaceName	""
dataCollectionType	iri-only
gwType	all-gw
number-of-agent-instances	1
number-of-x3-events-dropped-no-interface	0
number-of-x3-events-sent	0
number-of-x3-event-encode-failures-no-buffer	0
number-of-x3-event-xer-encode-failures	0
number-of-x3-event-ber-encode-failures	0
number-of-x2-events-dropped-no-interface	0
number-of-x2-events-sent	0
number-of-x2-pdhir-per-packet-raw-events-sent	0
number-of-x2-pdhir-per-packet-mapped-events-sent	0
number-of-x2-packet-data-summary-report-events-sent	11
number-of-x2-encode-fail-no-buffer	0
number-of-x2-ber-encode-fail	0
number-of-x2-xer-encode-fail	0
number-of-x3-ber-encode-fail	0
number-of-x3-xer-encode-fail	0
calea target-status msisdn-target	
identity-type	msisdn
identity-value	17567547898922
identity-value-mac	
x2InterfaceName	test-iri
x3InterfaceName	test-cc
dataCollectionType	iri-plus-cc
gwType	all-gw
number-of-agent-instances	1
number-of-x3-events-dropped-no-interface	0
number-of-x3-events-sent	0
number-of-x3-event-encode-failures-no-buffer	0
number-of-x3-event-xer-encode-failures	0
number-of-x3-event-ber-encode-failures	0
number-of-x2-events-dropped-no-interface	0
number-of-x2-events-sent	0
number-of-x2-pdhir-per-packet-raw-events-sent	0
number-of-x2-pdhir-per-packet-mapped-events-sent	0
number-of-x2-packet-data-summary-report-events-sent	11
number-of-x2-encode-fail-no-buffer	0
number-of-x2-ber-encode-fail	0
number-of-x2-xer-encode-fail	0
number-of-x3-ber-encode-fail	0
number-of-x3-xer-encode-fail	0

Example 5: Configuring Persistent CALEA Targets

The following example shows how to configure targets that persist following a chassis reboot.

```
~/echo $ ssh calea@10.32.1.199
calea@10.32.1.199's password:
calea connected from 10.1.1.60 using ssh on an3000-mcm-slot7cpu1

an3000-mcm-slot7cpu1# calea target-table IMSI-Target-1 identityType
imsi identityVal 26801115678910 location-info true operator-id NEW
gwType sae-gw dataCollectionType iri-plus-cc iri-event-time-stamp-type
local-time cc-data-time-stamp-type utc-time x2InterfaceGroupName
X2-BER-HI-GRP x3InterfaceGroupName X3-BER-HI-GRP
packet-data-header-info-report PDHIR_DISABLE
an3000-mcm-slot7cpu1# commit
Commit complete.

an3000-mcm-slot7cpu1# calea target-table IMSI-Target-2 identityType
imsi identityVal 26801115678910 location-info true operator-id NEW
gwType sae-gw dataCollectionType iri-only iri-event-time-stamp-type
utc-time x2InterfaceGroupName X2-BER-HI-GRP
packet-data-header-info-report PDHIR_DISABLE
an3000-mcm-slot7cpu1# commit
Commit complete.

an3000-mcm-slot7cpu1# calea target-table IMEI-Target identityType imei
identityVal 9999999469696910 location-info true operator-id NEW gwType
sae-gw dataCollectionType iri-plus-cc iri-event-time-stamp-type
local-time cc-data-time-stamp-type local-time x2InterfaceGroupName
X2-BER-HI-GRP x3InterfaceGroupName X3-BER-HI-GRP
packet-data-header-info-report PDHIR_DISABLE
an3000-mcm-slot7cpu1# commit
Commit complete.

an3000-mcm-slot7cpu1# calea target-table MSISDN-Target identityType
msisdn identityVal 7777999469696910 location-info true operator-id NEW
gwType sae-gw dataCollectionType iri-plus-cc iri-event-time-stamp-type
local-time cc-data-time-stamp-type local-time x2InterfaceGroupName
X2-BER-HI-GRP x3InterfaceGroupName X3-BER-HI-GRP
packet-data-header-info-report PDHIR_DISABLE
an3000-mcm-slot7cpu1# commit
Commit complete.
```

```

an3000-mcm-slot7cpu1# show calea target-status
calea target-status IMEI-Target
identity-type                               imei
identity-value                             9999999469696910
identity-value-mac
x2-interface-group-name                    X2-BER-HI-GRP
x3-interface-group-name                    X3-BER-HI-GRP
data-collection-type                       iri-plus-cc
gateway-type                              sae-gw
number-of-agent-instances                  1
number-of-x3-events-dropped-no-interface   0
number-of-x3-events-sent                   0
number-of-x3-event-encode-failures-no-buffer 0
number-of-x3-event-xer-encode-failures     0
number-of-x3-event-ber-encode-failures     0
number-of-x2-events-dropped-no-interface   0
number-of-x2-events-sent                   0
number-of-x2-pdhir-per-packet-raw-events-sent 0
number-of-x2-pdhir-per-packet-mapped-events-sent 0
number-of-x2-packet-data-summary-report-events-sent 11
number-of-x2-encode-fail-no-buffer         0
number-of-x2-ber-encode-fail               0
number-of-x2-xer-encode-fail               0
number-of-x3-ber-encode-fail               0
number-of-x3-xer-encode-fail               0
storage-type                              persistent
total-cumulative-LI-Sessions               0
total-number-of-intercepted-packets-for-current-active-calls 0
number-of-x2-events-for-current-active-calls 0
number-of-x3-events-for-current-active-calls 0
total-number-of-all-terminated-LI-sessions 0
total-number-of-terminated-LI-sessions-due-to-call-disconnect 0
total-number-of-terminated-LI-sessions-due-to-deletion-of-context 0
calea target-status IMSI-Target-1
identity-type                               imsi
identity-value                             26801115678910
identity-value-mac
x2-interface-group-name                    X2-BER-HI-GRP
x3-interface-group-name                    X3-BER-HI-GRP
data-collection-type                       iri-plus-cc
gateway-type                              sae-gw
number-of-agent-instances                  1
number-of-x3-events-dropped-no-interface   0

```


Example 5: Configuring Persistent CALEA Targets

number-of-x3-events-sent	0
number-of-x3-event-encode-failures-no-buffer	0
number-of-x3-event-xer-encode-failures	0
number-of-x3-event-ber-encode-failures	0
number-of-x2-events-dropped-no-interface	0
number-of-x2-events-sent	0
number-of-x2-pdhir-per-packet-raw-events-sent	0
number-of-x2-pdhir-per-packet-mapped-events-sent	0
number-of-x2-packet-data-summary-report-events-sent	11
number-of-x2-encode-fail-no-buffer	0
number-of-x2-ber-encode-fail	0
number-of-x2-xer-encode-fail	0
number-of-x3-ber-encode-fail	0
number-of-x3-xer-encode-fail	0
storage-type	persistent
total-cumulative-LI-Sessions	0
total-number-of-intercepted-packets-for-current-active-calls	0
number-of-x2-events-for-current-active-calls	0
number-of-x3-events-for-current-active-calls	0
total-number-of-all-terminated-LI-sessions	0
total-number-of-terminated-LI-sessions-due-to-call-disconnect	0
total-number-of-terminated-LI-sessions-due-to-deletion-of-context	0
calea target-status IMSI-Target-2	
identity-type	imsi
identity-value	26801115678910
identity-value-mac	
x2-interface-group-name	X2-BER-HI-GRP
x3-interface-group-name	"
data-collection-type	iri-only
gateway-type	sae-gw
number-of-agent-instances	1
number-of-x3-events-dropped-no-interface	0
number-of-x3-events-sent	0
number-of-x3-event-encode-failures-no-buffer	0
number-of-x3-event-xer-encode-failures	0
number-of-x3-event-ber-encode-failures	0
number-of-x2-events-dropped-no-interface	0
number-of-x2-events-sent	0
number-of-x2-pdhir-per-packet-raw-events-sent	0
number-of-x2-pdhir-per-packet-mapped-events-sent	0
number-of-x2-packet-data-summary-report-events-sent	11
number-of-x2-encode-fail-no-buffer	0
number-of-x2-ber-encode-fail	0

number-of-x2-xer-encode-fail	0
number-of-x3-ber-encode-fail	0
number-of-x3-xer-encode-fail	0
storage-type	persistent
total-cumulative-LI-Sessions	0
total-number-of-intercepted-packets-for-current-active-calls	0
number-of-x2-events-for-current-active-calls	0
number-of-x3-events-for-current-active-calls	0
total-number-of-all-terminated-LI-sessions	0
total-number-of-terminated-LI-sessions-due-to-call-disconnect	0
total-number-of-terminated-LI-sessions-due-to-deletion-of-context	0
calea target-status MSISDN-Target	
identity-type	msisdn
identity-value	7777999469696910
identity-value-mac	
x2-interface-group-name	X2-BER-HI-GRP
x3-interface-group-name	X3-BER-HI-GRP
data-collection-type	iri-plus-cc
gateway-type	sae-gw
number-of-agent-instances	1
number-of-x3-events-dropped-no-interface	0
number-of-x3-events-sent	0
number-of-x3-event-encode-failures-no-buffer	0
number-of-x3-event-xer-encode-failures	0
number-of-x3-event-ber-encode-failures	0
number-of-x2-events-dropped-no-interface	0
number-of-x2-events-sent	0
number-of-x2-pdhir-per-packet-raw-events-sent	0
number-of-x2-pdhir-per-packet-mapped-events-sent	0
number-of-x2-packet-data-summary-report-events-sent	11
number-of-x2-encode-fail-no-buffer	0
number-of-x2-ber-encode-fail	0
number-of-x2-xer-encode-fail	0
number-of-x3-ber-encode-fail	0
number-of-x3-xer-encode-fail	0
storage-type	persistent
total-cumulative-LI-Sessions	0
total-number-of-intercepted-packets-for-current-active-calls	0
number-of-x2-events-for-current-active-calls	0
number-of-x3-events-for-current-active-calls	0
total-number-of-all-terminated-LI-sessions	0
total-number-of-terminated-LI-sessions-due-to-call-disconnect	0
total-number-of-terminated-LI-sessions-due-to-deletion-of-context	0

```

an3000-mcm-slot7cpu1# show running-config calea target-table
calea target-table $3$HaLVFFxjpIoYbca7A//+Cw==
identityType          msisdn
identityVal           $3$mGkxloCv4t/Oy2ica1wIbjX1c0+HbtN
x2InterfaceGroupName  X2-BER-HI-GRP
x3InterfaceGroupName  X3-BER-HI-GRP
dataCollectionType    iri-plus-cc
gwType                sae-gw
operator-id           NEW
location-info         true
iri-event-time-stamp-type local-time
cc-data-time-stamp-type local-time
packet-data-header-info-report PDHIR_DISABLE
!
calea target-table $3$PQYcmL/pXwBnk5+0KrpQyQ==
identityType          imsi
identityVal           $3$ocsXTxf55Ge672mgGS6+Zw==
x2InterfaceGroupName  X2-BER-HI-GRP
x3InterfaceGroupName  X3-BER-HI-GRP
dataCollectionType    iri-plus-cc
gwType                sae-gw
operator-id           NEW
location-info         true
iri-event-time-stamp-type local-time
cc-data-time-stamp-type utc-time
packet-data-header-info-report PDHIR_DISABLE
!
calea target-table $3$PQYcmL/pXwD7sWoX/PJ07A==
identityType          imsi
identityVal           $3$ocsXTxf55Ge672mgGS6+Zw==
x2InterfaceGroupName  X2-BER-HI-GRP
dataCollectionType    iri-only
gwType                sae-gw
operator-id           NEW
location-info         true
iri-event-time-stamp-type utc-time
cc-data-time-stamp-type local-time
packet-data-header-info-report PDHIR_DISABLE
!
calea target-table $3$miAkraF8PJ9ExGEUwMyjpw==
identityType          imei
identityVal           $3$a6Nliou0njwZj0hjQZtujB9X0zWKopug
x2InterfaceGroupName  X2-BER-HI-GRP

```

```

x3InterfaceGroupName      X3-BER-HI-GRP
dataCollectionType        iri-plus-cc
gwType                    sae-gw
operator-id               NEW
location-info             true
iri-event-time-stamp-type local-time
cc-data-time-stamp-type   local-time
packet-data-header-info-report PDHIR_DISABLE

```

Example 6: Monitoring CALEA Targets

The following example shows how to monitor targets to verify they are present following a reset indication:

```

~/echo $ ssh calea@10.32.1.199
calea@10.32.1.199's password:
calea connected from 10.1.1.60 using ssh on an3000-mcm-slot7cpu1

an3000-mcm-slot7cpu1# show calea target-status | select
x2-interface-group-name | select x3-interface-group-name | select
data-collection-type
calea target-status

```

	X2 INTERFACE	X3 INTERFACE	DATA
LIID	GROUP NAME	GROUP NAME	COLLECTION TYPE
xer-hi-liid-1	x2-xer-hi-grp	x3-xer-hi-grp	iri-plus-cc

Example 7: Monitoring Missing CALEA Targets

The following example shows how to monitor targets that are missing following a reset indication and then restore them:

- 1 List the targets to verify they are all present:

```
~/echo $ ssh calea@10.32.1.199
```

```
calea@10.32.1.199's password:
```

```
calea connected from 10.1.1.60 using ssh on an3000-mcm-slot7cpu1
```

```
an3000-mcm-slot7cpu1# show calea target-status | select
x2-interface-group-name | select x3-interface-group-name | select
data-collection-type
```

```
calea target-status
```

	X2	X3	DATA
	INTERFACE	INTERFACE	COLLECTION
LIID	NAME	NAME	TYPE

```
-----
imei-target    test-iri    test-cc    iri-plus-cc
imsi-target-1  test-iri    test-cc    iri-plus-cc
msisdn-target  test-iri    test-cc    iri-plus-cc
```

The listing indicates that imsi-target-2 is missing.

- 2 Restore imsi-target-2:

```
an3000-mcm-slot7cpu1# calea target-table IMSI-Target-1 identityType
imsi identityVal 26801115678910 location-info true operator-id NEW
gwType sae-gw dataCollectionType iri-plus-cc iri-event-time-stamp-type
local-time cc-data-time-stamp-type utc-time x2InterfaceGroupName
X2-BER-HI-GRP x3InterfaceGroupName X3-BER-HI-GRP
```

```
result success
```

```
reason CALEA Director: target added
```

- 3 List the targets to verify they are all present:

```
an3000-mcm-slot7cpu1# show calea target-status | select
x2-interface-group-name | select x3-interface-group-name | select
data-collection-type
```

```
calea target-status
```

	X2	X3	DATA
	INTERFACE	INTERFACE	COLLECTION
LIID	NAME	NAME	TYPE

```
-----
imei-target    test-iri    test-cc    iri-plus-cc
```

```

imsi-target-1  test-iri  test-cc  iri-plus-cc
imsi-target-2  test-iri           iri-only
msisdn-target  test-iri  test-cc  iri-plus-cc

```

Example 8: Monitoring Extra CALEA Targets

The following example shows how to monitor for targets that are extraneous following a reset indication and then delete them. For example, during the period of lost IP connectivity between the system and ADMF, targets are deleted. Once connectivity is restored, the system still has the targets, but the ADMF no longer has them.

- 1 List the targets to verify no extra ones are present:

```
~/echo $ ssh calea@10.32.1.199
```

```
calea@10.32.1.199's password:
```

```
calea connected from 10.1.1.60 using ssh on an3000-mcm-slot7cpu1
```

```
an3000-mcm-slot7cpu1# show calea target-status | select
x2InterfaceName | select x3InterfaceName | select dataCollectionType
```

```
calea target-status
```

	X2	X3	DATA
	INTERFACE	INTERFACE	COLLECTION
LIID	NAME	NAME	TYPE
extra-target	test-iri		iri-only
imei-target	test-iri	test-cc	iri-plus-cc
imsi-target-1	test-iri	test-cc	iri-plus-cc
imsi-target-2	test-iri		iri-only
msisdn-target	test-iri	test-cc	iri-plus-cc

The listing indicates that extra-target is extraneous.

- 2 Delete extra-target:

```
an3000-mcm-slot7cpu1# no calea target-table extra-target
```

```
result success
```

```
reason CALEA Director: target deleted
```

Example 9: Configuring a CALEA interface-group-source-profile

CALEA interface-group-source-profiles are required for the MCC CALEA features [Call Content from UPF Dynamic LI Targets](#) and [Multiple X2/X3 Delivery Destinations](#).

To configure an interface-group-source-profile, login as calea user and issue a command such as the following:

```
an3000-mcm-slot7cpu1(config)# calea interface-group-source-profile
X3-IFC-GRP-SRC-PROFILE network-context GN source-ip-address-name
CALEAL00P source-port 1030 number-of-source-ports 8 interface-type x3
interfaceFormat-type HI_format ;top;commit;
Commit complete.
```

```
an3000-mcm-slot7cpu1(config)# exit
```

To view the resulting configuration, use the **show running-config** command as follows:

```
an3000-mcm-slot7cpu1# show running-config calea
interface-group-source-profile
calea interface-group-source-profile X3-IFC-GRP-SRC-PROFILE
network-context          GN
source-ip-address-name    CALEAL00P
source-port               1030
number-of-source-ports    8
interface-type            x3
interfaceFormat-type      HI_format
transport-type            tcp
output-type               ber
buffer-utilization        70
failover-buffer-threshold 50
cc-buffer-size            102400
tcp-keep-alive            disabled
tcp-connection-retry-interval 10
tcp-user-timeout          60
tcp-send-sock-buffer-size 1024
tpkt-keep-alive           disabled
dscp-value                dscp_0_be
!
```

Example 10: Configuring a Target with Explicit LEMFs Using the Multiple HI2/HI3 Delivery Destinations Feature

To configure a target using the CALEA feature [Multiple X2/X3 Delivery Destinations](#), login as calea user and issue a command such as the following:

```
an3000-mcm-slot7cpu1(config)# calea target-table Mobster_1
identityType imsi identityVal 17567547897049 dataCollectionType
iri-plus-cc location-info true operator-id att gwType sae-gw
x2-primary-ip-address 10.33.39.49 x2-primary-port 7422
x3-primary-ip-address 10.33.39.49 x3-primary-port 7433;top;commit;
Commit complete.
```

```
an3000-mcm-slot7cpu1(config)# exit
```

To view the resulting configuration, use the **show running-config** command as follows:

```
an3000-mcm-slot7cpu1# show running-config calea target-table Mobster_1
calea target-table $3$r/zkEp20u73wQoByA3qE7A==
identityType          imsi
identityVal           $3$JX4DsZ6Twd6dwte5gobZug==
dataCollectionType    iri-plus-cc
gwType                sae-gw
wlanGwType            none
operator-id           att
location-info         true
iri-event-time-stamp-type local-time
cc-data-time-stamp-type local-time
packet-data-header-info-report PDHIR_DISABLE
x2-primary-ip-address 10.33.39.49
x2-primary-port       7422
x3-primary-ip-address 10.33.39.49
x3-primary-port       7433
!
```

As shown in this example, the Lawful Intercept Identifier and IdentityVal values are displayed in encrypted form.

Example 11: Configuring a Target for packet-data-header-info-report Features

When a target is configured with packet-data-header-info-report setting other than PDHIR_DISABLED, additional statistics are displayed in the **show calea target-status** command.

The following additional fields are displayed for a target on which packet-data-header-info-report has been enabled:

Field Name	Value
packet-data-header-info-report	PDHIR_PER_PACKET_MAPPED_HEADER PDHIR_PER_PACKET_RAW_HEADER PDHIR_SUMMARY
number-of-x2-pdhir-per-packet-raw-events-sent	numeric counter
number-of-x2-pdhir-per-packet-mapped-events-sent	numeric counter
number-of-x2-packet-data-summary-report-events-sent	numeric counter

For example:

- 1 Configure a target with packet-data-header-info-report PDHIR_SUMMARY:

```
an3000-mcm-slot7cpu1(config)# calea target-table Mobster_1
identityType imsi identityVal 17567547897049 dataCollectionType
iri-only location-info true operator-id att gwType sae-gw
x2InterfaceGroupName x2-ber-hi-grp packet-data-header-info-report
PDHIR_SUMMARY ;top;commit;
```

Commit complete.

- 2 Because a packet-data-header-info-report feature is enabled (PDHIR_SUMMARY), the **show calea target-status** displays the additional fields, highlighted below:

```
an3000-mcm-slot7cpu1# show calea target-status
calea target-status Mobster_1
number-of-agent-instances                2
identity-type                            imsi
identity-value                           17567547897049
identity-value-mac
x2-interface-group-name                  x2-ber-hi-grp
x3-interface-group-name                  ""
data-collection-type                     iri-only
```

gateway-type-3gpp	sae-gw
gateway-type-wlan	none
packet-data-header-info-report	PDHIR_SUMMARY
number-of-x3-events-dropped-no-interface	0
number-of-x3-events-sent	0
number-of-x3-event-encode-failures-no-buffer	0
number-of-x3-event-xer-encode-failures	0
number-of-x3-event-ber-encode-failures	0
number-of-x2-events-dropped-no-interface	0
number-of-x2-events-sent	11
number-of-x2-pdhir-per-packet-raw-events-sent	0
number-of-x2-pdhir-per-packet-mapped-events-sent	0
number-of-x2-packet-data-summary-report-events-sent	11
number-of-x2-encode-fail-no-buffer	0
number-of-x2-ber-encode-fail	0
number-of-x2-xer-encode-fail	0
number-of-x3-ber-encode-fail	0
number-of-x3-xer-encode-fail	0
storage-type	persistent
total-cumulative-LI-Sessions	2
total-number-of-intercepted-packets-for-current-active-calls	11
number-of-x2-events-for-current-active-calls	11
number-of-x3-events-for-current-active-calls	0
total-number-of-all-terminated-LI-sessions	0
total-number-of-terminated-LI-sessions-due-to-call-disconnect	0
total-number-of-terminated-LI-sessions-due-to-deletion-of-context	0

AGENT LIID	NUMBER OF IDENTITY INSTANCES	IDENTITY TYPE	IDENTITY VALUE	IDENTITY VALUE MAC	x2 INTERFACE NAME
liid_1	2	imei	9991234567850011		X2
liid_2	2	imsi	111345634		X2
liid_3	2	imei	099912345678502		X2